

CIL MT - Paper-II

Questions and answers only. No explanations.

518 questions across 35 chapters

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Asymptotic Complexity and Sorting Algorithms

Algorithms - 5 questions

1. What is the worst-case time complexity of the Merge Sort algorithm for sorting an array of size n ?
(A) $O(n)$
(B) $O(n^2 \log n)$
(C) $O(n \log n)$
(D) $O(n^2)$
Answer: (C) $O(n \log n)$
2. What is meant by a 'stable' sorting algorithm?
(A) A sorting algorithm is stable if it preserves the order of all keys
(B) A sorting algorithm is stable if it preserves the order of non-duplicate keys
(C) A sorting algorithm is stable if it preserves the relative order of duplicate keys
(D) A sorting algorithm is stable if it doesn't preserve the order of duplicate keys
Answer: (C) A sorting algorithm is stable if it preserves the relative order of duplicate keys
3. Consider a Binary Search Tree (BST) containing n elements. What is the average time complexity and worst-case time complexity required to search for a given element in this tree?
(A) Average $O(\log n)$, Worst $O(n)$
(B) Average $O(n \log n)$, Worst $O(n^2)$
(C) Average $O(\log n)$, Worst $O(\log n)$
(D) Average $O(n)$, Worst $O(n)$
Answer: (A) Average $O(\log n)$, Worst $O(n)$
4. Which of the following statements regarding sorting stability is correct?
(A) Insertion Sort and Merge Sort are stable. Heap Sort is inherently unstable.
(B) Selection Sort is stable. Quick Sort is stable.
(C) Heap Sort and Quick Sort are stable. Insertion Sort is unstable.
(D) All comparison-based sorting algorithms are stable.
Answer: (A) Insertion Sort and Merge Sort are stable. Heap Sort is inherently unstable.
5. Using Master's Theorem, solve the following recurrence relation: $T(n) = 4T(n/2) + n^2$.
(A) (n^3)
(B) $(n^2 \log n)$
(C) (n^2)
(D) $(\log n)$
Answer: (B) $(n^2 \log n)$

Graph Algorithms and Hashing

Algorithms - 20 questions

6. Which auxiliary data structure is used to implement Breadth-First Search (BFS) traversal of a graph?
(A) Stack
(B) Queue
(C) Binary Heap
(D) Linked List
Answer: (B) Queue
7. A vertex v in an undirected connected graph G is called an 'articulation point' if:
(A) Removing v and its associated edges increases the number of connected components in G
(B) Removing v and its associated edges decreases the number of connected components in G
(C) v is connected to all other vertices in G
(D) v lies on a cycle in G
Answer: (A) Removing v and its associated edges increases the number of connected components in G

8. Which of the following describes Prim's algorithm for finding the Minimum Spanning Tree (MST) of a connected, weighted undirected graph?
- (A) It starts with a single vertex and grows the MST by repeatedly adding the cheapest edge connecting a vertex in the tree to a vertex outside the tree.
 - (B) It sorts all edges in ascending order of weights and repeatedly adds the smallest edge that does not form a cycle.
 - (C) It is a dynamic programming algorithm that finds shortest paths.
 - (D) It computes shortest paths starting from a source vertex.
- Answer: (A)** It starts with a single vertex and grows the MST by repeatedly adding the cheapest edge connecting a vertex in the tree to a vertex outside the tree.
9. What is the worst-case time complexity of Dijkstra's single-source shortest path algorithm on a graph with V vertices and E edges, when implemented using a binary heap?
- (A) $O(V^2)$
 - (B) $O(E \log V)$
 - (C) $O(V \log V + E)$
 - (D) $O(V + E)$
- Answer: (B)** $O(E \log V)$
10. In hashing with open addressing, if a collision occurs when trying to insert an item, what is 'quadratic probing'?
- (A) Probing successive slots according to a linear function: $h(k, i) = (h'(k) + i) \bmod m$
 - (B) Probing successive slots according to a quadratic function: $h(k, i) = (h'(k) + c_1*i + c_2*i^2) \bmod m$
 - (C) Creating a linked list at the collided index to hold duplicate elements
 - (D) Computing a second hash function to find the next slot
- Answer: (B)** Probing successive slots according to a quadratic function: $h(k, i) = (h'(k) + c_1*i + c_2*i^2) \bmod m$
11. Compare the space complexities of Breadth-First Search (BFS) and Depth-First Search (DFS) when traversing a balanced d -ary tree of depth h and branching factor b . What are the correct bounds for the auxiliary memory required by BFS and DFS respectively?
- (A) BFS: $O(b^h)$, DFS: $O(bh)$
 - (B) BFS: $O(bh)$, DFS: $O(b^h)$
 - (C) BFS: $O(d)$, DFS: $O(h)$
 - (D) BFS: $O(V + E)$, DFS: $O(V + E)$
- Answer: (A)** BFS: $O(b^h)$, DFS: $O(bh)$
12. In graph theory, which of the following rules correctly matches the structural condition used to detect cycles in Directed and Undirected graphs respectively during Depth-First Search (DFS)?
- (A) Directed: Back-edge (visiting a node currently in the recursion stack); Undirected: Visiting an already visited node that is not the immediate parent.
 - (B) Directed: Forward-edge (visiting a descendant); Undirected: Back-edge (visiting an ancestor in the stack).
 - (C) Directed: Cross-edge (visiting a sibling subtree); Undirected: Visiting a leaf node twice.
 - (D) Both Directed and Undirected use the exact same condition: detecting any edge leading to an already visited vertex.
- Answer: (A)** Directed: Back-edge (visiting a node currently in the recursion stack); Undirected: Visiting an already visited node that is not the immediate parent.
13. A Disjoint Set Union (DSU) data structure is used to maintain partition states of n elements. If we implement the DSU with both 'Path Compression' and 'Union by Rank/Size' optimizations, what is the worst-case amortized time complexity of executing a sequence of m find and union operations?
- (A) $O(m \log n)$
 - (B) $O(m * \alpha(n))$
 - (C) $O(n + m \log^* n)$
 - (D) $O(m)$
- Answer: (B)** $O(m * \alpha(n))$
14. Consider a directed graph G with V vertices and E edges containing negative edge weights. The Bellman-Ford single-source shortest path algorithm is executed to find shortest distances. During which step can we mathematically guarantee the detection of a negative-weight cycle in G ?
- (A) If any distance value gets updated during the $(V-1)$ -th iteration of relaxing all E edges.
 - (B) If any distance value gets updated during the V -th iteration of relaxing all E edges, after completing $V-1$ iterations.
 - (C) If all distance values remain unchanged during the first iteration.
 - (D) If the sum of all edge weights is negative.
- Answer: (B)** If any distance value gets updated during the V -th iteration of relaxing all E edges, after completing $V-1$ iterations.

15. The Floyd-Warshall all-pairs shortest path algorithm uses a dynamic programming approach with three nested loops. To ensure correctness, which variable must be iterated in the outermost loop, and what are the respective time and space complexities?
- (A) The outermost loop must iterate over the intermediate vertex k ; Time: $O(V^3)$, Space: $O(V^2)$
 (B) The outermost loop must iterate over the source vertex i ; Time: $O(V^3)$, Space: $O(V^3)$
 (C) The outermost loop must iterate over the destination vertex j ; Time: $O(V^2 \log V)$, Space: $O(V^2)$
 (D) The outermost loop can be i , j , or k interchangeably; Time: $O(V^3)$, Space: $O(V^2)$
- Answer: (A) The outermost loop must iterate over the intermediate vertex k ; Time: $O(V^3)$, Space: $O(V^2)$**
16. According to the fundamental properties of Minimum Spanning Trees (MST) of a weighted connected graph G , which of the following statements is mathematically true?
- (A) Cycle Property: For any cycle in G , the edge with the strictly largest weight cannot be part of any MST of G .
 (B) Cut Property: For any partition (cut) of G , the maximum weight edge crossing the cut must be in the MST.
 (C) An MST of a graph G with V vertices always contains exactly V edges and $V-1$ vertices.
 (D) If all edge weights in G are distinct, there can still be multiple different MSTs.
- Answer: (A) Cycle Property: For any cycle in G , the edge with the strictly largest weight cannot be part of any MST of G .**
17. Topological sorting of a directed graph G represents a linear ordering of its vertices such that for every directed edge $u \rightarrow v$, vertex u comes before v . Under which graph structure is topological sort possible, and which algorithm computes it using in-degrees of nodes?
- (A) Possible in any Directed Acyclic Graph (DAG); Computed via Kahn's Algorithm.
 (B) Possible in any Connected Directed Graph; Computed via Prim's Algorithm.
 (C) Possible only in Directed Graphs with negative cycles; Computed via Bellman-Ford.
 (D) Possible only in Bipartite Directed Graphs; Computed via Kosaraju's Algorithm.
- Answer: (A) Possible in any Directed Acyclic Graph (DAG); Computed via Kahn's Algorithm.**
18. A double hashing scheme resolves collisions in a hash table of size $M = 11$ using the primary hash function $h_1(k) = k \bmod 11$ and a secondary hash function $h_2(k) = 7 - (k \bmod 7)$. If we insert the key $k = 27$ into an initially empty table, and find that slot $h_1(27)$ is already occupied, what is the next probe slot checked by the double hashing algorithm?
- (A) Slot 6
 (B) Slot 7
 (C) Slot 8
 (D) Slot 5
- Answer: (A) Slot 6**
19. In hashing with open addressing, 'Linear Probing' is highly cache-friendly but suffers from a major performance degradation issue called 'Primary Clustering'. Which of the following describes this issue?
- (A) Long consecutive blocks of occupied slots build up, making collision search times approach $O(N)$ as the table fills.
 (B) Keys with different initial hash values follow identical quadratic probe sequences, causing secondary collisions.
 (C) The linked lists at each hash index grow too long, degrading search to linear scan.
 (D) The secondary hash function evaluates to 0, locking the search in an infinite loop.
- Answer: (A) Long consecutive blocks of occupied slots build up, making collision search times approach $O(N)$ as the table fills.**
20. A hash table has $M = 100$ slots and stores $n = 70$ elements, resulting in a load factor $\alpha = 0.70$. According to the expected search complexity formulas, how do Separate Chaining and Open Addressing (Linear Probing) compare regarding the expected number of comparisons for a successful search?
- (A) Separate Chaining: 1.35 comparisons; Linear Probing: 3.33 comparisons
 (B) Separate Chaining: 1.70 comparisons; Linear Probing: 1.35 comparisons
 (C) Separate Chaining: 3.33 comparisons; Linear Probing: 1.35 comparisons
 (D) Both require exactly 1.70 comparisons on average.
- Answer: (A) Separate Chaining: 1.35 comparisons; Linear Probing: 3.33 comparisons**
21. Kosaraju's algorithm is an efficient DFS-based technique used to find the Strongly Connected Components (SCCs) of a directed graph G . Which sequence of steps correctly describes this algorithm, and what is its time complexity?
- (A) 1) Perform DFS on G and push vertices to stack by finishing times; 2) Reverse all directed edges in G ; 3) Pop from stack and perform DFS on the reversed graph to find SCCs. Time: $O(V + E)$
 (B) 1) Sort all edge weights; 2) Run BFS from the source vertex; 3) Build a spanning tree avoiding cycles. Time: $O(E \log V)$

(C) 1) Partition graph into bipartite sets; 2) Run Dijkstra's algorithm from each source; 3) Combine shortest paths. Time: $O(V^3)$

(D) 1) Build an adjacency matrix; 2) Perform matrix multiplication; 3) Identify zero rows. Time: $O(V^2)$

Answer: (A) 1) Perform DFS on G and push vertices to stack by finishing times; 2) Reverse all directed edges in G; 3) Pop from stack and perform DFS on the reversed graph to find SCCs. Time: $O(V + E)$

22. Boruvka's algorithm is a historical Minimum Spanning Tree (MST) algorithm. What is its key operational mechanism, and why is it considered highly advantageous in modern high-performance computing architectures?

(A) It grows several components simultaneously by connecting each component to its cheapest neighbor; highly advantageous because it is naturally parallelizable.

(B) It always selects edges starting from a single root node; advantageous because it uses less stack space than DFS.

(C) It replaces the graph representation with a dense matrix; advantageous because it avoids pointer chasing in cache memories.

(D) It utilizes a second hash function to rehash edge lists; advantageous because it avoids cycles in $O(1)$ time.

Answer: (A) It grows several components simultaneously by connecting each component to its cheapest neighbor; highly advantageous because it is naturally parallelizable.

23. A bipartite graph is a graph whose vertices can be divided into two disjoint sets such that no two adjacent vertices belong to the same set. Which cycle condition characterizes bipartite graphs, and what is the time complexity to verify this property?

(A) The graph must contain no cycles of odd length; verified in $O(V + E)$ using BFS or DFS.

(B) The graph must contain only cycles of even length; verified in $O(V^2)$ using Floyd-Warshall.

(C) The graph must contain no cycles of any length; verified in $O(V \log V)$ using disjoint sets.

(D) The graph must be a tree structure; verified in $O(E \log E)$ using Kruskal's.

Answer: (A) The graph must contain no cycles of odd length; verified in $O(V + E)$ using BFS or DFS.

24. In hash function designs, why is the 'Division Method' formula $h(k) = k \bmod M$ implemented ideally with M chosen as a prime number that is not close to a power of 2?

(A) Because if M has small factors, keys that share those common factors will all map to the same slots, causing severe clustering.

(B) Because modulo division with prime numbers executes faster in standard CPU ALUs.

(C) Because prime table sizes automatically eliminate all collisions by the Pigeonhole Principle.

(D) Because prime numbers prevent the load factor from ever exceeding 0.70.

Answer: (A) Because if M has small factors, keys that share those common factors will all map to the same slots, causing severe clustering.

25. In hashing with open addressing, what is the operation of 'Rehashing', and when is it executed?

(A) Allocating a new table of size roughly $2x$, and reinserting all existing keys using a modified hash function; executed when the load factor α exceeds a threshold (typically 0.70).

(B) Applying a second hash function to the collided slot to locate another index; executed on every collision.

(C) Converting a separate chaining hash table into a red-black tree; executed when a linked list length exceeds 8.

(D) Clearing all 'deleted' tombstone markers from the table; executed when the table is empty.

Answer: (A) Allocating a new table of size roughly $2x$, and reinserting all existing keys using a modified hash function; executed when the load factor α exceeds a threshold (typically 0.70).

Greedy, Dynamic Programming, and Divide & Conquer

Algorithms - 20 questions

26. What is the optimal substructure and overlapping subproblems property that distinguishes Dynamic Programming from Greedy algorithms?

(A) Greedy algorithms solve subproblems recursively, while Dynamic Programming is always iterative.

(B) Dynamic Programming requires that subproblems are independent, whereas Greedy requires overlapping subproblems.

(C) Dynamic Programming solves all overlapping subproblems and memoizes/caches results, whereas a Greedy algorithm makes a single locally optimal choice without backtracking.

(D) Dynamic Programming requires the Greedy Choice property to hold at each step.

Answer: (C) Dynamic Programming solves all overlapping subproblems and memoizes/caches results, whereas a Greedy algorithm makes a single locally optimal choice without backtracking.

27. Consider a set of five characters {a, b, c, d, e} with frequencies 0.35, 0.25, 0.20, 0.15, and 0.05 respectively. If Huffman Coding is applied to find optimal prefix codes, what is the expected average code length per character?
- (A) 2.05 bits
(B) 2.10 bits
(C) 2.20 bits
(D) 2.35 bits
- Answer: (C) 2.20 bits**
28. What is the tight asymptotic upper bound of the recurrence relation $T(n) = 3T(n/4) + n \log n$ according to Master's Theorem?
- (A) (n)
(B) $(n \log n)$
(C) $(n^{\log_4(3)})$
(D) (n^2)
- Answer: (B) $(n \log n)$**
29. In the context of Dynamic Programming, how many entries of the DP table are computed to solve the Matrix Chain Multiplication problem for 4 matrices with dimensions 10x20, 20x30, 30x40, and 40x30?
- (A) 6
(B) 10
(C) 16
(D) 20
- Answer: (A) 6**
30. Which of the following problems does NOT have an optimal greedy solution, requiring Dynamic Programming or branch-and-bound for an exact solution?
- (A) Fractional Knapsack Problem
(B) Single-Source Shortest Path (Dijkstra's with non-negative weights)
(C) 0/1 Knapsack Problem
(D) Activity Selection Problem
- Answer: (C) 0/1 Knapsack Problem**
31. An algorithm for simultaneously locating both the minimum and maximum elements in an unsorted array of size n is implemented. Instead of performing two separate linear passes which takes $2n - 2$ comparisons, a Divide and Conquer approach compares elements in pairs. What is the exact minimum number of comparisons needed under this optimal Divide and Conquer approach?
- (A) $3n/2 - 2$ comparisons
(B) $2n - 3$ comparisons
(C) $n - 1$ comparisons
(D) $n \log n$ comparisons
- Answer: (A) $3n/2 - 2$ comparisons**
32. Consider the Activity Selection Problem with 6 activities having start and finish times as follows: A1: (1, 4), A2: (3, 5), A3: (0, 6), A4: (5, 7), A5: (3, 9), A6: (5, 9). What is the maximum number of non-overlapping activities that a single machine can execute, and which greedy parameter drives the selection?
- (A) 3 activities; driven by selecting the activity with the earliest finish time.
(B) 4 activities; driven by selecting the activity with the shortest duration.
(C) 3 activities; driven by selecting the activity with the earliest start time.
(D) 2 activities; driven by selecting the activity with the maximum profit.
- Answer: (A) 3 activities; driven by selecting the activity with the earliest finish time.**
33. A student needs to schedule five jobs with deadlines and profits to maximize total profit using the Job Sequencing with Deadlines algorithm. The jobs are: J1: (d=2, p=100), J2: (d=1, p=19), J3: (d=2, p=27), J4: (d=1, p=25), J5: (d=3, p=15). Each job takes 1 unit of time. What is the optimal job schedule and the maximum profit achieved respectively?
- (A) Optimal schedule: [J3, J1, J5]; Profit: 142
(B) Optimal schedule: [J1, J3, J4]; Profit: 152
(C) Optimal schedule: [J1, J2, J5]; Profit: 134
(D) Optimal schedule: [J4, J1, J3]; Profit: 152
- Answer: (A) Optimal schedule: [J3, J1, J5]; Profit: 142**

34. A fractional knapsack has a total weight capacity of $W = 50$. Three items are available with values and weights as follows: Item 1: ($v=60, w=10$), Item 2: ($v=100, w=20$), Item 3: ($v=120, w=30$). What is the maximum possible value that can be carried in the knapsack, and what is the key greedy choice parameter?
- (A) Value: 240; chosen greedily by sorting items by Value/Weight ratio.
 (B) Value: 220; chosen greedily by sorting items by lowest weight first.
 (C) Value: 280; chosen greedily by sorting items by maximum value first.
 (D) Value: 200; chosen greedily by sorting items by weight in descending order.
- Answer: (A)** Value: 240; chosen greedily by sorting items by Value/Weight ratio.
35. In the Coin Change Problem, a greedy strategy always selects the largest coin denomination less than or equal to the remaining amount. Under which of the following coin denomination sets does this greedy strategy FAIL to yield the minimum number of coins, requiring a Dynamic Programming solution instead?
- (A) Coins = {1, 2, 5, 10, 20, 50, 100}, Target = 41
 (B) Coins = {1, 5, 10, 25}, Target = 30
 (C) Coins = {1, 3, 4}, Target = 6
 (D) Coins = {1, 2, 4, 8}, Target = 12
- Answer: (C)** Coins = {1, 3, 4}, Target = 6
36. A dynamic programming approach is used to solve the 0/1 Knapsack Problem. If n is the number of items and W is the maximum capacity of the knapsack, what is the recurrence relation used to fill the DP table, and what is the time and space complexity of the standard tabulation method?
- (A) $dp[i][w] = \max(dp[i-1][w], dp[i-1][w-w_i] + v_i)$ if $w_i \leq w$; Time: $O(nW)$, Space: $O(nW)$
 (B) $dp[i][w] = dp[i-1][w] + dp[i][w-w_i] + v_i$; Time: $O(n^2)$, Space: $O(W)$
 (C) $dp[i][w] = \max(dp[i][w], dp[i-1][w-1] + v_i)$; Time: $O(2^n)$, Space: $O(n)$
 (D) $dp[w] = \max(dp[w], dp[w-w_i] + v_i)$; Time: $O(nW)$, Space: $O(1)$
- Answer: (A)** $dp[i][w] = \max(dp[i-1][w], dp[i-1][w-w_i] + v_i)$ if $w_i \leq w$; Time: $O(nW)$, Space: $O(nW)$
37. Consider the Longest Common Subsequence (LCS) problem for two strings $X = \text{"ABCBDAB"}$ and $Y = \text{"BDCABA"}$. What is the length of the LCS of these strings, and how does the dynamic programming recurrence handle matching and non-matching characters?
- (A) Length: 4; If $X[i] == Y[j]$, $dp[i][j] = dp[i-1][j-1] + 1$, else $dp[i][j] = \max(dp[i-1][j], dp[i][j-1])$.
 (B) Length: 5; If $X[i] == Y[j]$, $dp[i][j] = dp[i-1][j-1] + 1$, else $dp[i][j] = dp[i-1][j] + dp[i][j-1]$.
 (C) Length: 3; If $X[i] == Y[j]$, $dp[i][j] = dp[i][j-1] + 1$, else $dp[i][j] = \min(dp[i-1][j], dp[i][j-1])$.
 (D) Length: 4; If $X[i] == Y[j]$, $dp[i][j] = dp[i-1][j-1]$, else $dp[i][j] = \max(dp[i-1][j], dp[i][j-1]) + 1$.
- Answer: (A)** Length: 4; If $X[i] == Y[j]$, $dp[i][j] = dp[i-1][j-1] + 1$, else $dp[i][j] = \max(dp[i-1][j], dp[i][j-1])$.
38. A dynamic programming algorithm computes the Longest Increasing Subsequence (LIS) of an array of size n . What is the time complexity of the standard DP approach using the recurrence $dp[i] = \max(dp[j] + 1)$ for $j < i$ and $arr[j] < arr[i]$, and what is the optimized time complexity when using a binary search tail-expansion strategy?
- (A) Standard DP: $O(n^2)$; Optimized: $O(n \log n)$
 (B) Standard DP: $O(n \log n)$; Optimized: $O(n)$
 (C) Standard DP: $O(n^2)$; Optimized: $O(n^2)$
 (D) Standard DP: $O(2^n)$; Optimized: $O(n \log n)$
- Answer: (A)** Standard DP: $O(n^2)$; Optimized: $O(n \log n)$
39. The Levenshtein Edit Distance computes the minimum number of insertion, deletion, or replacement operations to convert a string X into Y . If $X = \text{"SUNDAY"}$ and $Y = \text{"SATURDAY"}$, what is the computed Edit Distance, and how does the mismatch recurrence handle operations?
- (A) Edit Distance: 3; Mismatch cost = $1 + \min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1])$ which represent Delete, Insert, and Replace respectively.
 (B) Edit Distance: 4; Mismatch cost = $1 + \min(dp[i-1][j], dp[i][j-1])$ which represent Delete and Insert.
 (C) Edit Distance: 3; Mismatch cost = $1 + \max(dp[i-1][j], dp[i][j-1], dp[i-1][j-1])$.
 (D) Edit Distance: 2; Mismatch cost = $\min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1])$.
- Answer: (A)** Edit Distance: 3; Mismatch cost = $1 + \min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1])$ which represent Delete, Insert, and Replace respectively.

40. A grid DP problem requires finding the number of unique paths from the top-left cell (0,0) to the bottom-right cell (m-1, n-1) of an m x n grid with no obstacles, moving only right or down. Which combinatorial formula gives the exact number of unique paths, and what is the answer for a 3 x 3 grid?
- (A) Formula: $C(m+n-2, m-1)$; 3x3 grid = 6 paths
 (B) Formula: $C(m+n, m)$; 3x3 grid = 20 paths
 (C) Formula: $(m * n) / 2$; 3x3 grid = 4 paths
 (D) Formula: $2^{(m+n-2)}$; 3x3 grid = 16 paths
- Answer: (A) Formula: $C(m+n-2, m-1)$; 3x3 grid = 6 paths**
41. In the Matrix Chain Multiplication problem, we are given a chain of three matrices A1, A2, and A3 with dimensions [10, 30, 5, 60] respectively. What is the optimal parenthesization of these matrices to minimize the number of scalar multiplications, and what is the minimum cost?
- (A) Optimal parenthesization: $(A1 * A2) * A3$; Minimum Cost: 4,500 multiplications
 (B) Optimal parenthesization: $A1 * (A2 * A3)$; Minimum Cost: 27,000 multiplications
 (C) Optimal parenthesization: $(A1 * A2) * A3$; Minimum Cost: 1,500 multiplications
 (D) Both parenthesizations yield identical costs of 15,000 multiplications.
- Answer: (A) Optimal parenthesization: $(A1 * A2) * A3$; Minimum Cost: 4,500 multiplications**
42. Strassen's Matrix Multiplication algorithm is a classic Divide and Conquer strategy for multiplying two n x n matrices. By reducing the number of recursive matrix multiplications, it lowers the asymptotic time complexity. How many recursive multiplications are performed, and what is the resulting time complexity?
- (A) 7 recursive multiplications; Time complexity: $O(n^{2.807})$
 (B) 8 recursive multiplications; Time complexity: $O(n^3)$
 (C) 6 recursive multiplications; Time complexity: $O(n^{2.585})$
 (D) 7 recursive multiplications; Time complexity: $O(n^{\log_2 8})$
- Answer: (A) 7 recursive multiplications; Time complexity: $O(n^{2.807})$**
43. The Karatsuba Algorithm is an efficient Divide and Conquer method used to multiply two large n-digit integers. What is the number of single-digit multiplication operations required, and what is its asymptotic time complexity?
- (A) 3 recursive multiplications; Time complexity: $O(n^{1.585})$
 (B) 4 recursive multiplications; Time complexity: $O(n^2)$
 (C) 3 recursive multiplications; Time complexity: $O(n \log n)$
 (D) 2 recursive multiplications; Time complexity: $O(n^{1.5})$
- Answer: (A) 3 recursive multiplications; Time complexity: $O(n^{1.585})$**
44. The Tower of Hanoi is a classic mathematical problem modeled by an exponential recurrence relation. If $T(n)$ represents the minimum number of moves to transfer n disks from one peg to another, what is the recurrence relation with its base case, and what is the closed-form solution for n = 10 disks?
- (A) $T(n) = 2T(n-1) + 1$, $T(1) = 1$; Closed-form: 1,023 moves
 (B) $T(n) = T(n-1) + 2$, $T(1) = 1$; Closed-form: 20 moves
 (C) $T(n) = 2T(n/2) + n$, $T(1) = 1$; Closed-form: 1,024 moves
 (D) $T(n) = 2T(n-1) + 1$, $T(1) = 1$; Closed-form: 511 moves
- Answer: (A) $T(n) = 2T(n-1) + 1$, $T(1) = 1$; Closed-form: 1,023 moves**
45. Under the Master Theorem, we are given a recurrence relation $T(n) = 2T(n/2) + n \log n$. Which case of the Master Theorem applies, and what is the tight asymptotic bound?
- (A) Case 2 (Extended); Tight bound is $(n \log^2 n)$
 (B) Case 2 (Standard); Tight bound is $(n \log n)$
 (C) Case 3 (Standard); Tight bound is (n^2)
 (D) Case 1 (Standard); Tight bound is (n)
- Answer: (A) Case 2 (Extended); Tight bound is $(n \log^2 n)$**

Lexical Analysis and Parsing

Compiler Design - 15 questions

46. How many tokens are present in the following C statement?
- ```
printf("Value of x = %d", &x);
```
- (A) 5  
 (B) 7  
 (C) 8  
 (D) 9
- Answer: (C) 8**

47. An LL(1) parser is a:

- (A) Top-down parser with one symbol lookahead
- (B) Bottom-up parser with one symbol lookahead
- (C) Bottom-up parser with multiple symbol lookahead
- (D) Top-down parser with multiple symbol lookahead

**Answer: (A) Top-down parser with one symbol lookahead**

48. Which of the following describes the power/capability order of bottom-up LR parsers?

- (A)  $LR(0) < SLR(1) < LALR(1) < CLR(1)$
- (B)  $LR(0) < LALR(1) < SLR(1) < CLR(1)$
- (C)  $SLR(1) < LR(0) < LALR(1) < CLR(1)$
- (D)  $CLR(1) < LALR(1) < SLR(1) < LR(0)$

**Answer: (A)  $LR(0) < SLR(1) < LALR(1) < CLR(1)$**

49. Consider the grammar rules:

$S \rightarrow Aa \mid b$

$A \rightarrow b \mid$

What is Follow(A) for this grammar?

- (A) {a}
- (B) {a, b}
- (C) {a, \$}
- (D) {b, \$}

**Answer: (A) {a}**

50. In compilation, what is the role of the Symbol Table?

- (A) To perform syntax validation and print syntax errors
- (B) To record details about variable names, types, scope, and memory offsets across all compilation phases
- (C) To generate machine-independent assembly code
- (D) To optimize the register allocation in the code generation phase

**Answer: (B) To record details about variable names, types, scope, and memory offsets across all compilation phases**

51. In a two-buffer input scheme for a lexical analyzer (where each buffer is of size N), the primary advantage of placing a sentinel character (EOF) at the end of each buffer is to:

- (A) Allow the scanner to support unlimited backtracking
- (B) Eliminate the need for double-boundary tests on every character scanned
- (C) Decrease the size of the buffer needed on disk
- (D) Facilitate direct parallel parsing of the token stream

**Answer: (B) Eliminate the need for double-boundary tests on every character scanned**

52. If a lexical analyzer encounters an erroneous input sequence '3apple' where a valid identifier is expected, which of the following recovery strategies would delete the starting digit '3' to resolve the error?

- (A) Transposition
- (B) Deletion
- (C) Replacement
- (D) Panic Mode

**Answer: (B) Deletion**

53. Consider the following grammar with direct left recursion:

$S \rightarrow S a \mid b \mid c$

Which of the following represents the equivalent grammar with left recursion eliminated?

- (A)  $S \rightarrow (b \mid c) S'$ ,  $S' \rightarrow a S' \mid$
- (B)  $S \rightarrow a S' \mid$ ,  $S' \rightarrow b \mid c$
- (C)  $S \rightarrow S' (b \mid c)$ ,  $S' \rightarrow S' a \mid$
- (D)  $S \rightarrow a S \mid b S \mid c$

**Answer: (A)  $S \rightarrow (b \mid c) S'$ ,  $S' \rightarrow a S' \mid$**

54. Consider the grammar rules:

$S \rightarrow i E t S \mid i E t S e S \mid a$

This grammar has a common prefix which prevents an LL(1) parser from choosing a production. The left-factored equivalent grammar is:

- (A)  $S \rightarrow i E t S S' \mid a$ ,  $S' \rightarrow e S \mid$
- (B)  $S \rightarrow i E t S S' \mid a$ ,  $S' \rightarrow e \mid$
- (C)  $S \rightarrow S' a \mid i E t S$ ,  $S' \rightarrow e S \mid$



(D)  $S \rightarrow iEtS \mid a, S' \rightarrow e \mid$

**Answer: (A)**  $S \rightarrow iEtSS' \mid a, S' \rightarrow eS \mid$

55. Consider the grammar:

$S \rightarrow ABC$

$A \rightarrow a \mid$

$B \rightarrow b \mid$

$C \rightarrow c$

What are the  $FIRST(S)$  and  $FOLLOW(B)$  sets respectively?

(A)  $FIRST(S) = \{a, b, c\}$ ,  $FOLLOW(B) = \{c\}$

(B)  $FIRST(S) = \{a, b, c\}$ ,  $FOLLOW(B) = \{b, c\}$

(C)  $FIRST(S) = \{a, b\}$ ,  $FOLLOW(B) = \{c, \$\}$

(D)  $FIRST(S) = \{a, b, c\}$ ,  $FOLLOW(B) = \{a, b, c\}$

**Answer: (A)**  $FIRST(S) = \{a, b, c\}$ ,  $FOLLOW(B) = \{c\}$

56. A grammar is given by:

$S \rightarrow aS \mid a$

Why is this grammar NOT LL(1)?

(A) It is left-recursive

(B)  $FIRST(aS) \cap FIRST(a) = \{a\} \neq \emptyset$

(C) It is ambiguous

(D) It contains an  $\epsilon$ -production which clashes with the start symbol

**Answer: (B)**  $FIRST(aS) \cap FIRST(a) = \{a\} \neq \emptyset$

57. If a grammar has  $N$  states in its LR(0) collection of items, what is the exact relationship between the state counts of its LR(0), SLR(1), LALR(1), and CLR(1) parsing tables?

(A)  $LR(0) = SLR(1) = LALR(1) \leq CLR(1)$

(B)  $LR(0) < SLR(1) < LALR(1) < CLR(1)$

(C)  $LR(0) = SLR(1) < LALR(1) = CLR(1)$

(D)  $LR(0) = SLR(1) = CLR(1) < LALR(1)$

**Answer: (A)**  $LR(0) = SLR(1) = LALR(1) \leq CLR(1)$

58. A bottom-up parser's action table contains a 'Shift-Reduce' conflict in state  $I$  on lookahead 'a'. This means:

(A) The parser has a choice between shifting 'a' onto the stack and reducing the stack using some production rule on lookahead 'a'

(B) The parser has a choice between reducing using two different production rules

(C) The stack has reached its maximum size limit and must be popped

(D) The compiler cannot decide whether to record an identifier or a keyword

**Answer: (A)** The parser has a choice between shifting 'a' onto the stack and reducing the stack using some production rule on lookahead 'a'

59. In an LR(0) automaton, which of the following items is classified as a 'Kernel Item'?

(A)  $A \rightarrow XYZ$

(B)  $S' \rightarrow S$  (where  $S'$  is the augmented start symbol)

(C)  $B \rightarrow$

(D)  $C \rightarrow XY$

**Answer: (D)**  $C \rightarrow XY$

60. Which of the following describes the grammatical restrictions of Operator Precedence Grammars and their corresponding bottom-up parsers?

(A) They allow arbitrary  $\epsilon$ -productions but forbid recursive rules

(B) No production can have an empty RHS ( $\epsilon$ ) and no two non-terminals can be adjacent on the RHS of any production

(C) All productions must be left-recursive and have at least two terminals

(D) They only support regular grammars with no variables on the RHS

**Answer: (B)** No production can have an empty RHS ( $\epsilon$ ) and no two non-terminals can be adjacent on the RHS of any production

## SDT, Intermediate Code, and Optimization

Compiler Design - 15 questions

61. An S-attributed Syntax Directed Translation (SDT) can be evaluated during bottom-up parsing because S-attributed definitions use:
- (A) Only inherited attributes
  - (B) Only synthesized attributes
  - (C) Both inherited and synthesized attributes
  - (D) No attributes
- Answer: (B) Only synthesized attributes**
62. Which of the following compiler optimization techniques replaces a multiplication operation with a sequence of additions or bit-shifts?
- (A) Loop Unrolling
  - (B) Constant Folding
  - (C) Strength Reduction
  - (D) Common Subexpression Elimination
- Answer: (C) Strength Reduction**
63. Consider the following statements regarding L-attributed definitions:
- I. Every L-attributed definition is also an S-attributed definition.
  - II. Every S-attributed definition is also an L-attributed definition.
- Which of the statements is/are TRUE?
- (A) Only I
  - (B) Only II
  - (C) Both I and II
  - (D) Neither I nor II
- Answer: (B) Only II**
64. In intermediate code generation, a Three-Address Code (3AC) instruction contains at most how many operands?
- (A) 1
  - (B) 2
  - (C) 3
  - (D) 4
- Answer: (C) 3**
65. What does 'constant folding' optimization do?
- (A) It folds loops with constant ranges into a single block.
  - (B) It evaluates expressions with constant values at compile-time instead of running them at runtime.
  - (C) It allocates global variables to constant registers.
  - (D) It removes code blocks that are never executed.
- Answer: (B) It evaluates expressions with constant values at compile-time instead of running them at runtime.**
66. If the attribute dependency graph of a parsed syntax tree has a cycle, which of the following is true regarding attribute evaluation?
- (A) Attributes can be evaluated using a post-order traversal
  - (B) Attributes cannot be evaluated for that parse tree
  - (C) The evaluation must be performed from right-to-left
  - (D) Only inherited attributes can be evaluated
- Answer: (B) Attributes cannot be evaluated for that parse tree**
67. In an L-attributed Syntax Directed Translation Scheme (SDTS) evaluated during top-down LL(1) parsing, where can the semantic actions be placed?
- (A) Only at the extreme right end of the production RHS
  - (B) Anywhere within the RHS of the production
  - (C) Only before any terminal symbols
  - (D) Only at the left end of the production before the non-terminals
- Answer: (B) Anywhere within the RHS of the production**

68. In the standard virtual memory layout of a running program, how do the Stack and Heap segments grow relative to each other, and why?
- (A) Both segments grow upward to prevent address collision
  - (B) The stack grows downward (high address to low address) and the heap grows upward (low address to high address) towards each other to maximize free memory utility
  - (C) The stack grows upward and the heap grows downward away from each other
  - (D) Both segments are allocated static, fixed sizes that do not grow
- Answer: (B)** The stack grows downward (high address to low address) and the heap grows upward (low address to high address) towards each other to maximize free memory utility
69. Which field in a function's activation record (stack frame) is used to point to the activation record of the lexically enclosing (statically parent) scope to access non-local variables?
- (A) Control Link (Dynamic Link)
  - (B) Access Link (Static Link)
  - (C) Saved Machine State
  - (D) Frame Pointer
- Answer: (B)** Access Link (Static Link)
70. If a programming language supports both recursion (functions calling themselves) and dynamic arrays whose dimensions are determined at runtime, which of the following storage allocation strategies must the runtime environment support?
- (A) Static allocation only
  - (B) Stack allocation only
  - (C) Heap allocation only
  - (D) Both Stack and Heap allocation
- Answer: (D)** Both Stack and Heap allocation
71. In parameter passing, under which mechanism is the actual parameter expression evaluated lazily and re-evaluated every time the formal parameter is referenced in the procedure body?
- (A) Call by Value
  - (B) Call by Reference
  - (C) Call by Value-Result
  - (D) Call by Name
- Answer: (D)** Call by Name
72. In intermediate representation, what is the main difference between Quadruples and Triples?
- (A) Quadruples require explicit temporary variable names to store results, whereas Triples refer to intermediate results implicitly using the instruction indices
  - (B) Triples require 4 addresses while Quadruples require 3 addresses
  - (C) Quadruples are harder to re-order for optimization than Triples
  - (D) Triples do not support unary operator instructions
- Answer: (A)** Quadruples require explicit temporary variable names to store results, whereas Triples refer to intermediate results implicitly using the instruction indices
73. What is the total number of nodes (including leaves and internal operators) in the minimal Directed Acyclic Graph (DAG) for the expression:  
 $x = (a + b) * (a + b) + c$
- (A) 5
  - (B) 6
  - (C) 7
  - (D) 9
- Answer: (B)** 6
74. Which of the following compiler optimization techniques moves a computation that yields the same value in every iteration of a loop to a point outside the loop?
- (A) Constant Folding
  - (B) Loop Invariant Code Motion (Code Motion)
  - (C) Strength Reduction
  - (D) Dead Code Elimination
- Answer: (B)** Loop Invariant Code Motion (Code Motion)

75. In Static Single Assignment (SSA) form, every variable is assigned exactly once. What mathematical helper function is used to select the correct version of a variable when two distinct control flow paths merge?
- (A) mu-function (mu-function)
  - (B) lambda-function (lambda-function)
  - (C) -function (phi-function)
  - (D) theta-function (theta-function)
- Answer: (C) -function (phi-function)**

## OSI Model and TCP IP

Computer Networks - 20 questions

76. Which layer of the OSI reference model is responsible for translation, encryption, and compression of data?
- (A) Session Layer
  - (B) Presentation Layer
  - (C) Application Layer
  - (D) Transport Layer
- Answer: (B) Presentation Layer**
77. A protocol data unit (PDU) at the Transport layer of the OSI model is called a:
- (A) Packet
  - (B) Frame
  - (C) Segment or Datagram
  - (D) Bit
- Answer: (C) Segment or Datagram**
78. Which of the following field in IPv4 header is used to prevent routing loops by limiting packet lifetime?
- (A) Header Checksum
  - (B) Time to Live (TTL)
  - (C) Protocol
  - (D) Identification
- Answer: (B) Time to Live (TTL)**
79. In the TCP/IP suite, which protocol is responsible for resolving a logical IP address to a physical MAC address?
- (A) DHCP
  - (B) DNS
  - (C) ARP
  - (D) ICMP
- Answer: (C) ARP**
80. What is the size of the TCP Header without any optional fields?
- (A) 10 bytes
  - (B) 20 bytes
  - (C) 32 bytes
  - (D) 40 bytes
- Answer: (B) 20 bytes**
81. A packet is sent from source host S to destination host D through 5 intermediate routers. How many times is the Network layer visited in total during this transmission?
- (A) 5 times
  - (B) 6 times
  - (C) 7 times
  - (D) 10 times
- Answer: (C) 7 times**
82. A packet is sent from a source host to a destination host through 4 intermediate routers. How many times is the Data Link layer visited during the entire journey?
- (A) 4 times
  - (B) 5 times
  - (C) 8 times
  - (D) 10 times
- Answer: (D) 10 times**

**83.** Which of the following functions is primarily handled by the Session Layer of the OSI reference model?

- (A) Process-to-process reliable data delivery
- (B) User log-in, log-out operations, session synchronization, and token management
- (C) Syntax conversion, character code translation, and data compression
- (D) Mechanical and electrical interface specifications

**Answer: (B)** User log-in, log-out operations, session synchronization, and token management

**84.** In contrast to IPv4, how does an IPv6 router handle an incoming packet that exceeds the Maximum Transmission Unit (MTU) of the next outgoing link?

- (A) The router fragments the packet into smaller fragments and forwards them
- (B) The router buffers the packet and waits for link congestion to clear
- (C) The router discards the packet and sends an 'ICMPv6 Packet Too Big' message back to the source
- (D) The router converts the packet format to IPv4 and retries transmission

**Answer: (C)** The router discards the packet and sends an 'ICMPv6 Packet Too Big' message back to the source

**85.** What is the primary purpose of the 20-bit 'Flow Label' field in the IPv6 header?

- (A) To specify the total size of any active extension headers
- (B) To identify and handle delay-sensitive real-time traffic flows without analyzing inner payloads
- (C) To determine the maximum number of hops a packet can traverse before being dropped
- (D) To encrypt the packet payload using symmetric cryptography

**Answer: (B)** To identify and handle delay-sensitive real-time traffic flows without analyzing inner payloads

**86.** If the 4-bit 'Data Offset' (Header Length) field in a TCP header has the binary value '1010', what is the size of the options field in the TCP header?

- (A) 0 bytes
- (B) 10 bytes
- (C) 20 bytes
- (D) 40 bytes

**Answer: (C)** 20 bytes

**87.** Why does the UDP checksum calculation include a 'Pseudo-Header' derived from the IP layer?

- (A) To encrypt the UDP payload before sending it over physical links
- (B) To allow the UDP layer to determine the physical MAC address of the destination
- (C) To detect misrouted packets by verifying that the source and destination IP addresses match
- (D) To compress the UDP header size for low-overhead links

**Answer: (C)** To detect misrouted packets by verifying that the source and destination IP addresses match

**88.** In a cryptographic Challenge-Response authentication protocol, how does a claimant prove her identity to a verifier?

- (A) By sending her secret key encrypted with a symmetric cipher
- (B) By solving a mathematical challenge using her secret key without ever sending the key itself
- (C) By transmitting her digital certificate over a secure TLS session
- (D) By generating a one-time random password and emailing it to the verifier

**Answer: (B)** By solving a mathematical challenge using her secret key without ever sending the key itself

**89.** In server-side socket programming using TCP, which sequence of system primitives is correct for setting up a connection-oriented listener?

- (A) socket() -> listen() -> bind() -> accept()
- (B) socket() -> bind() -> listen() -> accept()
- (C) socket() -> connect() -> accept() -> bind()
- (D) socket() -> accept() -> listen() -> bind()

**Answer: (B)** socket() -> bind() -> listen() -> accept()

**90.** As a data payload moves down the TCP/IP stack from the Application layer to the Physical layer, which layer is responsible for appending BOTH a header and a trailer?

- (A) Transport Layer
- (B) Network Layer
- (C) Data Link Layer
- (D) Physical Layer

**Answer: (C)** Data Link Layer



91. Which of the following describes the difference between the service models of the Data Link layer and the Transport layer?
- (A) Data Link layer provides process-to-process delivery; Transport layer provides node-to-node delivery
  - (B) Data Link layer provides node-to-node frame delivery; Transport layer provides end-to-end process delivery
  - (C) Data Link layer handles logical addressing; Transport layer handles physical MAC addressing
  - (D) Data Link layer is connection-oriented; Transport layer is strictly connectionless
- Answer: (B) Data Link layer provides node-to-node frame delivery; Transport layer provides end-to-end process delivery**
92. When an intermediate router receives a packet and routes it towards its destination, which layers of the network stack are active in the router?
- (A) Physical, Data Link, and Network layers
  - (B) Physical and Data Link layers only
  - (C) Physical, Data Link, Network, and Transport layers
  - (D) All 5 layers of the TCP/IP stack
- Answer: (A) Physical, Data Link, and Network layers**
93. How does the IPv6 header handle optional fields that were previously present in the IPv4 header's variable-size options?
- (A) By placing them inside a variable-length field inside the base 40-byte header
  - (B) By using chained 'Extension Headers' pointed to by the 'Next Header' field
  - (C) By sending them as a separate UDP packet before the main packet
  - (D) IPv6 completely removed support for optional fields
- Answer: (B) By using chained 'Extension Headers' pointed to by the 'Next Header' field**
94. During a graceful TCP connection teardown (Four-way handshake), which control flags are exchanged to coordinate the close of both data transmission directions?
- (A) SYN and ACK
  - (B) FIN and ACK
  - (C) RST and SYN
  - (D) URG and PSH
- Answer: (B) FIN and ACK**
95. Which of the following lists contains ONLY devices that operate strictly at the Physical layer (Layer 1) of the OSI model?
- (A) Hub, Repeater, and Passive Tap
  - (B) Switch, Router, and Hub
  - (C) Bridge, Gateway, and Repeater
  - (D) Modem, Access Point, and L3 Switch
- Answer: (A) Hub, Repeater, and Passive Tap**

## Routing Algorithms and Protocols

Computer Networks - 15 questions

96. Which of the following routing protocols uses the Bellman-Ford algorithm to find the shortest path?
- (A) OSPF
  - (B) RIP
  - (C) BGP
  - (D) IS-IS
- Answer: (B) RIP**
97. Which routing protocol is categorized as an Exterior Gateway Protocol (EGP) and is used to route traffic between Autonomous Systems (AS)?
- (A) OSPF
  - (B) BGP
  - (C) RIP
  - (D) EIGRP
- Answer: (B) BGP**

**98.** In a Link State routing protocol, what mechanism is used to distribute topology information to all routers in the routing domain?

- (A) Distance vector flooding
- (B) Reliable Flooding of Link State Advertisements (LSA)
- (C) Split Horizon
- (D) Route Poisoning

**Answer: (B) Reliable Flooding of Link State Advertisements (LSA)**

**99.** What is the primary reason for OSPF to establish Router ID (RID)?

- (A) To encrypt connection packets
- (B) To uniquely identify a router within an OSPF routing domain
- (C) To determine the network bandwidth limit
- (D) To map IP addresses to virtual LANs

**Answer: (B) To uniquely identify a router within an OSPF routing domain**

**100**How does the 'Split Horizon' mechanism prevent routing loops in Distance Vector routing protocols?

- (A) It limits the maximum hop count of any path to 15
- (B) It prevents a router from advertising routing information back out of the same interface from which it learned that information
- (C) It immediately broadcasts an infinite distance metric to all neighbors when a link fails
- (D) It forces the router to run Dijkstra's algorithm to verify paths

**Answer: (B) It prevents a router from advertising routing information back out of the same interface from which it learned that information**

**101**What is the key technical difference between 'Split Horizon' and 'Split Horizon with Poison Reverse' loop-prevention mechanisms?

- (A) Poison Reverse disables advertisements entirely, whereas Split Horizon permits them
- (B) Poison Reverse actively advertises a failed route with an infinite metric (e.g., hop count 16) back to the node it learned it from, rather than omitting it from the advertisement
- (C) Poison Reverse is used in Link State protocols, while Split Horizon is used in Distance Vector protocols
- (D) There is no difference; they are synonymous terms in routing theory

**Answer: (B) Poison Reverse actively advertises a failed route with an infinite metric (e.g., hop count 16) back to the node it learned it from, rather than omitting it from the advertisement**

**102**Which of the following parameters can be utilized as edge costs (weights) in Dijkstra's shortest path routing algorithm to calculate the optimal path?

- (A) Hop count only
- (B) Hop count, link bandwidth, queue delay, and link congestion
- (C) Autonomous System Path vector only
- (D) MAC address hop length

**Answer: (B) Hop count, link bandwidth, queue delay, and link congestion**

**103**Which of the following routing loop scenarios in Distance Vector routing is characterized by neighboring nodes repeatedly incrementing their hop counts to each other in a loop when a destination link breaks?

- (A) BGP convergence lag
- (B) Count-to-Infinity problem
- (C) OSPF flooding storm
- (D) Dijkstra oscillation

**Answer: (B) Count-to-Infinity problem**

**104**Why does Open Shortest Path First (OSPF) converge significantly faster than RIP on large network topologies?

- (A) OSPF uses the Bellman-Ford algorithm, which converges faster than Dijkstra
- (B) OSPF routers maintain a complete synchronized map of the network topology and run Dijkstra locally when a link-state change occurs, rather than waiting for periodic vector updates
- (C) OSPF limits the maximum hop count of any path to 15
- (D) OSPF does not use any acknowledgments for packets

**Answer: (B) OSPF routers maintain a complete synchronized map of the network topology and run Dijkstra locally when a link-state change occurs, rather than waiting for periodic vector updates**

- 105**In Routing Information Protocol (RIP), why is infinity represented by a relatively small number like 16?
- (A) To save bandwidth in the RIP packet headers
  - (B) To limit the maximum time a packet can spend in a loop by capping the convergence iterations during a count-to-infinity event
  - (C) Because 16 is the maximum size of a MAC address table
  - (D) To align RIP packets with 32-bit alignments
- Answer: (B)** To limit the maximum time a packet can spend in a loop by capping the convergence iterations during a count-to-infinity event
- 106**In Link State routing, what specific information is contained within a Link State Packet (LSP) built and flooded by a router?
- (A) The entire routing table of the router with calculated next-hops
  - (B) The ID of the router, its directly connected neighbors, and the cost/status of those direct links
  - (C) A list of all MAC addresses active on the local subnet
  - (D) A set of public keys for OSPF encryption
- Answer: (B)** The ID of the router, its directly connected neighbors, and the cost/status of those direct links
- 107**Why is Border Gateway Protocol (BGP) classified as a Path Vector routing protocol rather than a standard Distance Vector protocol?
- (A) Because BGP utilizes physical angles of transmission on fiber optic lines
  - (B) Because BGP advertises the entire list of Autonomous Systems (AS-Path) traversed to reach a destination to prevent routing loops
  - (C) Because BGP does not use TCP for its connections
  - (D) Because BGP is limited to 15 AS hops
- Answer: (B)** Because BGP advertises the entire list of Autonomous Systems (AS-Path) traversed to reach a destination to prevent routing loops
- 108**Which of the following is a key advantage of Non-Adaptive (Static) routing over Adaptive (Dynamic) routing?
- (A) It can easily route around link failures automatically
  - (B) It has zero CPU and memory overhead for computing routing tables and consumes no network bandwidth for routing updates
  - (C) It is highly scalable for extremely large networks like the global internet
  - (D) It automatically learns new subnets added to the network
- Answer: (B)** It has zero CPU and memory overhead for computing routing tables and consumes no network bandwidth for routing updates
- 109**Why does OSPF partition a large routing domain into distinct hierarchical 'Areas' with a backbone Area 0?
- (A) To limit the maximum transmission speed of packets
  - (B) To restrict IP address allocation to specific classes
  - (C) To limit the flooding of Link State Advertisements (LSAs) and reduce the size of the link-state database on individual routers
  - (D) To enforce different routing protocols in different departments
- Answer: (C)** To limit the flooding of Link State Advertisements (LSAs) and reduce the size of the link-state database on individual routers
- 110**What is a major limitation of Dijkstra's shortest path algorithm that makes it unsuitable for Distance Vector routing protocols?
- (A) It requires a global, synchronized view of the entire network topology, which Distance Vector routers do not have
  - (B) It cannot handle multiple paths to the same destination
  - (C) It is computationally slower than the Bellman-Ford algorithm on all graphs
  - (D) It only operates on directed graphs
- Answer: (A)** It requires a global, synchronized view of the entire network topology, which Distance Vector routers do not have

## IP Addressing, Subnetting, and Support Protocols

Computer Networks - 15 questions

**111** Consider classful addressing. The IP address 128.252.144.84 denotes which of the following?

- (A) 0.0.0.0 as network ID and 128.252.252.84 as node ID
- (B) 128.0.0.0 as network ID and 128.252.127.84 as node ID
- (C) 128.252.144.0 as network ID and 128.252.144.84 as node ID
- (D) 128.252.0.0 as network ID and 128.252.144.84 as node ID

**Answer: (D) 128.252.0.0 as network ID and 128.252.144.84 as node ID**

**112** An organization is assigned the classless block 192.168.10.0/26. What is the total number of usable host IP addresses available in this subnet?

- (A) 64
- (B) 62
- (C) 32
- (D) 30

**Answer: (B) 62**

**113** An IP router receives an IPv4 datagram with a total length of 1500 bytes (including a 20-byte IP header). The router must forward this packet over a link with a Maximum Transmission Unit (MTU) of 620 bytes. What is the fragment offset value in the second fragment generated by the router?

- (A) 0
- (B) 75
- (C) 150
- (D) 600

**Answer: (B) 75**

**114** Which protocol is responsible for automatically assigning dynamic IP addresses, default gateways, and DNS server details to clients within a local area network?

- (A) ARP
- (B) ICMP
- (C) DHCP
- (D) DNS

**Answer: (C) DHCP**

**115** What type of ICMP message is generated and sent back to the sender when a router discards a packet because its Time to Live (TTL) field has reached zero?

- (A) Destination Unreachable (Type 3)
- (B) Source Quench (Type 4)
- (C) Redirect (Type 5)
- (D) Time Exceeded (Type 11)

**Answer: (D) Time Exceeded (Type 11)**

**116** In Network Address Translation (NAT), which of the following is correct regarding PAT (Port Address Translation) / Network Address Port Translation (NAPT)?

- (A) It maps multiple private IP addresses to a single public IP address by changing the transport layer port numbers.
- (B) It maps a single private IP address to multiple public IP addresses on a round-robin basis.
- (C) It translation occurs strictly at the physical layer to encrypt MAC addresses.
- (D) It requires every local network device to have a unique registered public IP address.

**Answer: (A) It maps multiple private IP addresses to a single public IP address by changing the transport layer port numbers.**

**117** An organization has a Class B Network (default mask 255.255.0.0) and wishes to form subnets to support 60 departments. Find the required subnet mask.

- (A) 255.255.192.0
- (B) 255.255.240.0
- (C) 255.255.252.0
- (D) 255.255.255.192

**Answer: (C) 255.255.252.0**

**118** In the IPv4 header, what does the value of the 4-bit Internet Header Length (IHL) field represent, and how is it calculated?

- (A) It represents the total size of the IP packet in bytes
- (B) It represents the length of the IP header measured in 32-bit (4-byte) words
- (C) It represents the size of the payload in bits
- (D) It represents the number of intermediate router hops allowed

**Answer: (B) It represents the length of the IP header measured in 32-bit (4-byte) words**

**119**In the IPv4 header, the 13-bit 'Fragment Offset' field is used to specify the position of the fragmented data relative to the start of the original unfragmented datagram. What is the scale factor of this field?

- (A) 1 byte
- (B) 4 bytes
- (C) 8 bytes
- (D) 16 bytes

**Answer: (C) 8 bytes**

**120**The 3-bit flags field in the IPv4 header controls fragmentation. What are the specific roles of Bit 1 (DF) and Bit 2 (MF)?

- (A) DF = Dynamic Fragment, MF = Maximum Fragment
- (B) DF = Don't Fragment (if 1, router discards if packet exceeds MTU), MF = More Fragments (if 1, more fragments follow; if 0, this is the last fragment)
- (C) DF = Delay Flow, MF = Multiplexed Flow
- (D) DF = Destination Filter, MF = Mask Filter

**Answer: (B) DF = Don't Fragment (if 1, router discards if packet exceeds MTU), MF = More Fragments (if 1, more fragments follow; if 0, this is the last fragment)**

**121**Which leading bits determine that an IP address belongs to Class C in Classful Addressing, and what is its default subnet mask?

- (A) Leading bits '0', mask 255.0.0.0
- (B) Leading bits '10', mask 255.255.0.0
- (C) Leading bits '110', mask 255.255.255.0
- (D) Leading bits '1110', mask 224.0.0.0

**Answer: (C) Leading bits '110', mask 255.255.255.0**

**122**On which UDP port numbers do DHCP clients and DHCP servers listen for messages during address allocation?

- (A) Client on Port 53, Server on Port 53
- (B) Client on Port 67, Server on Port 68
- (C) Client on Port 68, Server on Port 67
- (D) Client on Port 80, Server on Port 443

**Answer: (C) Client on Port 68, Server on Port 67**

**123**What is the purpose of an ICMP Redirect (Type 5) message sent by a gateway router to a source host?

- (A) To inform the host that the link is congested and it should slow down
- (B) To notify the host of a better first-hop router (gateway) on the same subnet to reach the destination
- (C) To report that the destination port is closed on the target host
- (D) To request the host to retransmit its last fragmented packet

**Answer: (B) To notify the host of a better first-hop router (gateway) on the same subnet to reach the destination**

**124**A router's routing table contains the following entries: 10.0.0.0/8, 10.1.0.0/16, and 10.1.1.0/24. If an incoming packet arrives with destination IP 10.1.1.5, which entry will the router select to forward the packet?

- (A) 10.0.0.0/8
- (B) 10.1.0.0/16
- (C) 10.1.1.0/24
- (D) The packet will be discarded due to duplicate overlap entries

**Answer: (C) 10.1.1.0/24**

**125**Which of the following IPv4 address ranges is designated as private address space by RFC 1918 and is not routable on the public Internet?

- (A) 10.0.0.0 to 10.255.255.255, 172.16.0.0 to 172.31.255.255, and 192.168.0.0 to 192.168.255.255
- (B) 11.0.0.0 to 11.255.255.255 and 192.169.0.0 to 192.169.255.255
- (C) 127.0.0.0 to 127.255.255.255
- (D) 224.0.0.0 to 239.255.255.255

**Answer: (A) 10.0.0.0 to 10.255.255.255, 172.16.0.0 to 172.31.255.255, and 192.168.0.0 to 192.168.255.255**

## Data Link Layer and Transmission Control

Computer Networks - 15 questions

**126** Calculate the link utilization (throughput efficiency) for a Stop-and-Wait flow control mechanism if the frame size is 4800 bits, the transmission bit rate is 9600 bps, and the distance between the two devices is 2000 km. Assume the propagation speed of the signal is 200,000 km/s.

- (A) 0.92
- (B) 0.82
- (C) 0.96
- (D) 0.86

**Answer: (C) 0.96**

**127** The maximum throughput of a Slotted ALOHA wireless network is approximately:

- (A) 18.4%
- (B) 36.8%
- (C) 50.0%
- (D) 35.8%

**Answer: (B) 36.8%**

**128** In a CSMA/CD (Carrier Sense Multiple Access with Collision Detection) network, if the propagation speed is  $v$  m/s and the distance between the two furthest nodes is  $d$  meters, what is the minimum frame transmission time ( $T_t$ ) required to ensure collision detection?

- (A)  $T_t \geq d / v$
- (B)  $T_t \geq 2d / v$
- (C)  $T_t \leq d / v$
- (D)  $T_t \leq 2d / v$

**Answer: (B)  $T_t \geq 2d / v$**

**129** An error-detecting code is designed with a generator polynomial  $G(x) = x^3 + x + 1$ . If a sender wishes to transmit the message 1101, what are the Cyclic Redundancy Check (CRC) bits generated by this system?

- (A) 001
- (B) 101
- (C) 100
- (D) 111

**Answer: (A) 001**

**130** If the window size of a sliding window protocol at the sender is  $W$ , and we are using Go-Back-N (GBN) flow control over a noisy channel with  $n$ -bit sequence numbers, what is the maximum permissible window size  $W$  to avoid sequence number ambiguity?

- (A)  $2^n$
- (B)  $2^n - 1$
- (C)  $2^{(n-1)}$
- (D)  $2^n + 1$

**Answer: (B)  $2^n - 1$**

**131** How does an Ethernet learning bridge handle a received frame whose destination MAC address is not present in its forwarding directory table?

- (A) It discards the frame immediately to prevent network loops.
- (B) It buffers the frame and sends an ARP query to discover the MAC address.
- (C) It floods the frame out of all ports except the port on which it was received.
- (D) It forwards the frame strictly to the default gateway router port.

**Answer: (C) It floods the frame out of all ports except the port on which it was received.**

**132** In the Binary Exponential Backoff algorithm used in Ethernet (CSMA/CD), what is the range from which a station selects its random backoff slot time ' $r$ ' after the 4th collision?

- (A)  $r$  in  $[0, 1]$
- (B)  $r$  in  $[0, 7]$
- (C)  $r$  in  $[0, 15]$
- (D)  $r$  in  $[0, 1023]$

**Answer: (C)  $r$  in  $[0, 15]$**



**133**If a sender wants to transmit a 4-bit data word ( $m = 4$ ) using a single-bit error-correcting Hamming Code, what is the minimum number of redundant parity bits ( $r$ ) required to satisfy the fundamental Hamming inequality?

- (A)  $r = 2$
- (B)  $r = 3$
- (C)  $r = 4$
- (D)  $r = 5$

**Answer: (B)  $r = 3$**

**134**To design a code that is capable of correcting up to ' $p$ ' single-bit errors and detecting up to ' $d$ ' single-bit errors (where  $d > p$ ), what is the minimum Hamming distance ( $d_{\min}$ ) required between any two valid codewords?

- (A)  $d_{\min} \geq d + p$
- (B)  $d_{\min} \geq d + p + 1$
- (C)  $d_{\min} \geq 2d + 1$
- (D)  $d_{\min} \geq 2p + 1$

**Answer: (D)  $d_{\min} \geq 2p + 1$**

**135**A sliding window protocol uses Selective Repeat ARQ over a channel with sequence numbers represented by 4 bits. What is the maximum sender window size ( $W_s$ ) and receiver window size ( $W_r$ ) that can be used to prevent duplicate frames from overlapping?

- (A)  $W_s = 15, W_r = 1$
- (B)  $W_s = 8, W_r = 8$
- (C)  $W_s = 16, W_r = 16$
- (D)  $W_s = 7, W_r = 7$

**Answer: (B)  $W_s = 8, W_r = 8$**

**136**What is the maximum number of corrupted bit errors that a Two-Dimensional Parity Check can reliably detect, and how many bit errors can it correct?

- (A) Detect 1 error, correct 0 errors
- (B) Detect all 1, 2, and 3-bit errors, and correct single-bit errors by identifying the intersection of row and column parity mismatches
- (C) Detect any burst error of length 8, correct 2 errors
- (D) Detect 4-bit errors, correct 3-bit errors

**Answer: (B) Detect all 1, 2, and 3-bit errors, and correct single-bit errors by identifying the intersection of row and column parity mismatches**

**137**Which of the following correctly describes how acknowledgments (ACKs) are handled in Go-Back-N (GBN) vs. Selective Repeat (SR) protocols?

- (A) GBN uses cumulative ACKs; SR uses individual selective ACKs
- (B) GBN uses individual selective ACKs; SR uses cumulative ACKs
- (C) Both GBN and SR use strictly cumulative ACKs
- (D) Both GBN and SR use strictly individual selective ACKs

**Answer: (A) GBN uses cumulative ACKs; SR uses individual selective ACKs**

**138**In an Ethernet network with bandwidth  $B = 10$  Mbps and propagation speed  $v = 200,000$  km/s, if the distance between the two furthest stations is 2.5 km, what is the minimum Ethernet frame size required to guarantee collision detection?

- (A) 64 bytes
- (B) 125 bytes
- (C) 31.25 bytes
- (D) 500 bytes

**Answer: (C) 31.25 bytes**

**139**Why is the collision vulnerability window of Slotted ALOHA half that of Pure ALOHA?

- (A) Because Slotted ALOHA operates at twice the frequency band
- (B) Because Slotted ALOHA restricts transmissions to discrete slot boundaries, making the collision window equal to  $T_t$  instead of  $2 * T_t$
- (C) Because Slotted ALOHA uses a token ring approach
- (D) Because Slotted ALOHA uses carrier sensing before transmitting

**Answer: (B) Because Slotted ALOHA restricts transmissions to discrete slot boundaries, making the collision window equal to  $T_t$  instead of  $2 * T_t$**

**140**In the analysis of ARQ protocols, the parameter 'a' is defined as the ratio of propagation delay ( $T_p$ ) to transmission delay ( $T_t$ ). How does a very high value of 'a' ( $a \gg 1$ ) affect the performance of Stop-and-Wait ARQ?

- (A) It makes Stop-and-Wait extremely efficient (approaching 100%)
- (B) It degrades Stop-and-Wait efficiency to near zero, as the channel remains idle during the long propagation delay
- (C) It causes routing loops to form
- (D) It increases the transmission bandwidth of the link

**Answer: (B)** It degrades Stop-and-Wait efficiency to near zero, as the channel remains idle during the long propagation delay

## Transport Layer and Congestion Control

Computer Networks - 15 questions

**141**In TCP Congestion Control, suppose the current congestion window (cwnd) is 16 KB and the slow start threshold (ssthresh) is 8 KB. If the sender is in the Congestion Avoidance phase and successfully transmits one full window of segments, what will be the new value of cwnd?

- (A) 17 KB
- (B) 32 KB
- (C) 8 KB
- (D) 24 KB

**Answer: (A)** 17 KB

**142**A TCP connection is undergoing slow start. Suppose its cwnd is 32 segments and a packet timeout occurs. What are the new values of ssthresh and cwnd respectively (assuming standard TCP Tahoe)?

- (A) ssthresh = 16, cwnd = 1
- (B) ssthresh = 16, cwnd = 1
- (C) ssthresh = 32, cwnd = 1
- (D) ssthresh = 8, cwnd = 1

**Answer: (A)** ssthresh = 16, cwnd = 1

**143**Which of the following transport layer phenomena describes 'Silly Window Syndrome' (SWS)?

- (A) The sender transmits huge chunks of data that overflow the receiver's main buffer.
- (B) The receiver continually advertises tiny window sizes (e.g., a few bytes) and the sender transmits tiny segments to match, leading to high header overhead.
- (C) The network routers drop packets because the TCP window size is configured too high.
- (D) A sequence number wrap-around causes TCP segments to arrive completely out of order.

**Answer: (B)** The receiver continually advertises tiny window sizes (e.g., a few bytes) and the sender transmits tiny segments to match, leading to high header overhead.

**144**Host A is opening a TCP connection to Host B. Host A sends a SYN segment with Sequence Number = 1000. Host B accepts the connection and responds with a SYN-ACK segment. What is the Acknowledgment Number in Host B's response?

- (A) 1000
- (B) 1001
- (C) 2000
- (D) 1002

**Answer: (B)** 1001

**145**What is the key difference between User Datagram Protocol (UDP) and Transmission Control Protocol (TCP) headers?

- (A) UDP headers do not contain any port numbers.
- (B) UDP has a fixed 8-byte header size with no options, whereas TCP has a minimum 20-byte header.
- (C) UDP headers are dynamic and can expand up to 100 bytes using option fields.
- (D) UDP headers do not include a checksum field.

**Answer: (B)** UDP has a fixed 8-byte header size with no options, whereas TCP has a minimum 20-byte header.

**146**In network socket programming, which system call is used on the server-side to associate a local IP address and port number with a socket descriptor?

- (A) socket()
- (B) listen()
- (C) connect()
- (D) bind()

**Answer: (D)** bind()

**147**In TCP Reno congestion control, what happens to the congestion window (cwnd) and slow start threshold (sssthresh) when a loss is detected via 3 Duplicate ACKs (Fast Retransmit)?

- (A) cwnd is reset to 1 segment, sssthresh is halved
- (B) cwnd is halved, sssthresh is set to this halved value (Fast Recovery), and linear increase resumes
- (C) cwnd is doubled, sssthresh remains unchanged
- (D) No action is taken; the sender waits for a timeout

**Answer: (B)** cwnd is halved, sssthresh is set to this halved value (Fast Recovery), and linear increase resumes

**148**Which of the following correctly describes the behavioral difference between a Leaky Bucket traffic shaper and a Token Bucket traffic shaper?

- (A) Leaky Bucket permits bursty data transmission; Token Bucket releases packets at a strictly constant, uniform rate
- (B) Leaky Bucket drops all packets; Token Bucket compresses packets
- (C) Leaky Bucket releases packets at a strictly constant, uniform rate; Token Bucket generates tokens at a constant rate, permitting bursty data transmission when tokens are accumulated
- (D) There is no functional difference; they are identical traffic shaping models

**Answer: (C)** Leaky Bucket releases packets at a strictly constant, uniform rate; Token Bucket generates tokens at a constant rate, permitting bursty data transmission when tokens are accumulated

**149**During a TCP connection teardown (Four-way handshake), why does the host that initiates the active close enter the TIME-WAIT state and wait for  $2 * \text{MSL}$  (Maximum Segment Lifetime) before fully closing the socket?

- (A) To allow any duplicate, delayed packets in the network to die out and to guarantee that the final ACK was received by the peer
- (B) To download any remaining payload packets
- (C) To calculate the final checksum of the session
- (D) To request a new IP address from the DHCP server

**Answer: (A)** To allow any duplicate, delayed packets in the network to die out and to guarantee that the final ACK was received by the peer

**150**Which of the following statements is technically accurate regarding the UDP Checksum field in IPv4 vs. IPv6?

- (A) UDP Checksum is mandatory in both IPv4 and IPv6
- (B) UDP Checksum is optional in both IPv4 and IPv6
- (C) UDP Checksum is optional in IPv4 (if not calculated, set to all 0s), but is strictly mandatory in IPv6
- (D) UDP Checksum is mandatory in IPv4, but is optional in IPv6

**Answer: (C)** UDP Checksum is optional in IPv4 (if not calculated, set to all 0s), but is strictly mandatory in IPv6

**151**In server-side socket programming, what does the second parameter (backlog) of the listen(int sockfd, int backlog) system primitive represent?

- (A) The maximum speed of the link in Mbps
- (B) The maximum number of fully established, pending connections that can be queued waiting for the application to accept() them
- (C) The size of the socket's transmission buffer in bytes
- (D) The timeout duration in seconds for receiving packets

**Answer: (B)** The maximum number of fully established, pending connections that can be queued waiting for the application to accept() them

**152**How do TCP and UDP compare in terms of their flow control mechanisms?

- (A) TCP uses a sliding window size advertised by the receiver; UDP has zero flow control and depends on the application layer to prevent buffer overflow
- (B) Both TCP and UDP use identical sliding window flow control
- (C) TCP has no flow control; UDP uses credit-based flow control
- (D) TCP uses rate-limiting; UDP uses window-based flow control

**Answer: (A)** TCP uses a sliding window size advertised by the receiver; UDP has zero flow control and depends on the application layer to prevent buffer overflow

**153**What is the relationship between the TCP Maximum Segment Size (MSS) and the IP Maximum Transmission Unit (MTU) on a standard link?

- (A) MSS is always twice the size of the MTU
- (B) MSS is equal to the MTU of the link
- (C)  $\text{MSS} = \text{MTU} - (\text{IP Header Size} + \text{TCP Header Size})$ , which is typically  $\text{MTU} - 40$  bytes
- (D) MSS has no relationship to MTU

**Answer: (C)**  $\text{MSS} = \text{MTU} - (\text{IP Header Size} + \text{TCP Header Size})$ , which is typically  $\text{MTU} - 40$  bytes

**154**In which of the following scenarios will a host send a TCP segment with the RST (Reset) control flag set to 1?

- (A) To request the peer to slow down its transmission rate due to network congestion
- (B) When a client attempts to connect to a port on a server where no application is listening
- (C) During a normal, graceful four-way teardown
- (D) To acknowledge the receipt of an urgent out-of-band data segment

**Answer: (B)** When a client attempts to connect to a port on a server where no application is listening

**155**How does the TCP Selective Acknowledgment (SACK) option improve over the standard Cumulative Acknowledgment scheme during multiple packet losses?

- (A) By encrypting the retransmitted segments
- (B) By allowing the receiver to inform the sender of all non-contiguous segments received correctly, preventing the sender from having to retransmit segments that have already arrived
- (C) By forcing the sender to use UDP instead of TCP
- (D) By increasing the IP packet size to 65,535 bytes

**Answer: (B)** By allowing the receiver to inform the sender of all non-contiguous segments received correctly, preventing the sender from having to retransmit segments that have already arrived

## Application Layer Protocols and Network Security

Computer Networks - 15 questions

**156**Consider the following statements regarding network firewalls:

- I. A Packet Filter Firewall analyzes network traffic at the transport layer.
- II. A Circuit Level Firewall operates at the transport and session layers of the OSI Model.

Which of the statements is/are TRUE?

- (A) Only I
- (B) Only II
- (C) None
- (D) I and II

**Answer: (D)** I and II

**157**In the RSA public-key cryptosystem, if the two prime numbers are  $p = 7$  and  $q = 11$ , and the encryption key is chosen as  $e = 13$ , what is the value of the decryption key  $d$ ?

- (A) 23
- (B) 40
- (C) 37
- (D) 13

**Answer: (C)** 37

**158**Which of the following is NOT a public key (asymmetric) cryptosystem?

- (A) El Gamal
- (B) Rabin
- (C) AES
- (D) RSA

**Answer: (C)** AES

**159**In the RSA public-key algorithm, how is the encryption key 'e' selected with respect to the totient function  $\phi(n)$ ?

- (A) e must be chosen as a relative prime to  $q$
- (B) e must be chosen as a relative prime to  $p$
- (C) e must be chosen as a relative prime to  $\phi(n)$
- (D) e must be chosen as a relative prime to  $n$

**Answer: (C)** e must be chosen as a relative prime to  $\phi(n)$

**160**Which of the following application layer protocol maps well-known default port 53 and uses UDP for standard queries, but can fall back to TCP for large zone transfers?

- (A) DHCP
- (B) DNS
- (C) SMTP
- (D) FTP

**Answer: (B)** DNS

**161**In the File Transfer Protocol (FTP), which port numbers are utilized for the Data Connection and the Control Connection respectively in active mode?

- (A) Port 20 for Control, Port 21 for Data
- (B) Port 21 for Control, Port 20 for Data
- (C) Port 80 for Control, Port 443 for Data
- (D) Port 25 for Control, Port 110 for Data

**Answer: (B) Port 21 for Control, Port 20 for Data**

**162**In a Diffie-Hellman Key Exchange, the public parameters are a prime  $p = 11$  and its primitive root  $g = 2$ . If Alice selects private key  $a = 3$  and Bob selects private key  $b = 4$ , what is the shared secret key  $K$  computed by both?

- (A)  $K = 1$
- (B)  $K = 2$
- (C)  $K = 3$
- (D)  $K = 4$

**Answer: (D)  $K = 4$**

**163**In a network containing 'N' users, what is the total number of unique keys required for secure pairwise communication under a Symmetric Key Cryptosystem vs. an Asymmetric Key Cryptosystem?

- (A) N keys for Symmetric,  $N^2$  keys for Asymmetric
- (B)  $N(N-1)/2$  keys for Symmetric, 2N keys for Asymmetric
- (C) 2N keys for Symmetric,  $N(N-1)/2$  keys for Asymmetric
- (D) N keys for both cryptosystems

**Answer: (B)  $N(N-1)/2$  keys for Symmetric, 2N keys for Asymmetric**

**164**Which of the following correctly describes the distinct roles of SMTP vs. POP3/IMAP in email communications?

- (A) SMTP is used to retrieve email from a server; POP3/IMAP are used to push email from a client to a server
- (B) SMTP is used to push email from a client to a server and between servers; POP3 and IMAP are pull protocols used by a client to retrieve email from a mail server
- (C) SMTP operates over UDP port 25; POP3/IMAP operate over UDP port 80
- (D) SMTP encrypts email; POP3/IMAP decrypt email

**Answer: (B) SMTP is used to push email from a client to a server and between servers; POP3 and IMAP are pull protocols used by a client to retrieve email from a mail server**

**165**An HTML web page contains text and 5 embedded images. To download this complete page from a web server with a round-trip time (RTT) of 20 ms, how much total time is required under Non-Persistent HTTP without parallel connections vs. Persistent HTTP with pipelining (ignoring transmission times)?

- (A) 120 ms for Non-Persistent, 40 ms for Persistent
- (B) 240 ms for Non-Persistent, 40 ms for Persistent
- (C) 240 ms for Non-Persistent, 60 ms for Persistent
- (D) 100 ms for Non-Persistent, 20 ms for Persistent

**Answer: (C) 240 ms for Non-Persistent, 60 ms for Persistent**

**166**In symmetric block ciphers, which operation mode processes each block independently, causing identical plaintext blocks to produce identical ciphertext blocks, and making it highly vulnerable to pattern analyses?

- (A) Cipher Block Chaining (CBC)
- (B) Electronic Codebook (ECB)
- (C) Cipher Feedback (CFB)
- (D) Output Feedback (OFB)

**Answer: (B) Electronic Codebook (ECB)**

**167**How does a sender 'Alice' cryptographically sign an email message to prove non-repudiation and integrity to receiver 'Bob'?

- (A) Alice encrypts the message with Bob's public key
- (B) Alice encrypts a hash of the message with her own private key
- (C) Alice encrypts the message with her own public key
- (D) Alice hashes the message with Bob's private key

**Answer: (B) Alice encrypts a hash of the message with her own private key**

**168**What is the difference between a recursive DNS query and an iterative DNS query?

- (A) Recursive queries are encrypted; iterative queries are plaintext
- (B) In a recursive query, the DNS client demands the local DNS server resolve the address completely, whereas in an iterative query, the server returns the IP address of the next DNS server to ask
- (C) Recursive queries use TCP; iterative queries use UDP
- (D) There is no difference; they are synonyms

**Answer: (B)** In a recursive query, the DNS client demands the local DNS server resolve the address completely, whereas in an iterative query, the server returns the IP address of the next DNS server to ask

**169**During an SSL/TLS session negotiation, what is the role of the 'Server Key Exchange' message, and when is it sent?

- (A) It is sent by the server to transfer its digital certificate containing its public key
- (B) It is sent by the server to deliver temporary key parameters (e.g., Diffie-Hellman parameters) when the server's public key in its certificate is not sufficient for key exchange
- (C) It is sent to request the client's credit card credentials
- (D) It is sent to compress the HTTP headers

**Answer: (B)** It is sent by the server to deliver temporary key parameters (e.g., Diffie-Hellman parameters) when the server's public key in its certificate is not sufficient for key exchange

**170**Which SMTP command is sent by the client to gracefully terminate an established SMTP mail exchange session?

- (A) QUIT
- (B) CLOSE
- (C) EXIT
- (D) BYE

**Answer: (A)** QUIT

## Instruction Sets, CPU Design, and Pipelining

Computer Organization - 20 questions

**171**During the 'Instruction Fetch' phase, what is the precise register-transfer sequence that moves an instruction from RAM to the CPU for decoding, and which hardware buses are utilized?

- (A) PC copies address to MAR -> MAR places it on Address Bus -> Memory places instruction on Data Bus -> MDR stores it -> MDR moves it to IR
- (B) PC places address on Data Bus -> MDR copies to MAR -> Memory reads and moves to IR directly
- (C) IR copies address to MAR -> MAR places it on Address Bus -> Memory places instruction on Control Bus -> PC stores it
- (D) PC copies address to MDR -> MDR places it on Address Bus -> Memory places instruction on Data Bus -> IR copies to MAR

**Answer: (A)** PC copies address to MAR -> MAR places it on Address Bus -> Memory places instruction on Data Bus -> MDR stores it -> MDR moves it to IR

**172**A CPU has a 0-Address instruction format. Which architectural component does this instruction format heavily rely on for operand retrieval, and how are operations like addition executed?

- (A) A Stack architecture; addition pops the top two elements, adds them, and pushes the result back onto the stack.
- (B) An Accumulator (AC) register; addition adds the memory content directly to the AC.
- (C) A General Purpose Register (GPR) file; addition retrieves operands via explicit register index fields.
- (D) A FIFO Queue; addition retrieves operands from the front and writes the result to the rear.

**Answer: (A)** A Stack architecture; addition pops the top two elements, adds them, and pushes the result back onto the stack.

**173**Consider the execution of the statement ' $X = (A + B) * C$ ' on a 1-Address (Accumulator-based) CPU architecture. Which of the following represents the correct sequence of assembly instructions to calculate and store the result?

- (A) LOAD A; ADD B; MULT C; STORE X
- (B) LOAD A; ADD B; STORE T; LOAD C; MULT T; STORE X
- (C) ADD A, B; MULT X, C
- (D) PUSH A; PUSH B; ADD; PUSH C; MULT; POP X

**Answer: (A)** LOAD A; ADD B; MULT C; STORE X



**174**An 8-bit Arithmetic Logic Unit (ALU) performs subtraction of two signed numbers in 2's complement representation:  $X = A - B$ . If  $A = 01110000$  (+112) and  $B = 10010000$  (-112), what is the resulting bit-pattern of  $X$ , and what are the statuses of the Carry (C) and Overflow (O/V) flags?

- (A)  $X = 00000000$ ; Carry (C) = 1, Overflow (O) = 1
- (B)  $X = 11100000$ ; Carry (C) = 0, Overflow (O) = 1
- (C)  $X = 11100000$ ; Carry (C) = 1, Overflow (O) = 0
- (D)  $X = 00000000$ ; Carry (C) = 0, Overflow (O) = 0

**Answer: (B)**  $X = 11100000$ ; Carry (C) = 0, Overflow (O) = 1

**175**Which of the following correct pairings matches the CPU Addressing Mode to its primary practical application and use case in computer systems?

- (A) Index Mode -> Array Access; Base Register Mode -> Program Relocation; Relative Mode -> Branch Control; Auto-increment -> Stack Push/Pop
- (B) Immediate Mode -> Program Relocation; Indirect Mode -> Array Access; Register Mode -> Branch Control; Index Mode -> Stack Push/Pop
- (C) Relative Mode -> Program Relocation; Index Mode -> Pointers; Direct Mode -> Branch Control; Base Register -> Stack Push/Pop
- (D) Register Indirect -> Program Relocation; Base Register -> Array Access; Relative Mode -> Pointers; Auto-decrement -> Branch Control

**Answer: (A)** Index Mode -> Array Access; Base Register Mode -> Program Relocation; Relative Mode -> Branch Control; Auto-increment -> Stack Push/Pop

**176**If an operand is located in memory, what is the exact difference in the number of memory accesses required to fetch the operand under Direct Addressing Mode versus Indirect Addressing Mode?

- (A) Direct Mode requires 1 memory access; Indirect Mode requires 2 memory accesses.
- (B) Direct Mode requires 2 memory accesses; Indirect Mode requires 1 memory access.
- (C) Direct Mode requires 0 memory accesses; Indirect Mode requires 1 memory access.
- (D) Both modes require identical numbers of memory accesses.

**Answer: (A)** Direct Mode requires 1 memory access; Indirect Mode requires 2 memory accesses.

**177**A loop is accessing an array starting at base memory address 1000. Under Index Addressing Mode, if the constant Base Address field in the instruction is set to 1000 and the Index Register currently contains the value 2, how does the CPU compute the Effective Address, and does it modify the original instruction?

- (A) Effective Address is calculated as  $1000 + 2 = 1002$  using hardware adder; the original instruction remains unchanged.
- (B) Effective Address is calculated as  $1000 * 2 = 2000$ ; the base address in the instruction is dynamically overwritten with 2000.
- (C) Effective Address is calculated as  $1000 - 2 = 998$ ; the index register is reset to 0 in memory.
- (D) Effective Address is calculated as Memory[1002]; the instruction's opfield is modified to direct mode.

**Answer: (A)** Effective Address is calculated as  $1000 + 2 = 1002$  using hardware adder; the original instruction remains unchanged.

**178**A Program Counter (PC) relative branch instruction is stored in memory at address 1000. The branch instruction has an offset field containing the signed value -24. Assuming that the PC is automatically incremented by 2 bytes during the instruction fetch phase (before the branch offset is evaluated), what is the correct branch target address computed by the CPU?

- (A) 978
- (B) 976
- (C) 1026
- (D) 1024

**Answer: (A)** 978

**179**What are the fundamental architectural differences between a 'Hardwired Control Unit' and a 'Microprogrammed Control Unit' in terms of implementation hardware, speed, and instruction set flexibility?

- (A) Hardwired uses combinational logic (decoders, flip-flops) and is extremely fast but rigid (RISC); Microprogrammed stores control signals in ROM (Control Memory) and is slower but highly flexible (CISC).
- (B) Hardwired stores control micro-instructions in high-speed RAM; Microprogrammed uses pure mechanical logic gates and is faster.
- (C) Hardwired is flexible and easily reprogrammed via firmware updates; Microprogrammed requires physical redesign of silicon to add instructions.
- (D) Hardwired is commonly used in CISC processors due to instruction complexity; Microprogrammed is used in RISC because of speed requirements.

**Answer: (A)** Hardwired uses combinational logic (decoders, flip-flops) and is extremely fast but rigid (RISC); Microprogrammed stores control signals in ROM (Control Memory) and is slower but highly flexible (CISC).

- 180** In a Microprogrammed Control Unit, what is the distinction between 'Horizontal' and 'Vertical' microinstruction formats with respect to control signal encoding, execution speed, and control memory width?
- (A) Horizontal uses no encoding (one bit per control line), leading to wide memory formats and fastest execution; Vertical encodes mutually exclusive signals, requiring decoders which adds propagation delay but saves significant memory space.  
 (B) Horizontal encodes signals to save memory; Vertical uses one bit per line and is faster.  
 (C) Horizontal requires external decoders; Vertical directly controls hardware lines without decoders.  
 (D) Both formats have identical memory footprints and execution speeds.
- Answer: (A)** Horizontal uses no encoding (one bit per control line), leading to wide memory formats and fastest execution; Vertical encodes mutually exclusive signals, requiring decoders which adds propagation delay but saves significant memory space.
- 181** A microprogrammed control unit must generate 36 control signals. These signals are divided into 6 mutually exclusive groups containing 6, 8, 4, 7, 5, and 6 signals respectively. If vertical microprogramming is used, what is the minimum number of bits required in the microinstruction control field to represent these signals?
- (A) 19 bits  
 (B) 22 bits  
 (C) 36 bits  
 (D) 12 bits
- Answer: (B)** 22 bits
- 182** A 5-stage instruction pipeline has stage delays of 150 ns, 120 ns, 180 ns, 140 ns, and 110 ns. Interface latches between stages introduce an additional delay of 10 ns. What is the clock cycle time of this pipelined processor, and how much time does it take to execute 100 independent instructions?
- (A) Clock cycle = 190 ns; Total execution time = 19,760 ns  
 (B) Clock cycle = 180 ns; Total execution time = 18,720 ns  
 (C) Clock cycle = 190 ns; Total execution time = 19,000 ns  
 (D) Clock cycle = 180 ns; Total execution time = 19,000 ns
- Answer: (A)** Clock cycle = 190 ns; Total execution time = 19,760 ns
- 183** A non-pipelined processor takes 12 ns to execute a single instruction. The same processor is redesigned with an 8-stage pipeline running at a clock period of 1.8 ns. What is the actual speedup achieved by the pipelined processor for executing 1000 instructions?
- (A) Speedup = 6.62  
 (B) Speedup = 8.00  
 (C) Speedup = 5.43  
 (D) Speedup = 4.88
- Answer: (A)** Speedup = 6.62
- 184** An instruction pipeline experiences a 'Structural Hazard'. What is the physical cause of this hazard type, and what is the standard architectural solution used in modern CPUs?
- (A) Resource conflict where multiple stages attempt to access the same physical hardware simultaneously; solved by Harvard Architecture (separating Instruction Cache and Data Cache).  
 (B) Data dependency where an instruction reads a register before a previous one writes it; solved by branch prediction.  
 (C) A conditional branch instruction changing the target PC; solved by inserting multiple NOP bubbles.  
 (D) Clock signal skew where latches fail to synchronize; solved by reducing clock speed.
- Answer: (A)** Resource conflict where multiple stages attempt to access the same physical hardware simultaneously; solved by Harvard Architecture (separating Instruction Cache and Data Cache).
- 185** Which of the following describes a 'Read-After-Write' (RAW) data hazard in a pipelined processor, and what is the standard hardware optimization that eliminates execution stall cycles?
- (A) An instruction attempts to read a source register before a preceding instruction writes to it; resolved by Operand Forwarding (bypassing the result directly from execution latches to ALU inputs).  
 (B) An instruction writes to a register before a subsequent instruction reads it; resolved by delayed branching.  
 (C) Multiple instructions attempt to read from the memory bus concurrently; resolved by Harvard Architecture.  
 (D) An instruction reads a register after it has been written; resolved by inserting a standard hardware stall.
- Answer: (A)** An instruction attempts to read a source register before a preceding instruction writes to it; resolved by Operand Forwarding (bypassing the result directly from execution latches to ALU inputs).

- 186** Consider three instructions in execution: I1: ADD R1, R2, R3; I2: SUB R4, R1, R5; I3: ADD R1, R6, R7. Which of the following correctly identifies the hazard types existing between these instructions?
- (A) RAW hazard between I1 and I2 on register R1; WAW hazard between I1 and I3 on register R1.
  - (B) WAR hazard between I1 and I2 on register R1; RAW hazard between I2 and I3 on register R4.
  - (C) Structural hazard between I1 and I3 on ALU; Control hazard between I2 and I3.
  - (D) RAW hazard between I2 and I3 on register R1; WAR hazard between I1 and I2 on register R4.

**Answer: (A)** RAW hazard between I1 and I2 on register R1; WAW hazard between I1 and I3 on register R1.

- 187** A pipeline processor has a branch instruction frequency of 20%. The branch predictor is 80% accurate. If a branch is predicted correctly, there is 0 stall cycles. If a branch is mispredicted, the pipeline experiences a 3-cycle branch penalty. If the base CPI under ideal execution is 1.0, what is the average CPI of this processor?
- (A) 1.12
  - (B) 1.24
  - (C) 1.60
  - (D) 1.06

**Answer: (A)** 1.12

- 188** In a pipeline processor, what does a 'Control Hazard' refer to, and which of the following is a compiler-level software technique used to resolve or minimize it?
- (A) A delay caused by branch/jump instructions changing the Program Counter; minimized by 'Delayed Branching' where the compiler fills branch delay slots with useful independent instructions.
  - (B) A clash over hardware buses; resolved by inserting NOP bubbles manually in the assembly file.
  - (C) A register dependency; resolved by using an out-of-order execution register renaming technique.
  - (D) An interrupt service routine call; resolved by disabling CPU interrupts during loop execution.

**Answer: (A)** A delay caused by branch/jump instructions changing the Program Counter; minimized by 'Delayed Branching' where the compiler fills branch delay slots with useful independent instructions.

- 189** Which of the following represents the correct sequential order of the phases of a complete instruction cycle as executed inside a central processing unit?
- (A) Fetch -> Decode -> Evaluate Address -> Execute -> Store
  - (B) Fetch -> Evaluate Address -> Decode -> Execute -> Store
  - (C) Decode -> Fetch -> Evaluate Address -> Execute -> Store
  - (D) Fetch -> Decode -> Execute -> Evaluate Address -> Store

**Answer: (A)** Fetch -> Decode -> Evaluate Address -> Execute -> Store

- 190** What is the theoretical maximum speedup that an 8-stage instruction pipeline can achieve over a non-pipelined processor under ideal conditions as the number of instructions (n) approaches infinity, and why can this maximum speedup never be physically reached in practice?
- (A) Maximum speedup is 8; cannot be reached due to pipeline hazards (structural, data, control) and latch register overheads.
  - (B) Maximum speedup is infinity; cannot be reached because CPU clock speeds are physically limited by heat dissipation.
  - (C) Maximum speedup is 1; cannot be reached because pipelined execution is inherently slower.
  - (D) Maximum speedup is 4; cannot be reached because memory bus widths restrict data transfers.

**Answer: (A)** Maximum speedup is 8; cannot be reached due to pipeline hazards (structural, data, control) and latch register overheads.

## Memory Hierarchy, Cache, and I/O DMA Control

Computer Organization - 20 questions

- 191** Arrange the following storage devices of the memory hierarchy in increasing order of speed and cost per byte, and in decreasing order of capacity and latency.
- (A) Secondary Storage (Magnetic/SSD) < Main Memory (DRAM) < Cache Memory (SRAM) < Registers
  - (B) Registers < Cache Memory (SRAM) < Main Memory (DRAM) < Secondary Storage (Magnetic/SSD)
  - (C) Main Memory (DRAM) < Secondary Storage (Magnetic/SSD) < Cache Memory (SRAM) < Registers
  - (D) Secondary Storage (Magnetic/SSD) < Cache Memory (SRAM) < Main Memory (DRAM) < Registers

**Answer: (A)** Secondary Storage (Magnetic/SSD) < Main Memory (DRAM) < Cache Memory (SRAM) < Registers

**192** Which of the following describes how the 'Principle of Locality' is exploited by system designers to successfully operate Cache Memory?

- (A) Temporal Locality is exploited by keeping recently accessed variables in Cache; Spatial Locality is exploited by fetching data as larger 'Blocks' rather than single bytes.
- (B) Temporal Locality is exploited by bringing adjacent memory cells into the registers; Spatial Locality is exploited by running multiple loops concurrently.
- (C) Both temporal and spatial locality are exploited by bypassing the cache memory entirely and writing directly to magnetic disks.
- (D) Temporal Locality is exploited by increasing the physical size of RAM; Spatial Locality is exploited by reducing the CPU clock period.

**Answer: (A)** Temporal Locality is exploited by keeping recently accessed variables in Cache; Spatial Locality is exploited by fetching data as larger 'Blocks' rather than single bytes.

**193** A main memory block 'j' is mapped into a Direct Mapped cache containing a total of 'C' cache lines. What is the mathematical mapping formula used to determine the cache line index 'i' where block 'j' can reside?

- (A)  $i = j \text{ modulo } C$
- (B)  $i = j \text{ modulo } 2^C$
- (C)  $i = C \text{ modulo } j$
- (D)  $i = (j * C) \text{ modulo } 2$

**Answer: (A)**  $i = j \text{ modulo } C$

**194** Why does Direct Cache Mapping suffer heavily from 'Conflict Misses' even if the total capacity of the cache is much larger than the working set of the program?

- (A) Because multiple distinct memory blocks hash to the exact same cache line, causing them to repeatedly evict each other.
- (B) Because the physical address lacks a Word offset field, preventing correct word lookup.
- (C) Because the cache can only perform sequential tag comparison, slowing down execution.
- (D) Because the CPU is forced to clear the entire cache on every context switch.

**Answer: (A)** Because multiple distinct memory blocks hash to the exact same cache line, causing them to repeatedly evict each other.

**195** Which cache mapping technique provides the highest hit ratio by allowing any main memory block to reside in any cache line, and what is its primary hardware drawback?

- (A) Fully Associative Mapping; requires expensive parallel hardware comparators to check all tags simultaneously.
- (B) Direct Mapping; requires complex decoders that increase latency.
- (C) Set-Associative Mapping; completely disables write-back caching.
- (D) Fully Associative Mapping; suffers from high conflict misses in loops.

**Answer: (A)** Fully Associative Mapping; requires expensive parallel hardware comparators to check all tags simultaneously.

**196** A Set-Associative cache is divided into 'v' sets, with each set consisting of 'k' lines. Under which exact conditions does this Set-Associative mapping scheme reduce to (I) Direct Mapping, and (II) Fully Associative Mapping?

- (A) (I) when  $k = 1$ ; (II) when  $v = 1$
- (B) (I) when  $v = 1$ ; (II) when  $k = 1$
- (C) (I) when  $k = C$  (total lines); (II) when  $v = C$
- (D) (I) when  $k = 1$  and  $v = 1$ ; (II) when  $k = m$

**Answer: (A)** (I) when  $k = 1$ ; (II) when  $v = 1$

**197** A system has a 32-bit physical address. It features a Direct Mapped Cache of capacity 64 KB with a cache block (line) size of 64 bytes. What is the correct division of the physical address into Tag, Line Index, and Word Offset fields?

- (A) Tag = 16 bits, Line = 10 bits, Word = 6 bits
- (B) Tag = 17 bits, Line = 9 bits, Word = 6 bits
- (C) Tag = 18 bits, Line = 10 bits, Word = 4 bits
- (D) Tag = 15 bits, Line = 11 bits, Word = 6 bits

**Answer: (A)** Tag = 16 bits, Line = 10 bits, Word = 6 bits

**198**A computer system uses a 32-bit physical address. It has a 4-Way Set-Associative Cache of capacity 128 KB with a block size of 32 bytes. What is the correct division of the physical address into Tag, Set Index, and Word Offset fields?

- (A) Tag = 17 bits, Set = 10 bits, Word = 5 bits
- (B) Tag = 18 bits, Set = 9 bits, Word = 5 bits
- (C) Tag = 16 bits, Set = 11 bits, Word = 5 bits
- (D) Tag = 19 bits, Set = 8 bits, Word = 5 bits

**Answer: (A)** Tag = 17 bits, Set = 10 bits, Word = 5 bits

**199**In a 2-way set-associative cache, the physical address is divided into Tag, Set, and Word fields. If the CPU splits a 32-bit address as Tag = 20 bits, Set = 7 bits, and Word = 5 bits, what are the block size and the total cache capacity respectively?

- (A) Block size = 32 bytes, Cache capacity = 8 KB
- (B) Block size = 64 bytes, Cache capacity = 16 KB
- (C) Block size = 32 bytes, Cache capacity = 16 KB
- (D) Block size = 16 bytes, Cache capacity = 4 KB

**Answer: (A)** Block size = 32 bytes, Cache capacity = 8 KB

**200**A processor has a Cache access time (Hit Time) of 2 ns. If the memory block is not in the cache, it takes 80 ns (Miss Penalty) to load the block from Main Memory into the Cache. If the Cache Hit Ratio is 95%, what is the Average Memory Access Time (AMAT)?

- (A) 6.0 ns
- (B) 5.8 ns
- (C) 4.0 ns
- (D) 2.0 ns

**Answer: (A)** 6.0 ns

**201**Consider a hierarchy with L1 Cache, L2 Cache, and Main Memory. The L1 access time is 1 ns with a hit rate of 90%. The L2 access time is 5 ns with a local hit rate of 80%. Main Memory access time is 100 ns. What is the Average Memory Access Time (AMAT) for this multi-level memory system?

- (A) 3.0 ns
- (B) 3.5 ns
- (C) 2.5 ns
- (D) 5.0 ns

**Answer: (A)** 3.0 ns

**202**What is the key functional difference between 'Vectored Interrupts' and 'Non-Vectored Interrupts' in an I/O system, and how does it affect CPU performance?

- (A) In Vectored Interrupts, the interrupting device provides a specific 'Vector Address' pointing directly to the memory location of its ISR, making it much faster than Non-Vectored where the CPU must poll devices sequentially.
- (B) In Vectored Interrupts, the CPU is completely blocked; in Non-Vectored, the CPU can run concurrent instructions.
- (C) Vectored Interrupts require manual software polling; Non-Vectored Interrupts are hardwired directly into the registers.
- (D) There is no difference; both require identical polling cycles to locate the active device.

**Answer: (A)** In Vectored Interrupts, the interrupting device provides a specific 'Vector Address' pointing directly to the memory location of its ISR, making it much faster than Non-Vectored where the CPU must poll devices sequentially.

**203**What primary system bottleneck is resolved by introducing 'Direct Memory Access' (DMA) for bulk data transfers, and which hardware components does the DMA Controller bypass?

- (A) It bypasses the CPU entirely during bulk transfers, resolving the bottleneck where the CPU is forced to waste millions of clock cycles transferring individual data words from disk to memory.
- (B) It bypasses the system address bus, resolving the bottleneck of memory addressing size limits.
- (C) It bypasses the main memory entirely, transferring files directly into the execution ALU registers.
- (D) It bypasses the hard disk controller, reading data directly from magnetic platter surfaces.

**Answer: (A)** It bypasses the CPU entirely during bulk transfers, resolving the bottleneck where the CPU is forced to waste millions of clock cycles transferring individual data words from disk to memory.



- 204** During a DMA data transfer, which of the following hardware signaling sequences represents the correct handshake protocol between the CPU and the DMA Controller (DMAC)?
- (A) DMAC asserts Bus Request (BR) -> CPU releases buses and asserts Bus Grant (BG) -> DMAC transfers data -> DMAC releases BR
  - (B) CPU asserts Bus Request (BR) -> DMAC asserts Bus Grant (BG) -> CPU transfers data -> CPU releases BG
  - (C) DMAC asserts Interrupt -> CPU reads Data Bus -> DMAC grants bus control to memory
  - (D) Memory asserts Bus Grant (BG) -> DMAC asserts Bus Request (BR) -> CPU reads data directly
- Answer: (A)** DMAC asserts Bus Request (BR) -> CPU releases buses and asserts Bus Grant (BG) -> DMAC transfers data -> DMAC releases BR
- 205** Which of the following describes the 'Burst Mode' (Block Transfer) DMA transfer scheme, and what is its effect on CPU performance?
- (A) The DMA controller acquires the system bus and transfers the entire block of data sequentially before releasing the bus; the CPU is completely blocked and idles during this transfer.
  - (B) The DMA controller transfers only one word, returns bus control to the CPU, and then requests it again; the CPU is never blocked.
  - (C) The DMA controller only uses the bus when the CPU is performing internal decoding; there is zero CPU performance hit.
  - (D) The DMA controller shares the bus dynamically on every clock transition; the CPU speed doubles.
- Answer: (A)** The DMA controller acquires the system bus and transfers the entire block of data sequentially before releasing the bus; the CPU is completely blocked and idles during this transfer.
- 206** How does the 'Cycle Stealing' DMA transfer mode operate, and what is its primary advantage over Burst Mode?
- (A) The DMA controller transfers exactly one word and returns the bus to the CPU for one cycle before requesting it again, allowing the CPU to continue executing program instructions concurrently.
  - (B) The DMA controller steals cache lines from the CPU L1 cache, doubling the memory hit ratio.
  - (C) The DMA controller steals CPU registers to store intermediate data, bypassing main memory completely.
  - (D) The DMA controller executes instructions on behalf of the CPU, avoiding pipeline branch penalties.
- Answer: (A)** The DMA controller transfers exactly one word and returns the bus to the CPU for one cycle before requesting it again, allowing the CPU to continue executing program instructions concurrently.
- 207** Which of the following describes 'Transparent Mode' (Hidden Mode) DMA, and what is its specific performance hit on CPU operations?
- (A) The DMA controller only utilizes the system bus when the CPU is not using it (e.g., during internal instruction decode phases); there is zero performance hit on CPU execution.
  - (B) The DMA controller completely hides RAM from the CPU, causing a 50% performance slowdown.
  - (C) The DMA controller runs silently in the background by sharing L1 cache; causing a minor 5% performance hit.
  - (D) The DMA controller operates only when the CPU is in sleep mode, doubling the CPU's power dissipation.
- Answer: (A)** The DMA controller only utilizes the system bus when the CPU is not using it (e.g., during internal instruction decode phases); there is zero performance hit on CPU execution.
- 208** What is the key difference between a 'Write-Through' cache policy and a 'Write-Back' cache policy when the CPU performs a memory write operation?
- (A) Write-Through updates both the cache and main memory simultaneously; Write-Back updates only the cache line and sets a 'dirty bit', updating main memory only when the line is evicted.
  - (B) Write-Through updates main memory first, then cache; Write-Back updates cache first, then registers.
  - (C) Write-Through is used for magnetic SSDs; Write-Back is used for cache registers.
  - (D) Both write-through and write-back update main memory instantly, but write-back bypasses L1 cache.
- Answer: (A)** Write-Through updates both the cache and main memory simultaneously; Write-Back updates only the cache line and sets a 'dirty bit', updating main memory only when the line is evicted.
- 209** A system has a Page Fault Service Time of 10 milliseconds (10,000,000 ns) and a Main Memory Access Time of 100 ns. If the page fault rate is 0.01% ( $P = 10^{-4}$ ), what is the Effective Memory Access Time (EMAT) of this system?
- (A) 1100 ns
  - (B) 1000 ns
  - (C) 1090 ns
  - (D) 990 ns
- Answer: (A)** 1100 ns



**210**At what specific time does a Central Processing Unit (CPU) check for pending hardware interrupts, and why does it not check mid-instruction?

- (A) At the end of the current instruction's execution cycle; checking mid-instruction would require saving massive intermediate pipeline and register states, severely degrading performance.
- (B) At the start of every clock sub-cycle; to ensure instantaneous response times to external devices.
- (C) Only during instruction decode phase; to prevent decoding of corrupt code.
- (D) During memory access phases; to sync bus transfers with interrupting devices.

**Answer: (A)** At the end of the current instruction's execution cycle; checking mid-instruction would require saving massive intermediate pipeline and register states, severely degrading performance.

## Entity Relationship Model

DBMS - 8 questions

**211**In an Entity-Relationship (ER) diagram, how are multi-valued attributes represented?

- (A) Single Ellipse
- (B) Double Ellipse
- (C) Dashed Ellipse
- (D) Rectangle

**Answer: (B)** Double Ellipse

**212**A weak entity set in an ER diagram is represented by a:

- (A) Double Rectangle
- (B) Double Diamond
- (C) Double Ellipse
- (D) Dashed Rectangle

**Answer: (A)** Double Rectangle

**213**Which of the following describes a relationship where an occurrence of entity A is associated with any number of occurrences of entity B, and an occurrence of B is associated with at most one occurrence of A?

- (A) Many-to-Many (M:N)
- (B) One-to-Many (1:N)
- (C) Many-to-One (N:1)
- (D) One-to-One (1:1)

**Answer: (B)** One-to-Many (1:N)

**214**In the ER model, what is the 'degree' of a relationship set?

- (A) The number of entities in an entity set
- (B) The number of participating entity sets in the relationship set
- (C) The maximum number of relationship instances
- (D) The number of attributes associated with the relationship

**Answer: (B)** The number of participating entity sets in the relationship set

**215**Consider a ternary relationship set R involving entity sets A, B, and C. If we decompose R into three binary relationship sets Rab (between A and B), Rbc (between B and C), and Rca (between C and A), which of the following is correct?

- (A) The decomposition is always lossless and preserves all semantic constraints.
- (B) The decomposition generally causes a loss of the combined association, as binary relationships only represent pairwise connections.
- (C) The ternary relationship can always be reconstructed exactly via a simple natural join of the three binary relationships.
- (D) Ternary relationships are mathematically prohibited in the Relational Model and must be decomposed.

**Answer: (B)** The decomposition generally causes a loss of the combined association, as binary relationships only represent pairwise connections.

**216**In structural constraint modeling of ER diagrams, the (min, max) notation is used. Consider a 1:N relationship set 'PLACES' between 'Customer' and 'Order' (where a customer places zero or more orders, and each order is placed by exactly one customer). What are the correct (min, max) values placed on the edges connecting 'Customer' and 'Order' to the 'PLACES' relationship, respectively?

- (A) On Customer edge: (0, N); On Order edge: (1, 1)
- (B) On Customer edge: (1, 1); On Order edge: (0, N)
- (C) On Customer edge: (0, 1); On Order edge: (1, N)
- (D) On Customer edge: (1, N); On Order edge: (0, 1)

**Answer: (A)** On Customer edge: (0, N); On Order edge: (1, 1)

**217** Consider an ER diagram with a strong entity set 'Employee' having primary key 'EmpID', and a weak entity set 'Dependent' having partial key 'DepName' (with total participation in the identifying relationship 'HAS'). When mapping this weak entity set to a relational table, what is the primary key of the resulting 'Dependent' table?

- (A) DepName
- (B) EmpID
- (C) A composite primary key consisting of (EmpID, DepName)
- (D) A newly generated surrogate key

**Answer: (C)** A composite primary key consisting of (EmpID, DepName)

**218** An ER diagram has 3 strong entity sets E1 with simple key K1, E2 with simple key K2, and E3 with simple key K3. There is a many-to-many (M:N) binary relationship set R12 between E1 and E2, and a many-to-one (N:1) binary relationship set R23 between E2 and E3 (where participation is partial on both sides). Additionally, E1 has a multi-valued attribute M1. What is the minimum number of relational tables required to represent this schema without losing any information?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

**Answer: (C)** 5

## Normalization

DBMS - 8 questions

**219** A relation is in 2NF if it is in 1NF and there is no:

- (A) Transitive dependency
- (B) Partial dependency
- (C) Multi-valued dependency
- (D) Join dependency

**Answer: (B)** Partial dependency

**220** A relation schema R is in Third Normal Form (3NF) if, for every non-trivial functional dependency X → A, either X is a superkey of R or:

- (A) A is a non-prime attribute
- (B) A is a prime attribute
- (C) X is a prime attribute
- (D) X is a subset of some candidate key

**Answer: (B)** A is a prime attribute

**221** Which of the following is true regarding Boyce-Codd Normal Form (BCNF)?

- (A) Every BCNF relation is also in 3NF, and BCNF is dependency-preserving by default.
- (B) Every BCNF relation is also in 3NF, but BCNF is NOT always dependency-preserving.
- (C) BCNF is weaker than 3NF.
- (D) BCNF eliminates multi-valued dependencies.

**Answer: (B)** Every BCNF relation is also in 3NF, but BCNF is NOT always dependency-preserving.

**222** Which normal form is specifically designed to eliminate multi-valued dependencies?

- (A) 3NF
- (B) BCNF
- (C) 4NF
- (D) 5NF

**Answer: (C)** 4NF

**223** Consider a relation R(A, B, C, D, E) with the following functional dependencies: F = { AB → C, C → D, D → B, D → E }. Which of the following is a Candidate Key of R?

- (A) AB only
- (B) AB and AD only
- (C) AB, AD, and AC only
- (D) AC and AD only

**Answer: (C)** AB, AD, and AC only

**224** A relation schema  $R(A, B, C, D)$  with functional dependencies  $F = \{ A \rightarrow B, B \rightarrow C \}$  is decomposed into  $R_1(A, B)$  and  $R_2(A, C, D)$ . Which of the following statements is true regarding this decomposition?

- (A) The decomposition is both lossless-join and dependency-preserving.
- (B) The decomposition is lossless-join, but NOT dependency-preserving.
- (C) The decomposition is dependency-preserving, but NOT lossless-join.
- (D) The decomposition is neither lossless-join nor dependency-preserving.

**Answer: (A)** The decomposition is both lossless-join and dependency-preserving.

**225** Which of the following describes a fundamental design trade-off in relational database normalization when decomposing a relation schema?

- (A) It is always possible to achieve a 3NF decomposition that is both lossless-join and dependency-preserving, but for BCNF, we can only guarantee lossless-join (dependency preservation is not always possible).
- (B) It is always possible to achieve a BCNF decomposition that is both lossless-join and dependency-preserving, but for 3NF, dependency preservation is not always possible.
- (C) Both 3NF and BCNF can always guarantee lossless-join and dependency preservation simultaneously.
- (D) BCNF eliminates all redundancy but can never produce a lossless-join decomposition.

**Answer: (A)** It is always possible to achieve a 3NF decomposition that is both lossless-join and dependency-preserving, but for BCNF, we can only guarantee lossless-join (dependency preservation is not always possible).

**226** Consider a relation schema  $R(A, B, C, D, E, P, G)$  with functional dependencies  $F = \{ AB \rightarrow C, C \rightarrow DE, E \rightarrow P, P \rightarrow G \}$ . What is the highest normal form of  $R$ ?

- (A) 1NF
- (B) 2NF
- (C) 3NF
- (D) BCNF

**Answer: (B)** 2NF

## SQL and Relational Algebra

DBMS - 8 questions

**227** Which of the following relational algebra operations is NOT a basic/primitive operation?

- (A) Selection ()
- (B) Projection ()
- (C) Cartesian Product (x)
- (D) Natural Join ()

**Answer: (D)** Natural Join ()

**228** Consider a relation  $R(A, B, C)$  where  $A$  is the primary key. Which of the following SQL queries is syntactically invalid?

- (A) `SELECT B, COUNT(C) FROM R GROUP BY B;`
- (B) `SELECT B, C, COUNT(A) FROM R GROUP BY B;`
- (C) `SELECT B, MAX(C) FROM R GROUP BY B HAVING MAX(C) > 10;`
- (D) `SELECT B FROM R WHERE COUNT(C) > 5 GROUP BY B;`

**Answer: (D)** `SELECT B FROM R WHERE COUNT(C) > 5 GROUP BY B;`

**229** What is the result of the relational algebra expression  $R \div S$  (Division), where  $R(A, B)$  and  $S(B)$  are two relations?

- (A) The set of all tuples in  $R$  that do not match  $S$
- (B) The set of all values of  $A$  that are associated with some values of  $B$  in  $S$
- (C) The set of all values of  $A$  in  $R$  that are associated with ALL values of  $B$  in  $S$
- (D) The Cartesian product of  $A$  and  $B$  minus the intersection of  $R$  and  $S$

**Answer: (C)** The set of all values of  $A$  in  $R$  that are associated with ALL values of  $B$  in  $S$

**230** Consider the following SQL query over a table `Employee(id, name, salary, dept_id)`:

```
SELECT name FROM Employee e1 WHERE salary > (SELECT AVG(salary) FROM Employee e2
WHERE e1.dept_id = e2.dept_id);
```

What type of subquery is this, and what does it retrieve?

- (A) A non-correlated subquery; it retrieves employees earning more than the company's average salary.
- (B) A correlated subquery; it retrieves employees earning more than their respective department's average salary.
- (C) A nested view; it retrieves employees earning less than their department's average salary.
- (D) An invalid query due to referencing `e1` inside the subquery.

**Answer: (B)** A correlated subquery; it retrieves employees earning more than their respective department's average salary.

**231**In SQL, what is the behavior of the outer join operation (e.g., LEFT OUTER JOIN)?

- (A) It returns only the rows that have matching values in both tables.
- (B) It returns all rows from the left table, and the matched rows from the right table. For unmatched rows of the right table, NULL values are filled.
- (C) It returns all rows from the right table, and the matched rows from the left table.
- (D) It returns the Cartesian product of both tables regardless of the join condition.

**Answer: (B)** It returns all rows from the left table, and the matched rows from the right table. For unmatched rows of the right table, NULL values are filled.

**232**Consider the standard logical query execution order in an SQL engine. Given the query: SELECT dept\_id, AVG(salary) AS avg\_sal FROM Employee WHERE salary > 30000 GROUP BY dept\_id HAVING avg\_sal > 50000 ORDER BY avg\_sal; Which clause in this query will trigger a syntax/ compilation error in standard SQL, and why?

- (A) The WHERE clause, because aggregate functions cannot be used there.
- (B) The HAVING clause, because the column alias 'avg\_sal' is used before it is logically defined.
- (C) The ORDER BY clause, because it runs before the SELECT clause.
- (D) The GROUP BY clause, because you cannot group by dept\_id if it is also in the SELECT clause.

**Answer: (B)** The HAVING clause, because the column alias 'avg\_sal' is used before it is logically defined.

**233**In Tuple Relational Calculus (TRC), a query expression must be 'safe' to guarantee that it terminates and returns a finite result set. Which of the following TRC expressions is UNSAFE?

- (A)  $\{ t \mid (t \text{ Employee}) \}$
- (B)  $\{ t \mid t \text{ Employee } t.\text{salary} > 50000 \}$
- (C)  $\{ t \mid t \text{ Employee } s(s \text{ Department } s.\text{dept\_id} = t.\text{dept\_id}) \}$
- (D) All of the above are safe expressions.

**Answer: (A)**  $\{ t \mid (t \text{ Employee}) \}$

**234**The Division Operator ( $R \div S$ ) in Relational Algebra is used for queries involving 'for all' or 'every' conditions (e.g. 'Find students who have registered for every course'). Since standard SQL does not have a native FORALL operator, how is this logically expressed in SQL using existential quantifiers and double negation?

- (A) Find students where EXISTS a course that they have registered for.
- (B) Find students where NOT EXISTS a course for which there is a registration.
- (C) Find students where NOT EXISTS a course such that NOT EXISTS a registration record matching both the student and the course.
- (D) Find students where EXISTS a registration record for some course.

**Answer: (C)** Find students where NOT EXISTS a course such that NOT EXISTS a registration record matching both the student and the course.

## Transactions, Concurrency, and Indexing

DBMS - 8 questions

**235**Which of the following ACID properties ensures that once a transaction commits, its updates survive any subsequent system crashes?

- (A) Atomicity
- (B) Consistency
- (C) Isolation
- (D) Durability

**Answer: (D)** Durability

**236**A schedule S of transactions is conflict-serializable if and only if:

- (A) Its precedence (serialization) graph contains a cycle
- (B) Its precedence (serialization) graph is acyclic
- (C) It uses two-phase locking (2PL)
- (D) Every transaction in it commits successfully

**Answer: (B)** Its precedence (serialization) graph is acyclic

**237**The Two-Phase Locking (2PL) protocol ensures conflict-serializability but does NOT guarantee freedom from:

- (A) Non-serializable schedules
- (B) Deadlocks
- (C) Inconsistent database states
- (D) Cascade aborts (unless strict 2PL is used)

**Answer: (B)** Deadlocks

**238** What is the primary architectural difference between a B-tree and a B+ tree index structure?

- (A) B-trees do not allow duplicate keys, while B+ trees do.
- (B) In a B-tree, keys and data pointers are stored only in leaf nodes, whereas in a B+ tree they are stored in internal nodes.
- (C) In a B-tree, keys and data/record pointers can be stored in both internal and leaf nodes. In a B+ tree, all actual records/data pointers are stored in leaf nodes, and leaf nodes are linked as a linked list for rapid sequential access.
- (D) B+ trees are binary search trees, whereas B-trees can have any number of children.

**Answer: (C)** In a B-tree, keys and data/record pointers can be stored in both internal and leaf nodes. In a B+ tree, all actual records/data pointers are stored in leaf nodes, and leaf nodes are linked as a linked list for rapid sequential access.

**239** In database crash recovery, what is the 'write-ahead logging' (WAL) protocol?

- (A) Log records are written to disk after the corresponding database changes have been written to disk.
- (B) A database backup is created before writing logs.
- (C) The log record for a database modification must be written to stable storage (disk) before the actual data page is written to disk.
- (D) Logs are written to stable storage only when the transaction commits.

**Answer: (C)** The log record for a database modification must be written to stable storage (disk) before the actual data page is written to disk.

**240** Consider a database with  $N = 100,000$  records stored in a data file. The block size is  $B = 4096$  bytes, and each record size is  $R = 400$  bytes. We create a primary index on a key field of size  $V = 16$  bytes, with block pointers of size  $P = 8$  bytes. Assuming the data file is sorted on the key field, what is the total number of disk block accesses required to search for a record using this primary index with binary search?

- (A) 6
- (B) 7
- (C) 8
- (D) 9

**Answer: (B)** 7

**241** Consider a B+ tree index structure where the block size is  $B = 512$  bytes. The search key size is  $V = 12$  bytes, and the child block pointer size is  $P_r = 8$  bytes. What is the maximum order  $p$  (maximum number of child pointers) of an internal node in this B+ tree?

- (A) 25
- (B) 26
- (C) 27
- (D) 28

**Answer: (B)** 26

**242** Under the basic Timestamp Ordering (TO) Protocol for concurrency control, a transaction  $T_i$  with timestamp  $TS(T_i)$  attempting to perform a  $Write(Q)$  operation on data item  $Q$  is aborted and rolled back if  $TS(T_i) < R\_TS(Q)$  or  $TS(T_i) < W\_TS(Q)$ . However, the Thomas Write Rule modifies this protocol to increase concurrency. What does the Thomas Write Rule specify when  $TS(T_i) < W\_TS(Q)$ ?

- (A) The transaction is aborted and restarted with a newer timestamp.
- (B) The write operation is simply ignored, and the transaction is allowed to proceed without aborting.
- (C) The write operation is buffered and executed later after a lock is released.
- (D) The read timestamp of  $Q$  is updated to  $TS(T_i)$  and the transaction proceeds.

**Answer: (B)** The write operation is simply ignored, and the transaction is allowed to proceed without aborting.

## Boolean Algebra, Logic Gates, and Minimization

Digital Logic - 15 questions

**243** Which of the following correct combinations matches the Boolean algebraic postulate/axiom to its corresponding valid equation?

- (A) Identity Axiom:  $A + 0 = A$  and  $A - 1 = A$ ; Complementarity:  $A + A' = 1$  and  $A - A' = 0$ ; Involution:  $(A')' = A$
- (B) Idempotent Axiom:  $A + 1 = A$  and  $A - 0 = A$ ; Identity Axiom:  $A + A = A$ ; Involution:  $A + A' = 1$
- (C) Complementarity Axiom:  $A + 0 = A$ ; Identity Axiom:  $A - A' = 0$ ; Idempotent Axiom:  $(A')' = A$
- (D) Involution Axiom:  $A + A' = 1$ ; Complementarity:  $A + A = A$ ; Idempotent:  $A - 1 = A$

**Answer: (A)** Identity Axiom:  $A + 0 = A$  and  $A - 1 = A$ ; Complementarity:  $A + A' = 1$  and  $A - A' = 0$ ; Involution:  $(A')' = A$

- 244** According to the Consensus Theorem, the Boolean expression  $A-B + A'-C + B-C$  simplifies to which of the following expressions, and what is its primary practical application in digital design?
- (A)  $A-B + A'-C$ ; used to identify and eliminate static hazards in asynchronous logic design.
  - (B)  $A'-C + B-C$ ; used to implement high-speed carry lookahead adders.
  - (C)  $A-B + B-C$ ; used to convert POS expressions into SOP expressions quickly.
  - (D)  $A-C + B'-C$ ; used to minimize the number of universal gates in a multiplexer.
- Answer: (A)**  $A-B + A'-C$ ; used to identify and eliminate static hazards in asynchronous logic design.
- 245** Using the Transposition Law, convert the Sum-of-Products (SOP) expression  $A-B + A'-C$  into its equivalent Product-of-Sums (POS) form.
- (A)  $(A + C)(A' + B)$
  - (B)  $(A + B)(A' + C)$
  - (C)  $(A' + C)(A + B')$
  - (D)  $(A' + B')(A + C')$
- Answer: (A)**  $(A + C)(A' + B)$
- 246** Which of the following represents the correct dual of the valid Boolean expression:  $A + A'-B = A + B$ ?
- (A)  $A - (A' + B) = A - B$
  - (B)  $A' - (A + B') = A' - B'$
  - (C)  $A - (A + B') = A' - B$
  - (D)  $A' + A - B' = A' + B'$
- Answer: (A)**  $A - (A' + B) = A - B$
- 247** A VLSI designer wants to implement a 2-input XOR gate ( $A \oplus B$ ) using only 2-input universal gates. What is the minimum number of 2-input NAND gates and 2-input NOR gates required respectively?
- (A) 4 NAND gates, 5 NOR gates
  - (B) 4 NAND gates, 4 NOR gates
  - (C) 5 NAND gates, 4 NOR gates
  - (D) 3 NAND gates, 4 NOR gates
- Answer: (A)** 4 NAND gates, 5 NOR gates
- 248** Under the principle of 'Bubble Pushing' derived from De Morgan's Laws, which of the following is a correct logic gate equivalency?
- (A) A NAND gate is equivalent to an OR gate with inverted inputs (Bubbled OR); a NOR gate is equivalent to an AND gate with inverted inputs (Bubbled AND).
  - (B) A NAND gate is equivalent to an AND gate with inverted inputs; a NOR gate is equivalent to an OR gate with inverted inputs.
  - (C) A NAND gate is equivalent to a bubbled XOR gate; a NOR gate is equivalent to a bubbled XNOR gate.
  - (D) An OR gate is equivalent to a bubbled NOR gate; an AND gate is equivalent to a bubbled NAND gate.
- Answer: (A)** A NAND gate is equivalent to an OR gate with inverted inputs (Bubbled OR); a NOR gate is equivalent to an AND gate with inverted inputs (Bubbled AND).
- 249** A digital parity generator system utilizes cascading XOR gates to check for 1-bit errors in transmission. What mathematical property of the XOR gate allows it to count odd parity, and what is the output of a 3-input XOR gate for inputs A, B, and C?
- (A) The output of an XOR gate is true if and only if an odd number of inputs are true; Output  $Y = A \oplus B \oplus C$ .
  - (B) The output of an XOR gate is true if and only if an even number of inputs are true; Output  $Y = A'B'C' + ABC$ .
  - (C) The output is true if any input is true; Output  $Y = A + B + C$ .
  - (D) The output is true only when all inputs are different; Output  $Y = A'B + BC' + CA'$ .
- Answer: (A)** The output of an XOR gate is true if and only if an odd number of inputs are true; Output  $Y = A \oplus B \oplus C$ .
- 250** Simplify the Boolean expression  $F = A + A'-B + A-B'$  using the Absorption Law and standard Boolean postulates.
- (A)  $A + B$
  - (B)  $A - B$
  - (C)  $A' + B'$
  - (D)  $A' - B$
- Answer: (A)**  $A + B$



**251** Which of the following describes the difference between the 'canonical Sum-of-Products (SOP)' form and standard SOP forms of a Boolean function?

- (A) In canonical SOP form, every single product term (minterm) must contain all the variables of the function; in standard SOP, terms can have fewer variables.
- (B) Canonical SOP form uses maxterms which yield a '0' output, while standard SOP uses minterms.
- (C) Canonical SOP form is implemented using only NOR gates, whereas standard SOP is implemented with NAND.
- (D) There is no difference; both require every variable to be present in every term.

**Answer: (A)** In canonical SOP form, every single product term (minterm) must contain all the variables of the function; in standard SOP, terms can have fewer variables.

**252** A Karnaugh Map (K-Map) is a visual, grid-based tool for logic minimization. Why are K-map cells ordered in Gray Code rather than standard binary code, and how does this affect cell adjacency?

- (A) Cells are ordered using Gray Code so that adjacent cells differ by exactly one Boolean variable, allowing adjacent 1s to be visually grouped and variables cancelled.
- (B) Cells are ordered in Gray Code to increase the number of cells in each set, enabling larger groupings.
- (C) Gray Code is used because it is a weighted code that speeds up address evaluation in hardware adders.
- (D) Gray Code prevents conflict misses by routing the cell addresses through parallel hardware comparators.

**Answer: (A)** Cells are ordered using Gray Code so that adjacent cells differ by exactly one Boolean variable, allowing adjacent 1s to be visually grouped and variables cancelled.

**253** According to K-map grouping rules, how many variables are eliminated when grouping cells in groups of 8 (octet), 4 (quad), and 2 (pair) respectively, in a 4-variable K-map?

- (A) Octet eliminates 3 variables; Quad eliminates 2 variables; Pair eliminates 1 variable.
- (B) Octet eliminates 4 variables; Quad eliminates 3 variables; Pair eliminates 2 variables.
- (C) Octet eliminates 2 variables; Quad eliminates 4 variables; Pair eliminates 1 variable.
- (D) Octet eliminates 3 variables; Quad eliminates 1 variable; Pair eliminates 2 variables.

**Answer: (A)** Octet eliminates 3 variables; Quad eliminates 2 variables; Pair eliminates 1 variable.

**254** A Boolean function is minimized using a 4-variable K-map. Minterms are located at cell indices 5, 7, 13, and 15. Assuming no other 1s exist in the map, what is the minimized logical expression obtained by grouping these cells, and which variables are eliminated?

- (A) Expression = B-D; variables A and C are eliminated.
- (B) Expression = A-C; variables B and D are eliminated.
- (C) Expression = B'-D'; variables A and C are eliminated.
- (D) Expression = A'-C'; variables B and D are eliminated.

**Answer: (A)** Expression = B-D; variables A and C are eliminated.

**255** In K-Map minimization, what are the exact definitions of a Prime Implicant (PI) versus an Essential Prime Implicant (EPI), and what are the rules regarding their inclusion in the final minimized logical expression?

- (A) A PI is any group of 1s that cannot be combined to form a larger group; an EPI is a PI that covers at least one '1' not covered by any other PI. All EPIs MUST be included in the final expression.
- (B) A PI must be included in the final expression; an EPI can be treated as a don't care and ignored.
- (C) An EPI is a group of 1s of size 2; a PI is a group of 1s of size 4. Only PIs are included in the final expression.
- (D) A PI is a group of don't cares; an EPI is a group containing only 1s. Both are optional.

**Answer: (A)** A PI is any group of 1s that cannot be combined to form a larger group; an EPI is a PI that covers at least one '1' not covered by any other PI. All EPIs MUST be included in the final expression.

**256** How should 'Don't Care' conditions (often denoted by X or d) be treated during K-map minimization to achieve the most simplified Boolean expression?

- (A) A Don't Care (X) can be treated as a 1 if it helps form a larger group to eliminate more variables, or treated as a 0 and ignored if it does not help.
- (B) A Don't Care must always be treated as a 1 and included in all groups.
- (C) A Don't Care must always be treated as a 0 and completely ignored.
- (D) Don't Cares are only valid in product-of-sums (POS) formats and are forbidden in SOP.

**Answer: (A)** A Don't Care (X) can be treated as a 1 if it helps form a larger group to eliminate more variables, or treated as a 0 and ignored if it does not help.

**257** Why becomes the Quine-McCluskey method preferred over K-Maps for logic minimization in industrial software synthesizers, and how does it systematically find the minimal cover?

- (A) K-Maps are difficult to visualize for 6 or more variables; Quine-McCluskey is a tabular, algorithmic approach that systematically identifies all prime implicants and uses a Prime Implicant Chart to find the minimal cover.
- (B) Quine-McCluskey is implemented using physical analog circuits, which are faster than digital K-maps.
- (C) Quine-McCluskey completely eliminates the need for prime implicants, generating POS forms directly.
- (D) Quine-McCluskey uses Gray Code cell ordering to bypass parallel comparators.

**Answer: (A)** K-Maps are difficult to visualize for 6 or more variables; Quine-McCluskey is a tabular, algorithmic approach that systematically identifies all prime implicants and uses a Prime Implicant Chart to find the minimal cover.

## Combinational Circuits, Adders, and Multiplexers

Digital Logic - 15 questions

**258** Which of the following represents the defining architectural characteristic of a combinational logic circuit, distinguishing it from sequential circuits?

- (A) The outputs at any given instant depend entirely on the inputs present at that exact instant, with no memory elements, feedback loops, or clock signals involved.
- (B) The outputs depend on both the present inputs and the past states, synchronized by a master clock signal.
- (C) The circuit uses cross-coupled NAND gates to store binary data asynchronously.
- (D) The propagation delay is always zero, allowing instantaneous output changes.

**Answer: (A)** The outputs at any given instant depend entirely on the inputs present at that exact instant, with no memory elements, feedback loops, or clock signals involved.

**259** A Half Adder is designed to add two 1-bit binary digits. What are the correct Boolean equations for the Sum and Carry outputs, and what is the primary structural limitation of a Half Adder?

- (A)  $\text{Sum} = A \oplus B$ ,  $\text{Carry} = A \cdot B$ ; Limitation: It cannot handle a carry-in from a previous stage in multi-bit addition.
- (B)  $\text{Sum} = A + B$ ,  $\text{Carry} = A + B$ ; Limitation: It requires an external master clock to stabilize outputs.
- (C)  $\text{Sum} = A'B + AB'$ ,  $\text{Carry} = A' - B'$ ; Limitation: It can only add signed 2's complement numbers.
- (D)  $\text{Sum} = A \oplus B$ ,  $\text{Carry} = AB + BC_{in} + AC_{in}$ ; Limitation: It requires expensive parallel comparators.

**Answer: (A)**  $\text{Sum} = A \oplus B$ ,  $\text{Carry} = A \cdot B$ ; Limitation: It cannot handle a carry-in from a previous stage in multi-bit addition.

**260** A Full Adder is designed to add three 1-bit numbers: A, B, and Carry-in ( $C_{in}$ ). What are the correct mathematical expressions for its Sum and Carry-out ( $C_{out}$ ) outputs?

- (A)  $\text{Sum} = A \oplus B \oplus C_{in}$ ;  $C_{out} = A \cdot B + B \cdot C_{in} + A \cdot C_{in}$
- (B)  $\text{Sum} = A + B + C_{in}$ ;  $C_{out} = A + B + C_{in}$
- (C)  $\text{Sum} = A \oplus B \oplus C_{in}$ ;  $C_{out} = A \cdot B + C_{in}$
- (D)  $\text{Sum} = A + B + C_{in}$ ;  $C_{out} = A'B' + BC_{in} + AC_{in}$

**Answer: (A)**  $\text{Sum} = A \oplus B \oplus C_{in}$ ;  $C_{out} = A \cdot B + B \cdot C_{in} + A \cdot C_{in}$

**261** How can a 1-bit Full Adder be structurally constructed using lower-level combinational modules, and which components are required?

- (A) By connecting two Half Adders in cascade and routing their intermediate carries through a single OR gate.
- (B) By connecting two Half Adders in parallel and routing their sums through a 2:1 multiplexer.
- (C) By using three Half Adders and one AND gate to synchronize the carry outputs.
- (D) By using a single 2-to-4 active-low decoder with external NAND gates.

**Answer: (A)** By connecting two Half Adders in cascade and routing their intermediate carries through a single OR gate.

**262** In an n-bit Ripple Carry Adder, what is the primary propagation delay bottleneck, and how does it scale with the number of bits (n)?

- (A) The carry propagation delay: Each stage must wait for the carry-out from the previous stage to stabilize; the delay scales linearly as  $O(n)$ .
- (B) The sum calculation delay: The sum must wait for the multiplexer select lines; the delay scales exponentially as  $O(2^n)$ .
- (C) The decoder delay: The decoder must activate rows in parallel; the delay scales as  $O(\log n)$ .
- (D) The clock skew delay: The delay scales as  $O(n^2)$  due to feedback registers.

**Answer: (A)** The carry propagation delay: Each stage must wait for the carry-out from the previous stage to stabilize; the delay scales linearly as  $O(n)$ .

**263** A multiplexer (MUX) is often referred to as a 'Data Selector' in digital systems. If a multiplexer has 3 select lines, how many data inputs can it select from, and how many output lines does it have?

- (A) 8 data inputs, 1 output line
- (B) 6 data inputs, 1 output line
- (C) 3 data inputs, 8 output lines
- (D) 16 data inputs, 3 output lines

**Answer: (A)** 8 data inputs, 1 output line

**264** A 4:1 Multiplexer is shown in Figure 3.1 with select lines S1 and S0, and inputs I0, I1, I2, I3. What is the correct Boolean expression for the output Y?

- (A)  $Y = S1'S0'I0 + S1'S0'I1 + S1'S0'I2 + S1'S0'I3$
- (B)  $Y = S1-S0-I0 + S1'-S0'-I1 + S1-S0'-I2 + S1'-S0-I3$
- (C)  $Y = S1'I0 + S0'I1 + S1-I2 + S0-I3$
- (D)  $Y = (S1 + S0)-I0 + (S1' + S0')-I1 + S1-I2 + S0-I3$

**Answer: (A)**  $Y = S1'S0'I0 + S1'S0'I1 + S1'S0'I2 + S1'S0'I3$

**265** According to the PDF, what is the 'Universal Logic Design' rule for implementing any arbitrary n-variable Boolean function using a multiplexer?

- (A) An n-variable Boolean function can be implemented using a  $2^{(n-1)} : 1$  MUX by connecting (n-1) variables to the select lines, and mapping the remaining variable (and its complement/constants) to the data input lines.
- (B) An n-variable Boolean function requires a  $2^n : 1$  MUX with all select lines connected to ground.
- (C) An n-variable Boolean function can only be implemented using a 2-to-1 MUX cascading tree of depth  $n^2$ .
- (D) A multiplexer cannot implement non-linear Boolean functions like XOR without external logic gates.

**Answer: (A)** An n-variable Boolean function can be implemented using a  $2^{(n-1)} : 1$  MUX by connecting (n-1) variables to the select lines, and mapping the remaining variable (and its complement/constants) to the data input lines.

**266** A designer wants to implement the 3-variable Boolean function  $f(A, B, C) = \text{summ}(1, 3, 5, 6)$  using a 4:1 multiplexer. If variables A and B are connected to select lines S1 and S0 respectively, what are the correct values/signals that must be connected to the data inputs I0, I1, I2, I3?

- (A)  $I0 = C, I1 = C, I2 = C, I3 = C'$
- (B)  $I0 = C, I1 = C, I2 = C', I3 = C$
- (C)  $I0 = C', I1 = C, I2 = C, I3 = C$
- (D)  $I0 = C, I1 = C', I2 = C, I3 = C$

**Answer: (A)**  $I0 = C, I1 = C, I2 = C, I3 = C'$

**267** What is the primary function of an n-bit Binary Decoder, and what is its most common physical application in computer systems?

- (A) It takes an n-bit binary input code and activates exactly one of the  $2^n$  output lines; used in memory addressing to select specific rows or memory chips.
- (B) It compresses  $2^n$  input channels into an n-bit code; used in keyboard encoders to compress key strokes.
- (C) It resolves simultaneous active interrupt requests; used in microprocessor interrupt units.
- (D) It performs binary division using shift-and-subtract operations.

**Answer: (A)** It takes an n-bit binary input code and activates exactly one of the  $2^n$  output lines; used in memory addressing to select specific rows or memory chips.

**268** A designer wants to implement a Full Adder Sum function using an active-high 3-to-8 decoder. Which of the following represents the correct external connection?

- (A) Connect the three inputs A, B, Cin to the decoder select inputs, and feed the decoder outputs Y1, Y2, Y4, Y7 into a single 4-input OR gate.
- (B) Connect the three inputs A, B, Cin to the decoder select inputs, and feed the decoder outputs Y3, Y5, Y6, Y7 into a single 4-input AND gate.
- (C) Connect the three inputs A, B, Cin to the decoder select inputs, and feed the decoder outputs Y0, Y2, Y3, Y5 into a single 4-input OR gate.
- (D) Connect the three inputs A, B, Cin to the decoder select inputs, and feed the decoder outputs Y0, Y1, Y4, Y6 into a single 4-input NOR gate.

**Answer: (A)** Connect the three inputs A, B, Cin to the decoder select inputs, and feed the decoder outputs Y1, Y2, Y4, Y7 into a single 4-input OR gate.

**269**An Encoder is a combinational circuit that performs the reverse operation of a Decoder. What is the standard configuration of a binary encoder?

- (A) It has  $2^n$  input lines and produces an n-bit binary output code.
- (B) It has n input lines and produces  $2^n$  output lines.
- (C) It has  $2^n$  input lines and produces a single output line.
- (D) It has 1 input line and produces  $2^n$  output lines.

**Answer: (A)** It has  $2^n$  input lines and produces an n-bit binary output code.

**270**In priority encoders, how are simultaneous active inputs resolved, and what is its primary application in microprocessors?

- (A) By assigning priority to the higher-order inputs such that if multiple inputs are active at once, the output reflects the input with the highest priority; used in microprocessor interrupt handling.
- (B) By disabling all inputs and outputting a NaN state; used in arithmetic logic units.
- (C) By averaging the active inputs using parallel comparators; used in program relocation.
- (D) By using an internal feedback loop to cycle through active inputs sequentially; used in ring counters.

**Answer: (A)** By assigning priority to the higher-order inputs such that if multiple inputs are active at once, the output reflects the input with the highest priority; used in microprocessor interrupt handling.

**271**A 1-to-4 Demultiplexer (DEMUX) takes a single input data line (D) and distributes it to one of four output channels (Y0, Y1, Y2, Y3) based on two select lines (S1, S0). What are the correct Boolean equations for the four outputs?

- (A)  $Y0 = S1' \cdot S0' \cdot D$ ;  $Y1 = S1' \cdot S0 \cdot D$ ;  $Y2 = S1 \cdot S0' \cdot D$ ;  $Y3 = S1 \cdot S0 \cdot D$
- (B)  $Y0 = S1 \cdot S0 \cdot D$ ;  $Y1 = S1' \cdot S0' \cdot D$ ;  $Y2 = S1 \cdot S0' \cdot D$ ;  $Y3 = S1' \cdot S0 \cdot D$
- (C)  $Y0 = Y1 = Y2 = Y3 = (S1 \cdot S0) \cdot D$
- (D)  $Y0 = S1' \cdot D$ ;  $Y1 = S0' \cdot D$ ;  $Y2 = S1 \cdot D$ ;  $Y3 = S0 \cdot D$

**Answer: (A)**  $Y0 = S1' \cdot S0' \cdot D$ ;  $Y1 = S1' \cdot S0 \cdot D$ ;  $Y2 = S1 \cdot S0' \cdot D$ ;  $Y3 = S1 \cdot S0 \cdot D$

**272**An active-low 2-to-4 decoder has outputs Y0', Y1', Y2', Y3'. If the input select bits are S1S0 = 10, what are the output states of the four pins respectively?

- (A) Y0' = 1, Y1' = 1, Y2' = 0, Y3' = 1
- (B) Y0' = 0, Y1' = 0, Y2' = 1, Y3' = 0
- (C) Y0' = 1, Y1' = 1, Y2' = 1, Y3' = 0
- (D) Y0' = 0, Y1' = 1, Y2' = 1, Y3' = 1

**Answer: (A)** Y0' = 1, Y1' = 1, Y2' = 0, Y3' = 1

## Sequential Circuits, Flip-Flops, and Counters

Digital Logic - 15 questions

**273**Which of the following defines the primary difference between combinational and sequential logic circuits, and what core elements are present in sequential circuits?

- (A) Sequential circuits contain memory elements and a clock signal, meaning their outputs depend on both the present inputs and the past outputs (present state).
- (B) Sequential circuits use only logic gates without any feedback, meaning their outputs depend only on the present inputs.
- (C) Sequential circuits have zero propagation delay, whereas combinational circuits have a linear propagation delay.
- (D) Sequential circuits are used exclusively for floating-point calculations, while combinational are used for integer math.

**Answer: (A)** Sequential circuits contain memory elements and a clock signal, meaning their outputs depend on both the present inputs and the past outputs (present state).

**274**What is the key distinction between a 'Latch' and a 'Flip-Flop' in terms of triggering mechanisms, and what problem do flip-flops resolve?

- (A) A latch is asynchronous and level-triggered; a flip-flop is synchronous and edge-triggered. Flip-flops solve the transparency problem inherent in simple latches.
- (B) A latch is edge-triggered; a flip-flop is level-triggered. Flip-flops solve the race-around condition of ripple counters.
- (C) A latch requires an external decoder; a flip-flop utilizes an internal Master-Slave priority encoder.
- (D) There is no difference; both are level-triggered and change state as long as enable is high.

**Answer: (A)** A latch is asynchronous and level-triggered; a flip-flop is synchronous and edge-triggered. Flip-flops solve the transparency problem inherent in simple latches.

**275** An SR (Set-Reset) Flip-Flop is modeled by the characteristic equation  $Q(t+1) = S + R'Q(t)$ . Under which input condition does the SR flip-flop enter an invalid/forbidden state, and why is this state forbidden?

- (A)  $S = 1$  and  $R = 1$  simultaneously; it is forbidden because both outputs  $Q$  and  $Q'$  are forced to 0, violating the complementarity principle.
- (B)  $S = 0$  and  $R = 0$  simultaneously; it is forbidden because it causes the clock frequency to drop to zero.
- (C)  $S = 1$  and  $R = 0$  simultaneously; it is forbidden because it causes an automatic system reset.
- (D)  $S = 0$  and  $R = 1$  simultaneously; it is forbidden because it triggers a race-around condition.

**Answer: (A)**  $S = 1$  and  $R = 1$  simultaneously; it is forbidden because both outputs  $Q$  and  $Q'$  are forced to 0, violating the complementarity principle.

**276** Which universal flip-flop resolves the invalid/forbidden state of the SR flip-flop, what is its characteristic equation, and what action occurs when both inputs are asserted simultaneously?

- (A) The JK flip-flop; Characteristic Equation:  $Q(t+1) = JQ'(t) + K'Q(t)$ ; It toggles its output state when  $J = 1$  and  $K = 1$ .
- (B) The D flip-flop; Characteristic Equation:  $Q(t+1) = D$ ; It holds the previous state when  $D = 1$ .
- (C) The T flip-flop; Characteristic Equation:  $Q(t+1) = T Q(t)$ ; It resets the output to 0 when  $T = 1$ .
- (D) The Master-Slave SR flip-flop; Characteristic Equation:  $Q(t+1) = S-R$ ; It bypasses the clock entirely.

**Answer: (A)** The JK flip-flop; Characteristic Equation:  $Q(t+1) = JQ'(t) + K'Q(t)$ ; It toggles its output state when  $J = 1$  and  $K = 1$ .

**277** A D (Data/Delay) Flip-Flop has the characteristic equation  $Q(t+1) = D$ . What is its main structural significance, and in which systems is it most extensively utilized?

- (A) It eliminates the invalid state completely; used extensively in Shift Registers and Buffer Memory.
- (B) It toggles the output state on every clock pulse; used extensively in digital counters.
- (C) It uses level-triggering to speed up addition; used extensively in carry lookahead adders.
- (D) It acts as a priority encoder; used in interrupt controllers.

**Answer: (A)** It eliminates the invalid state completely; used extensively in Shift Registers and Buffer Memory.

**278** A T (Toggle) Flip-Flop has the characteristic equation  $Q(t+1) = T Q(t)$ . What is its output behavior when input  $T$  is connected to 1, and what is its primary practical application?

- (A) It toggles its state on every clock pulse; used extensively in building digital Counters.
- (B) It holds its state indefinitely; used extensively in buffer memories.
- (C) It enters an invalid state; used in error detection.
- (D) It acts as a shift register; used to load data in parallel.

**Answer: (A)** It toggles its state on every clock pulse; used extensively in building digital Counters.

**279** In a level-triggered JK flip-flop, what physical conditions trigger the 'Race-Around Condition', and which of the following represents a correct hardware solution to completely eliminate it?

- (A) It occurs when  $J = K = 1$  and the clock pulse width is greater than the propagation delay of the flip-flop; resolved by using a Master-Slave JK Flip-Flop configuration or by strictly ensuring edge-triggering.
- (B) It occurs when  $J = K = 0$  and the clock frequency is extremely fast; resolved by using an external Priority Encoder.
- (C) It occurs when the setup time is larger than the hold time; resolved by adding a series of buffer gates.
- (D) It occurs when separate data and instruction caches clash; resolved by using Harvard Architecture.

**Answer: (A)** It occurs when  $J = K = 1$  and the clock pulse width is greater than the propagation delay of the flip-flop; resolved by using a Master-Slave JK Flip-Flop configuration or by strictly ensuring edge-triggering.

**280** An  $n$ -bit shift register can be configured for Serial-In Serial-Out (SISO) or Parallel-In Parallel-Out (PIPO) operations. For a 4-bit shift register, what is the exact number of clock cycles required to load a 4-bit number, and subsequently read it out under SISO versus PIPO operations?

- (A) SISO: 4 clock cycles to load, 3 additional cycles to read; PIPO: 1 clock cycle to load, 0 additional cycles to read.
- (B) SISO: 4 clock cycles to load, 4 additional cycles to read; PIPO: 4 clock cycles to load, 4 additional cycles to read.
- (C) SISO: 1 clock cycle to load, 1 additional cycle to read; PIPO: 4 clock cycles to load, 1 additional cycle to read.
- (D) SISO: 3 clock cycles to load, 4 additional cycles to read; PIPO: 1 clock cycle to load, 4 additional cycles to read.

**Answer: (A)** SISO: 4 clock cycles to load, 3 additional cycles to read; PIPO: 1 clock cycle to load, 0 additional cycles to read.



**281**What is the structural and speed difference between a Ripple (Asynchronous) Counter and a Synchronous Counter?

- (A) In a Ripple Counter, only the first flip-flop is clocked by the master clock, and subsequent stages are clocked by the outputs of preceding stages, making it slow due to cumulative delays. In a Synchronous Counter, all flip-flops are clocked simultaneously by the same clock, making it much faster.
- (B) In a Ripple Counter, all flip-flops are clocked simultaneously. In a Synchronous Counter, stages are clocked sequentially.
- (C) Ripple Counters have zero propagation delay, while Synchronous Counters suffer from cumulative delays.
- (D) Ripple Counters are used for program relocation, while Synchronous Counters are used for cache tag comparison.

**Answer: (A)** In a Ripple Counter, only the first flip-flop is clocked by the master clock, and subsequent stages are clocked by the outputs of preceding stages, making it slow due to cumulative delays. In a Synchronous Counter, all flip-flops are clocked simultaneously by the same clock, making it much faster.

**282**An n-bit binary ripple counter is constructed using flip-flops, each having a propagation delay of  $t_{pd}$ . What is the formula for the total propagation delay of the counter, and what is the maximum cumulative delay for a 16-bit binary ripple counter constructed using flip-flops with a 20 ns propagation delay?

- (A) Total Delay =  $n \times t_{pd}$ ; Maximum Cumulative Delay = 320 ns
- (B) Total Delay =  $t_{pd} / n$ ; Maximum Cumulative Delay = 1.25 ns
- (C) Total Delay =  $(n + 1) \times t_{pd}$ ; Maximum Cumulative Delay = 340 ns
- (D) Total Delay =  $n^2 \times t_{pd}$ ; Maximum Cumulative Delay = 5120 ns

**Answer: (A)** Total Delay =  $n \times t_{pd}$ ; Maximum Cumulative Delay = 320 ns

**283**In a 4-bit Synchronous Binary Counter, the flip-flop propagation delay is  $t_{pd} = 15$  ns, and the external combinational logic (AND gates) has a delay of  $t_{gate} = 5$  ns. If the setup time of the flip-flops is  $t_{setup} = 4$  ns, what is the absolute maximum clock frequency ( $f_{max}$ ) at which this synchronous counter can safely operate?

- (A) 41.67 MHz
- (B) 52.63 MHz
- (C) 66.67 MHz
- (D) 100.00 MHz

**Answer: (A)** 41.67 MHz

**284**Compare the number of distinct binary states and the counting sequences of an n-bit Ring Counter versus an n-bit Johnson Counter initialized with a single '1' and all '0's respectively.

- (A) Ring Counter: n states, initialized with single '1' (e.g. 1000 -> 0100 -> 0010 -> 0001); Johnson Counter:  $2n$  states, initialized with all '0's (e.g. 0000 -> 1000 -> 1100 -> 1110 -> 1111 -> 0111 -> 0011 -> 0001).
- (B) Ring Counter:  $2^n$  states, initialized with 1010; Johnson Counter: n states, initialized with all '1's.
- (C) Ring Counter: n states; Johnson Counter:  $2^n$  states. Both require a priority decoder to cycle.
- (D) Both counters provide identical  $2n$  states, but the Ring Counter is level-triggered.

**Answer: (A)** Ring Counter: n states, initialized with single '1' (e.g. 1000 -> 0100 -> 0010 -> 0001); Johnson Counter:  $2n$  states, initialized with all '0's (e.g. 0000 -> 1000 -> 1100 -> 1110 -> 1111 -> 0111 -> 0011 -> 0001).

**285**A bi-directional Up/Down Counter is a sequential circuit that can count either up or down depending on a control input M. Which of the following describes its steering logic?

- (A) When  $M = 1$ , it counts up by clocking stage i with Q of stage i-1; when  $M = 0$ , it counts down by clocking stage i with Q' of stage i-1.
- (B) When  $M = 1$ , it clocks stages simultaneously; when  $M = 0$ , it operates as a ripple counter.
- (C) When  $M = 1$ , it acts as a Ring Counter; when  $M = 0$ , it acts as a Johnson Counter.
- (D) It uses a priority encoder to count up, and a 2:1 multiplexer to count down.

**Answer: (A)** When  $M = 1$ , it counts up by clocking stage i with Q of stage i-1; when  $M = 0$ , it counts down by clocking stage i with Q' of stage i-1.

**286**To convert a JK flip-flop into a D flip-flop, what external logic must be added to the J and K inputs?

- (A) Connect J directly to D, and connect K to D through an inverter (NOT gate).
- (B) Connect J and K together directly to D.
- (C) Connect J to D, and connect K to ground (0).
- (D) Connect J to D, and connect K to Q.

**Answer: (A)** Connect J directly to D, and connect K to D through an inverter (NOT gate).



- 287** A MOD-10 decade counter counts from 0000 to 1001 (0 to 9 in decimal) and then resets back to 0000. How many flip-flops are required to construct this counter, and which combination of outputs must be decoded to trigger the early clear/reset signal?
- (A) Requires 4 flip-flops; output lines Q3 and Q1 must be connected to an active-low NAND gate to trigger the asynchronous CLEAR inputs of all flip-flops.
  - (B) Requires 10 flip-flops; output line Q9 must be connected directly to ground.
  - (C) Requires 3 flip-flops; output lines Q2 and Q0 must be connected to an AND gate.
  - (D) Requires 4 flip-flops; output lines Q2 and Q0 must be connected to a NOR gate.

**Answer: (A)** Requires 4 flip-flops; output lines Q3 and Q1 must be connected to an active-low NAND gate to trigger the asynchronous CLEAR inputs of all flip-flops.

## Number Representations, Fixed-Point and Floating-Point Computer Arithmetic

Digital Logic - 15 questions

- 288** Which of the following correct combinations matches the Binary Coded Decimal (BCD) representation properties and its invalid codes?
- (A) BCD is a weighted 8421 code; it requires more bits than standard binary (e.g., 355 requires 12 bits in BCD vs 9 in binary); codes 1010 through 1111 (10 to 15 decimal) are invalid.
  - (B) BCD is a non-weighted code; it requires fewer bits than binary; codes 0000 through 0101 are invalid.
  - (C) BCD is a 7-bit ASCII code; it requires exactly 7 bits per decimal digit; codes 1000 through 1001 are invalid.
  - (D) BCD represents binary numbers directly using Gray Code; it has no invalid codes.
- Answer: (A)** BCD is a weighted 8421 code; it requires more bits than standard binary (e.g., 355 requires 12 bits in BCD vs 9 in binary); codes 1010 through 1111 (10 to 15 decimal) are invalid.
- 289** Which of the following describes the key structural properties of the Excess-3 and Gray codes?
- (A) Excess-3 is a non-weighted self-complementing code obtained by adding 3 to each BCD digit; Gray Code is a non-weighted sequential code where adjacent codes differ by exactly 1 bit, preventing transient glitch states.
  - (B) Excess-3 is a weighted 8421 code; Gray Code is a weighted code used directly in ALU arithmetic.
  - (C) Excess-3 requires 7 bits per character; Gray Code has two representations of zero (+0 and -0).
  - (D) Excess-3 is used in floating-point exponent biasing; Gray Code is used only in active-low decoders.
- Answer: (A)** Excess-3 is a non-weighted self-complementing code obtained by adding 3 to each BCD digit; Gray Code is a non-weighted sequential code where adjacent codes differ by exactly 1 bit, preventing transient glitch states.
- 290** What is the exact ranges of signed integers that can be represented in an n-bit register using Sign-Magnitude, 1's Complement, and 2's Complement formats?
- (A) Sign-Magnitude:  $-(2^{(n-1)} - 1)$  to  $+(2^{(n-1)} - 1)$ ; 1's Complement:  $-(2^{(n-1)} - 1)$  to  $+(2^{(n-1)} - 1)$ ; 2's Complement:  $-2^{(n-1)}$  to  $+(2^{(n-1)} - 1)$ .
  - (B) Sign-Magnitude:  $-2^{(n-1)}$  to  $+2^{(n-1)}$ ; 1's Complement:  $-2^{(n-1)}$  to  $+2^{(n-1)}$ ; 2's Complement:  $-(2^{(n-1)} - 1)$  to  $+(2^{(n-1)} - 1)$ .
  - (C) Sign-Magnitude:  $-(2^{(n-1)} - 1)$  to  $+(2^{(n-1)} - 1)$ ; 1's Complement:  $-2^{(n-1)}$  to  $+2^{(n-1)}$ ; 2's Complement:  $-2^{(n-1)}$  to  $+2^{(n-1)}$ .
  - (D) All three formats represent identical ranges of  $-2^{(n-1)}$  to  $+(2^{(n-1)} - 1)$ .
- Answer: (A)** Sign-Magnitude:  $-(2^{(n-1)} - 1)$  to  $+(2^{(n-1)} - 1)$ ; 1's Complement:  $-(2^{(n-1)} - 1)$  to  $+(2^{(n-1)} - 1)$ ; 2's Complement:  $-2^{(n-1)}$  to  $+(2^{(n-1)} - 1)$ .
- 291** Which of the following represents the unique representations of decimal zero (0) in 8-bit Sign-Magnitude, 1's Complement, and 2's Complement formats respectively?
- (A) Sign-Magnitude has +0 (00000000) and -0 (10000000); 1's Complement has +0 (00000000) and -0 (11111111); 2's Complement has only one zero (00000000).
  - (B) Sign-Magnitude has only one zero (00000000); 1's Complement has only one zero (11111111); 2's Complement has +0 and -0.
  - (C) All three formats represent zero uniquely as 00000000.
  - (D) Sign-Magnitude has +0 (10000000) and -0 (00000000); 1's Complement has +0 (11111111) and -0 (00000000); 2's Complement has +0 (00000000) and -0 (11111111).
- Answer: (A)** Sign-Magnitude has +0 (00000000) and -0 (10000000); 1's Complement has +0 (00000000) and -0 (11111111); 2's Complement has only one zero (00000000).

**292** Perform the subtraction  $A - B$  in 8-bit signed 2's complement arithmetic, where  $A = +42$  and  $B = +68$ . What is the final 8-bit binary pattern in the register, and what decimal value does it represent?

- (A) Binary: 11100110; Decimal value: -26
- (B) Binary: 00011010; Decimal value: +26
- (C) Binary: 11100101; Decimal value: -27
- (D) Binary: 10011010; Decimal value: -102

**Answer: (A)** Binary: 11100110; Decimal value: -26

**293** In fixed-point computer arithmetic, when adding two signed  $n$ -bit 2's complement integers, what is the exact hardware condition used to detect arithmetic overflow, and how is it implemented using basic logic gates?

- (A) Overflow Condition:  $C_{in} \oplus C_{out} = 1$  (where  $C_{in}$  is the carry-in to the MSB and  $C_{out}$  is the carry-out of the MSB); implemented using a single 2-input XOR gate.
- (B) Overflow Condition:  $C_{out} = 1$ ; implemented using a single AND gate.
- (C) Overflow Condition:  $C_{in} = 1$ ; implemented using an inverter.
- (D) Overflow occurs only if the sign bit of both operands are different; implemented using an XNOR gate.

**Answer: (A)** Overflow Condition:  $C_{in} \oplus C_{out} = 1$  (where  $C_{in}$  is the carry-in to the MSB and  $C_{out}$  is the carry-out of the MSB); implemented using a single 2-input XOR gate.

**294** Which of the following describes the core operational characteristics and benefits of Booth's Multiplication Algorithm?

- (A) It handles signed 2's complement numbers directly without conversion; by examining adjacent pairs of bits, a string of consecutive 1s requires only one addition and one subtraction, reducing ALU cycles.
- (B) It performs binary multiplication by converting all negative numbers to sign-magnitude, requiring an extra clock cycle.
- (C) It computes product matrices using parallel carry-save subtractors, limiting execution to positive integers.
- (D) It processes bits from MSB to LSB, restoring partial products dynamically using an external priority encoder.

**Answer: (A)** It handles signed 2's complement numbers directly without conversion; by examining adjacent pairs of bits, a string of consecutive 1s requires only one addition and one subtraction, reducing ALU cycles.

**295** What is the primary difference between Restoring Division and Non-Restoring Division algorithms in computer fixed-point arithmetic?

- (A) In Restoring Division, the partial remainder is restored by adding the divisor back if the subtraction results in a negative value. Non-Restoring Division bypasses this restore cycle, making it faster.
- (B) Restoring Division is used only for signed numbers, while Non-Restoring is used only for unsigned numbers.
- (C) Restoring Division uses shift-right operations, whereas Non-Restoring division uses only Booth-encoded multiplication.
- (D) Restoring Division restores the exponent bias, whereas Non-Restoring Division uses normalized mantissas.

**Answer: (A)** In Restoring Division, the partial remainder is restored by adding the divisor back if the subtraction results in a negative value. Non-Restoring Division bypasses this restore cycle, making it faster.

**296** Under the IEEE 754 Floating-Point Standard, what are the three structural components used to represent a real number, and how is the value calculated?

- (A) Sign Bit ( $S$ ), Biased Exponent ( $E$ ), and Normalized Mantissa/Fraction ( $M$ ); Value =  $(-1)^S \times 1.M \times 2^{(E - \text{Bias})}$
- (B) Sign Bit ( $S$ ), Unsigned Magnitude ( $M$ ), and Gray-coded Exponent ( $E$ ); Value =  $M \times 10^E$
- (C) Excess-3 Exponent, BCD Mantissa, and Parity Bit; Value =  $(-1)^S \times M \times 10^{(E - 127)}$
- (D) Sign Bit ( $S$ ), Complement Exponent ( $E$ ), and Dual Mantissa ( $M$ ); Value =  $(-1)^S \times (1 - M) \times 2^E$

**Answer: (A)** Sign Bit ( $S$ ), Biased Exponent ( $E$ ), and Normalized Mantissa/Fraction ( $M$ ); Value =  $(-1)^S \times 1.M \times 2^{(E - \text{Bias})}$

**297** What is the exact bit allocations and exponent bias values for IEEE 754 Single Precision (32-bit) and Double Precision (64-bit) formats respectively?

- (A) Single Precision: Sign = 1, Exponent = 8, Mantissa = 23, Bias = 127; Double Precision: Sign = 1, Exponent = 11, Mantissa = 52, Bias = 1023.
- (B) Single Precision: Sign = 1, Exponent = 11, Mantissa = 20, Bias = 1023; Double Precision: Sign = 1, Exponent = 8, Mantissa = 55, Bias = 127.
- (C) Single Precision: Sign = 1, Exponent = 8, Mantissa = 23, Bias = 128; Double Precision: Sign = 1, Exponent = 12, Mantissa = 51, Bias = 2047.
- (D) Single Precision: Sign = 1, Exponent = 7, Mantissa = 24, Bias = 63; Double Precision: Sign = 1, Exponent = 10, Mantissa = 53, Bias = 511.

**Answer: (A)** Single Precision: Sign = 1, Exponent = 8, Mantissa = 23, Bias = 127; Double Precision: Sign = 1, Exponent = 11, Mantissa = 52, Bias = 1023.

**298**In the IEEE 754 standard, which combination of exponent (E) and mantissa (M) values represents positive or negative Infinity, and which represents NaN (Not a Number)?

- (A) Infinity: E = all 1s (255) and M = 0; NaN: E = all 1s (255) and M != 0.
- (B) Infinity: E = 0 and M = 0; NaN: E = 0 and M != 0.
- (C) Infinity: E = all 1s and M != 0; NaN: E = all 1s and M = 0.
- (D) Infinity: E = 127 and M = 0; NaN: E = 128 and M != 0.

**Answer: (A)** Infinity: E = all 1s (255) and M = 0; NaN: E = all 1s (255) and M != 0.

**299**In the IEEE 754 standard, which combination of exponent (E) and mantissa (M) values represents actual numerical Zero, and which represents Denormalized Numbers, and what is the physical purpose of denormalized numbers?

- (A) Zero: E = 0 and M = 0; Denormalized: E = 0 and M != 0. Purpose: Denormalized numbers represent extremely small values near zero, allowing gradual underflow.
- (B) Zero: E = all 1s and M = 0; Denormalized: E = all 1s and M != 0. Purpose: To accelerate binary division.
- (C) Zero: E = 0 and M != 0; Denormalized: E = 0 and M = 0. Purpose: To support dual representation of zero.
- (D) Zero: E = 127 and M = 0; Denormalized: E = 1023 and M != 0. Purpose: Bypassing K-map Gray code checks.

**Answer: (A)** Zero: E = 0 and M = 0; Denormalized: E = 0 and M != 0. Purpose: Denormalized numbers represent extremely small values near zero, allowing gradual underflow.

**300**When performing floating-point addition of two IEEE 754 numbers, why is it necessary to compare the exponents, and what action is performed on the mantissa of the number with the smaller exponent?

- (A) To perform addition, the exponents must be equal; therefore, we find the difference between exponents, and shift the mantissa of the number with the smaller exponent to the right by that difference to align their binary points.
- (B) We compare exponents to verify if overflow has occurred; we must shift the larger exponent mantissa to the left.
- (C) We compare exponents to swap their sign bits; we must add 127 bias to the smaller mantissa.
- (D) We compare exponents to convert the format into BCD; we must shift both mantissas to the left until the sign bits match.

**Answer: (A)** To perform addition, the exponents must be equal; therefore, we find the difference between exponents, and shift the mantissa of the number with the smaller exponent to the right by that difference to align their binary points.

**301**A 16-bit float format has 1 sign bit, 5 exponent bits (biased by 15), and 10 mantissa bits. What is the decimal representation of the float value stored as the binary pattern: 1 10010 1100000000?

- (A) -13.5
- (B) -6.75
- (C) -3.375
- (D) -27.0

**Answer: (A)** -13.5

**302**A student wants to convert the decimal real number -6.5 into IEEE 754 single-precision format. What is the correct hexadecimal representation of the 32-bit word stored in memory?

- (A) 0xC0D00000
- (B) 0xC0B00000
- (C) 0x40D00000
- (D) 0xC1100000

**Answer: (A)** 0xC0D00000

## Discrete Mathematics and Linear Algebra

Engineering Mathematics - 5 questions

**303**Which of the following propositions is a tautology?

- (A)  $(p \vee q) \rightarrow (p \wedge q)$
- (B)  $(p \rightarrow q) \vee (p \wedge q)$
- (C)  $(p \vee q) \rightarrow (p)$
- (D)  $(p \rightarrow q) \wedge (q \rightarrow p)$

**Answer: (B)**  $(p \rightarrow q) \vee (p \wedge q)$

**304** Let  $R$  be a relation on the set of all integers such that  $a R b$  if and only if  $(a - b)$  is divisible by 5. Which of the following properties are satisfied by  $R$ ?

- (A) Reflexive and symmetric, but not transitive
- (B) Reflexive, symmetric, and transitive (Equivalence relation)
- (C) Reflexive and transitive, but not symmetric
- (D) Symmetric and transitive, but not reflexive

**Answer: (B)** Reflexive, symmetric, and transitive (Equivalence relation)

**305** What is the chromatic number of a complete bipartite graph  $K_{3,3}$ ?

- (A) 1
- (B) 2
- (C) 3
- (D) 6

**Answer: (B)** 2

**306** Given a matrix  $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ . What are the eigenvalues of  $A$ ?

- (A) 1 and 2
- (B) 1 and 3
- (C) 2 and 3
- (D) 0 and 3

**Answer: (B)** 1 and 3

**307** A system of linear equations  $Ax = b$  is consistent if and only if:

- (A)  $\text{Rank}(A) = \text{Rank}([A|b])$
- (B)  $\text{Rank}(A) < \text{Rank}([A|b])$
- (C)  $\text{Rank}(A) > \text{Rank}([A|b])$
- (D)  $\text{Rank}(A)$  is equal to the number of variables

**Answer: (A)**  $\text{Rank}(A) = \text{Rank}([A|b])$

## Probability, Statistics, and Calculus

Engineering Mathematics - 5 questions

**308** A bag contains 5 red and 7 blue balls. Two balls are drawn at random one after another without replacement. What is the probability that the first ball is red and the second ball is blue?

- (A)  $35/144$
- (B)  $35/132$
- (C)  $5/12$
- (D)  $7/12$

**Answer: (B)**  $35/132$

**309** An observer knows that a certain diagnostic test is 99% accurate: if a patient has the disease, it tests positive with 99% probability, and if a patient does not have it, it tests negative with 99% probability. Suppose only 1% of the population has this disease. If a randomly selected person tests positive, what is the probability that they actually have the disease?

- (A) 0.99
- (B) 0.50
- (C) 0.01
- (D) 0.90

**Answer: (B)** 0.50

**310** The number of accidents occurring in a city follows a Poisson distribution with an average of 2 accidents per day. What is the probability that exactly 0 accidents occur tomorrow?

- (A)  $e^{-2}$
- (B)  $2 * e^{-2}$
- (C)  $1 - e^{-2}$
- (D)  $e^2$

**Answer: (A)**  $e^{-2}$

**311** Find the value of the limit:  $\lim_{x \rightarrow 0} [(\sin 3x) / x]$ .

- (A) 1
- (B) 3
- (C) 0
- (D) Does not exist

**Answer: (B)** 3

**312** Which of the following conditions is required for a function  $f(x)$  to have a local maximum at  $x = c$ ?

- (A)  $f'(c) = 0$  and  $f''(c) > 0$
- (B)  $f'(c) = 0$  and  $f''(c) < 0$
- (C)  $f'(c) > 0$  and  $f''(c) = 0$
- (D)  $f'(c) < 0$  and  $f''(c) < 0$

**Answer: (B)**  $f'(c) = 0$  and  $f''(c) < 0$

## CIL MT Mock Sheet-1 (Systems)

Mock Tests - 100 questions

**313** Which of the following CPU registers is used to interface with the main memory data bus?

- (A) Program Counter (PC)
- (B) Memory Address Register (MAR)
- (C) Memory Buffer Register (MBR)
- (D) Instruction Register (IR)

**Answer: (C)** Memory Buffer Register (MBR)

**314** The technique of executing multiple instructions in parallel by fetching them during different clock intervals is known as:

- (A) Paging
- (B) Pipelining
- (C) Multithreading
- (D) Vector Processing

**Answer: (B)** Pipelining

**315** What is the primary function of the Lexical Analyzer in a compiler?

- (A) Syntax tree generation
- (B) Type checking
- (C) Grouping characters into tokens
- (D) Code optimization

**Answer: (C)** Grouping characters into tokens

**316** Which data structure is used by the operating system to manage function calls and local variables?

- (A) Queue
- (B) Stack
- (C) Binary Tree
- (D) Heap

**Answer: (B)** Stack

**317** Which of the following sorting algorithms has a stable  $O(N \log N)$  worst-case time complexity?

- (A) Quick Sort
- (B) Bubble Sort
- (C) Merge Sort
- (D) Selection Sort

**Answer: (C)** Merge Sort

**318** In the relational model, which of the following is a candidate key?

- (A) Any superkey
- (B) A minimal superkey
- (C) Any primary key
- (D) A foreign key

**Answer: (B)** A minimal superkey

**319** Which layer of the OSI model handles end-to-end connection establishment, flow control, and error recovery?

- (A) Network Layer
- (B) Data Link Layer
- (C) Transport Layer
- (D) Session Layer

**Answer: (C)** Transport Layer

**320**What is the maximum number of nodes in a binary tree of height H (where root is at height 1)?

- (A)  $2^H$
- (B)  $2^H - 1$
- (C)  $2^{(H+1)} - 1$
- (D)  $H^2$

**Answer: (B)  $2^H - 1$**

**321**In standard SQL, which clause is used to filter groups formed by the GROUP BY clause?

- (A) WHERE
- (B) HAVING
- (C) ORDER BY
- (D) SELECT

**Answer: (B) HAVING**

**322**Which of the following is a non-preemptive CPU scheduling algorithm?

- (A) Round Robin
- (B) Shortest Job First (SJF) standard
- (C) First-Come, First-Served (FCFS)
- (D) Shortest Remaining Time First (SRTF)

**Answer: (C) First-Come, First-Served (FCFS)**

**323**Which memory management scheme eliminates external fragmentation completely?

- (A) Segmentation
- (B) Contiguous allocation
- (C) Paging
- (D) Dynamic partitioning

**Answer: (C) Paging**

**324**Which of the following represents the class of languages recognized by a Pushdown Automaton?

- (A) Regular Languages
- (B) Context-Free Languages
- (C) Context-Sensitive Languages
- (D) Recursively Enumerable

**Answer: (B) Context-Free Languages**

**325**Which protocol is used to map an IP address to its corresponding physical MAC address?

- (A) DHCP
- (B) DNS
- (C) ARP
- (D) NAT

**Answer: (C) ARP**

**326**In DBMS, the ACID property 'Durability' refers to:

- (A) All or nothing transaction execution.
- (B) Ensuring committed transaction modifications persist in the event of system failure.
- (C) Transactions executing in isolation from one another.
- (D) Maintaining database consistency before and after transaction.

**Answer: (B) Ensuring committed transaction modifications persist in the event of system failure.**

**327**What is the time complexity to insert an element into an AVL tree of N nodes in the worst case?

- (A)  $O(1)$
- (B)  $O(\log N)$
- (C)  $O(N)$
- (D)  $O(N \log N)$

**Answer: (B)  $O(\log N)$**

**328**Which of the following parser types is a Top-Down parser?

- (A) LR(1) Parser
- (B) LALR Parser
- (C) Recursive Descent Parser
- (D) SLR(1) Parser

**Answer: (C) Recursive Descent Parser**



**329**How many columns are there in a standard Karnaugh Map (K-map) for 4 variables?

- (A) 2
- (B) 4
- (C) 8
- (D) 16

**Answer: (B) 4**

**330**In Unix, which system call is used to create a new child process?

- (A) fork
- (B) exec
- (C) wait
- (D) signal

**Answer: (A) fork**

**331**What is the primary transport protocol used by Domain Name System (DNS) for query resolution?

- (A) TCP
- (B) UDP
- (C) SCTP
- (D) HTTP

**Answer: (B) UDP**

**332**In SQL, what is the default sorting order of the ORDER BY clause?

- (A) ASC
- (B) DESC
- (C) No sorting
- (D) Random

**Answer: (A) ASC**

**333**How many distinct states are present in an n-bit Johnson Counter?

- (A) n
- (B)  $2n$
- (C)  $2^n$
- (D)  $2n - 1$

**Answer: (B)  $2n$**

**334**In standard compiler operator precedence parsing, which grammar symbol possesses the highest precedence among the following options?

- (A) +
- (B) \*
- (C) id
- (D) \$

**Answer: (C) id**

**335**What is the binary representation of the decimal fraction 0.625?

- (A) 0.101
- (B) 0.110
- (C) 0.011
- (D) 0.111

**Answer: (A) 0.101**

**336**Which mechanism is utilized for flow control at the Transport Layer in the TCP protocol suite?

- (A) Stop-and-Wait ARQ
- (B) Go-Back-N ARQ
- (C) Sliding Window Protocol
- (D) Token Bucket Algorithm

**Answer: (C) Sliding Window Protocol**

**337**Consider a page reference string: 1, 2, 3, 1, 4. If the system uses the Least Recently Used (LRU) page replacement policy with 3 physical page frames (initially empty), how many page faults will occur?

- (A) 3 faults
- (B) 4 faults
- (C) 5 faults
- (D) 2 faults

**Answer: (B) 4 faults**

**338**What is the equivalent postfix notation for the infix expression:  $A + B * C - D$ ?

- (A)  $A B C * + D -$
- (B)  $A B C * D + -$
- (C)  $A B + C * D -$
- (D)  $A B C D * + -$

**Answer: (A)  $A B C * + D -$**

**339**Which of the following Unix/Linux commands is primarily used to continuously monitor active system processes and resource utilization in real-time?

- (A) `ps -ef`
- (B) `top`
- (C) `kill`
- (D) `grep`

**Answer: (B) `top`**

**340**In a B+ tree index, if a leaf node contains keys [10, 20] and its sibling pointer P2 points to the next leaf node, which range of keys can be fetched efficiently during a range search?

- (A) Keys less than 10
- (B) Keys between 10 and 20
- (C) Keys greater than 20
- (D) All keys in the database

**Answer: (B) Keys between 10 and 20**

**341**Which SQL clause corresponds directly to the Projection ( $\pi$ ) operator in relational algebra?

- (A) WHERE
- (B) SELECT
- (C) FROM
- (D) JOIN

**Answer: (B) SELECT**

**342**A multiplexer can be effectively used to implement which type of logic circuits?

- (A) Combinational logic only
- (B) Sequential logic only
- (C) Both combinational and sequential logic
- (D) Neither combinational nor sequential logic

**Answer: (C) Both combinational and sequential logic**

**343**Which level of the standard memory hierarchy possesses the longest access time but the highest capacity?

- (A) Cache Memory
- (B) Main Memory
- (C) Secondary Storage
- (D) CPU Registers

**Answer: (C) Secondary Storage**

**344**A full binary tree with N leaf nodes contains how many total nodes?

- (A)  $2N$
- (B)  $2N - 1$
- (C)  $2^N$
- (D)  $N + 1$

**Answer: (B)  $2N - 1$**

**345**Which of the following hashing collision resolution techniques suffers from the phenomenon of primary clustering?

- (A) Linear Probing
- (B) Quadratic Probing
- (C) Double Hashing
- (D) Chaining

**Answer: (A) Linear Probing**

**346** Context-Free Languages (CFL) are NOT closed under which of the following operations?

- (A) Union
- (B) Intersection
- (C) Concatenation
- (D) Kleene Star

**Answer: (B) Intersection**

**347** Which compiler code optimization technique involves replacing a highly expensive operation (like multiplication) with a cheaper one (like addition)?

- (A) Loop Invariant Method
- (B) Strength Reduction
- (C) Common Subexpression Elimination
- (D) Dead Code Elimination

**Answer: (B) Strength Reduction**

**348** During a Direct Memory Access (DMA) transfer, which entity assumes control over the system's address and data buses?

- (A) The CPU
- (B) The DMA Controller
- (C) The Main Memory
- (D) The I/O Device

**Answer: (B) The DMA Controller**

**349** Which of the following routing protocols is classified as a Link-State Routing protocol?

- (A) Routing Information Protocol (RIP)
- (B) Border Gateway Protocol (BGP)
- (C) Open Shortest Path First (OSPF)
- (D) Enhanced Interior Gateway Routing Protocol (EIGRP)

**Answer: (C) Open Shortest Path First (OSPF)**

**350** What is the worst-case time complexity to find the maximum element in a min-heap containing N elements?

- (A)  $O(1)$
- (B)  $O(\log N)$
- (C)  $O(N)$
- (D)  $O(N \log N)$

**Answer: (C)  $O(N)$**

**351** Which SQL transaction isolation level prevents dirty reads but allows non-repeatable reads to occur?

- (A) Read Uncommitted
- (B) Read Committed
- (C) Repeatable Read
- (D) Serializable

**Answer: (B) Read Committed**

**352** Which AVL tree rotation is required to restore height balance when a node becomes unbalanced due to insertion in the left subtree of its left child?

- (A) Left-Left (Single Right) Rotation
- (B) Right-Right (Single Left) Rotation
- (C) Left-Right Rotation
- (D) Right-Left Rotation

**Answer: (A) Left-Left (Single Right) Rotation**

**353** Using Boolean algebraic properties, simplify the expression:  $F = A(A + B) + B(A' + B)$ .

- (A) A
- (B) B
- (C)  $A + B$
- (D) AB

**Answer: (C)  $A + B$**

- 354** What process state transition occurs when the operating system's CPU scheduler preempts a running process to allocate CPU to another thread?
- (A) Running -> Ready
  - (B) Running -> Blocked
  - (C) Ready -> Running
  - (D) Blocked -> Ready
- Answer: (A) Running -> Ready**
- 355** Which of the following statements regarding Reduced Instruction Set Computer (RISC) architectures is FALSE?
- (A) RISC architectures have a large set of addressing modes.
  - (B) Instructions are typically of fixed size.
  - (C) Most instructions execute in a single clock cycle.
  - (D) RISC favors register-to-register load/store designs.
- Answer: (A) RISC architectures have a large set of addressing modes.**
- 356** A memory unit has a capacity of 1K x 8 bits. How many address lines and data lines are required for this memory unit?
- (A) 10 address lines, 8 data lines
  - (B) 8 address lines, 10 data lines
  - (C) 10 address lines, 10 data lines
  - (D) 1024 address lines, 8 data lines
- Answer: (A) 10 address lines, 8 data lines**
- 357** In a Pushdown Automaton, what type of stack operation is represented by the transition label 'a, Z<sub>0</sub> / X Z<sub>0</sub>'?
- (A) Push X onto the stack
  - (B) Pop Z<sub>0</sub> from the stack
  - (C) No stack operation
  - (D) Replace Z<sub>0</sub> with X
- Answer: (A) Push X onto the stack**
- 358** Which abstract machine model corresponds directly to Unrestricted Grammars (Chomsky Type 0)?
- (A) Finite Automata
  - (B) Pushdown Automata
  - (C) Linear Bound Automata
  - (D) Turing Machine
- Answer: (D) Turing Machine**
- 359** Which of the following statements about Turing machines is mathematically TRUE?
- (A) They can recognize any context-sensitive language.
  - (B) They have a finite tape memory.
  - (C) They are equivalent in power to finite automata.
  - (D) They cannot implement loops.
- Answer: (A) They can recognize any context-sensitive language.**
- 360** What is the worst-case time complexity to search for an item in a Hash Table of size N implemented using Chaining for collision resolution?
- (A) O(1)
  - (B) O(log N)
  - (C) O(N)
  - (D) O(1) on average, O(N) worst-case
- Answer: (D) O(1) on average, O(N) worst-case**
- 361** Which bus signal is asserted by the CPU to authorize the DMA controller to assume mastership of the system buses?
- (A) Bus Request (BR)
  - (B) Bus Grant (BG)
  - (C) Interrupt Request (IRQ)
  - (D) Address Enable
- Answer: (B) Bus Grant (BG)**

**362**How many input bits are processed by a standard binary Full Subtractor digital circuit?

- (A) 2 inputs
- (B) 3 inputs
- (C) 4 inputs
- (D) 5 inputs

**Answer: (B) 3 inputs**

**363**A relation R has attributes (A, B, C, D) and functional dependencies {A → B, B → C}. What is the highest normal form of this relation?

- (A) 1NF
- (B) 2NF
- (C) 3NF
- (D) BCNF

**Answer: (A) 1NF**

**364**Which of the following describes the Banker's Algorithm in modern operating systems?

- (A) Deadlock Prevention
- (B) Deadlock Avoidance
- (C) Deadlock Detection
- (D) Mutual Exclusion

**Answer: (B) Deadlock Avoidance**

**365**An IP address of 192.168.1.55 with a subnet mask of 255.255.255.240 belongs to which subnet?

- (A) 192.168.1.32
- (B) 192.168.1.48
- (C) 192.168.1.0
- (D) 192.168.1.64

**Answer: (B) 192.168.1.48**

**366**In a compiler, intermediate code generation is crucial because it:

- (A) Separates front-end analysis from back-end code generation.
- (B) Converts code directly to machine language.
- (C) Performs lexical analysis faster.
- (D) Reduces parsing complexity.

**Answer: (A) Separates front-end analysis from back-end code generation.**

**367**What is the primary difference between a process and a thread?

- (A) Threads share the process memory space, while processes have separate spaces.
- (B) Processes have their own stacks, while threads do not.
- (C) Threads cannot be scheduled by the CPU.
- (D) Processes execute faster than threads.

**Answer: (A) Threads share the process memory space, while processes have separate spaces.**

**368**Which of the following memory allocation algorithms suffers from external fragmentation?

- (A) Paging
- (B) Segmentation
- (C) Fixed Partitioning
- (D) Virtual Address Space

**Answer: (B) Segmentation**

**369**Which of the following data structures is most suitable for implementing a Priority Queue?

- (A) Linked List
- (B) Stack
- (C) Binary Heap
- (D) Queue

**Answer: (C) Binary Heap**

**370**Which of the following is an NP-Complete problem?

- (A) Shortest Path (Dijkstra)
- (B) Travelling Salesperson Problem
- (C) Minimum Spanning Tree (Kruskal)
- (D) Binary Search

**Answer: (B) Travelling Salesperson Problem**

**371** Which of the following database recovery protocols guarantees that no cascading rollbacks will occur?

- (A) Strict Two-Phase Locking (Strict 2PL)
- (B) Basic Two-Phase Locking
- (C) Deferred Update Protocol
- (D) Immediate Update Protocol

**Answer: (A) Strict Two-Phase Locking (Strict 2PL)**

**372** In standard IEEE 754 single-precision floating-point format, how many bits are allocated for the exponent?

- (A) 8 bits
- (B) 11 bits
- (C) 23 bits
- (D) 32 bits

**Answer: (A) 8 bits**

**373** What is the time complexity of the Floyd-Warshall all-pairs shortest path algorithm?

- (A)  $O(V^2)$
- (B)  $O(V^3)$
- (C)  $O(E \log V)$
- (D)  $O(V + E)$

**Answer: (B)  $O(V^3)$**

**374** Which MAC layer protocol is used in wireless networks (802.11) to avoid packet collisions?

- (A) CSMA/CD
- (B) CSMA/CA
- (C) Token Ring
- (D) ALOHA

**Answer: (B) CSMA/CA**

**375** The Halting Problem of Turing Machines is classified as:

- (A) Decidable
- (B) Undecidable but Semidecidable
- (C) Completely Unrecognizable
- (D) P-Space Complete

**Answer: (B) Undecidable but Semidecidable**

**376** Which of the following database normal forms is strictly stronger than 3NF and handles multi-valued dependencies?

- (A) BCNF
- (B) 4NF
- (C) 5NF
- (D) DKNF

**Answer: (B) 4NF**

**377** Which CPU scheduling algorithm can lead to the convoy effect?

- (A) First-Come, First-Served (FCFS)
- (B) Round Robin
- (C) Shortest Job First
- (D) Priority Scheduling

**Answer: (A) First-Come, First-Served (FCFS)**

**378** In a relational schema, what constraint ensures that every value in a foreign key column matches a value in the referenced primary key column?

- (A) Entity Integrity
- (B) Referential Integrity
- (C) Domain Constraint
- (D) Assertive Constraint

**Answer: (B) Referential Integrity**

**379** Which page replacement algorithm exhibits Belady's Anomaly?

- (A) LRU (Least Recently Used)
- (B) Optimal Page Replacement
- (C) FIFO (First-In, First-Out)
- (D) MRU (Most Recently Used)

**Answer: (C) FIFO (First-In, First-Out)**



**380** In the OSI model, which layer is responsible for translating, encrypting, and compressing data?

- (A) Application Layer
- (B) Presentation Layer
- (C) Session Layer
- (D) Transport Layer

**Answer: (B) Presentation Layer**

**381** Which algorithm is used to find the Minimum Spanning Tree of a connected, weighted graph by growing the tree vertex by vertex?

- (A) Kruskal's Algorithm
- (B) Prim's Algorithm
- (C) Dijkstra's Algorithm
- (D) Bellman-Ford Algorithm

**Answer: (B) Prim's Algorithm**

**382** Which of the following registers holds the address of the next instruction to be fetched from memory?

- (A) Instruction Register (IR)
- (B) Program Counter (PC)
- (C) Accumulator
- (D) Memory Data Register (MDR)

**Answer: (B) Program Counter (PC)**

**383** Which of the following concepts is central to understanding: Thread synchronization?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**384** Which of the following concepts is central to understanding: Deadlock detection?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**385** Which of the following concepts is central to understanding: TCP 3-way handshake?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**386** Which of the following concepts is central to understanding: B tree properties?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**387** Which of the following concepts is central to understanding: SQL view security?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**388** Which of the following concepts is central to understanding: IP subnet masking?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**389** Which of the following concepts is central to understanding: DNS MX record?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**390** Which of the following concepts is central to understanding: Dijkstra's complexity?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**391** Which of the following concepts is central to understanding: Greedy algorithms?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**392** Which of the following concepts is central to understanding: Dynamic programming?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**393** Which of the following concepts is central to understanding: RSA public key?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**394** Which of the following concepts is central to understanding: Symmetric encryption?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**395** Which of the following concepts is central to understanding: LL(1) parsing table?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**396** Which of the following concepts is central to understanding: LR(0) state machines?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**397** Which of the following concepts is central to understanding: K-map minimization?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**398** Which of the following concepts is central to understanding: Pipelining hazards?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**399** Which of the following concepts is central to understanding: IEEE 754 format?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**400** Which of the following concepts is central to understanding: Loop unrolling?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**401** Which of the following concepts is central to understanding: Register renaming?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**402** Which of the following concepts is central to understanding: Virtual page tables?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**403** Which of the following concepts is central to understanding: Inode file system?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**404** Which of the following concepts is central to understanding: RAID 5 parity?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**405** Which of the following concepts is central to understanding: Disk seek latency?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**406** Which of the following concepts is central to understanding: Ethernet collision detection?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**407** Which of the following concepts is central to understanding: HTTP GET vs POST?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**408** Which of the following concepts is central to understanding: NAT IP translation?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**409** Which of the following concepts is central to understanding: ARP mac binding?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**410** Which of the following concepts is central to understanding: BGP border routing?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**411** Which of the following concepts is central to understanding: SJF scheduling?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

**412** Which of the following concepts is central to understanding: Producer Consumer semaphore?

- (A) System resource utilization
- (B) Software scalability
- (C) Core hardware registers
- (D) Optimizing data access

**Answer: (A) System resource utilization**

## Processes, CPU Scheduling, and Deadlocks

Operating Systems - 8 questions

**413** Consider the following C program segment utilizing the fork() system call:

```
``c
#include <stdio.h>
#include <unistd.h>

int main() {
 fork();
 if (fork() == 0) {
 fork();
 }
 return 0;
}
```

Including the initial parent process, what is the total number of processes created during the execution of this program?

- (A) 4
- (B) 6
- (C) 8
- (D) 12

**Answer: (B) 6**

**414** An operating system uses the Shortest Job First (SJF) scheduling algorithm with exponential averaging to predict the next CPU burst. The initial predicted burst time is  $t_1 = 12$  milliseconds. The actual burst times for the first two CPU bursts are measured as  $t_1 = 8$  ms and  $t_2 = 14$  ms. If the smoothing factor is  $\alpha = 0.4$ , what is the predicted burst time  $t_3$  (in milliseconds) for the third CPU burst?

- (A) 10.40 ms
- (B) 11.20 ms
- (C) 11.84 ms
- (D) 12.00 ms

**Answer: (C) 11.84 ms**

**415** A counting semaphore  $S$  is initialized to 5. The system subsequently executes 18 P (wait) operations and 10 V (signal) operations. What is the current value of  $S$ , and how many processes are blocked on  $S$  respectively?

- (A) Value = -3, Blocked = 3
- (B) Value = 3, Blocked = 0
- (C) Value = -8, Blocked = 8
- (D) Value = -3, Blocked = 0

**Answer: (A) Value = -3, Blocked = 3**

**416** A system has five processes  $\{P_0, P_1, P_2, P_3, P_4\}$  and three resource types  $\{A, B, C\}$ . Resource type  $A$  has 10 instances,  $B$  has 5 instances, and  $C$  has 7 instances. Suppose the allocation and maximum need of the processes are as follows at time  $T_0$ :

**Allocation:**

-  $P_0$ : [0, 1, 0],  $P_1$ : [2, 0, 0],  $P_2$ : [3, 0, 2],  $P_3$ : [2, 1, 1],  $P_4$ : [0, 0, 2]

**Maximum (Max) Demand:**

-  $P_0$ : [7, 5, 3],  $P_1$ : [3, 2, 2],  $P_2$ : [9, 0, 2],  $P_3$ : [2, 2, 2],  $P_4$ : [4, 3, 3]

**Currently Available resources:** [3, 3, 2]

Which of the following represents a valid, safe execution sequence under the Banker's Algorithm?

- (A)  $\langle P_0, P_1, P_3, P_2, P_4 \rangle$
- (B)  $\langle P_2, P_1, P_3, P_4, P_0 \rangle$
- (C)  $\langle P_1, P_3, P_4, P_0, P_2 \rangle$
- (D)  $\langle P_1, P_0, P_3, P_4, P_2 \rangle$

**Answer: (C)  $\langle P_1, P_3, P_4, P_0, P_2 \rangle$**

**417** Which of the following statements is correct regarding User-Level Threads (ULT) versus Kernel-Level Threads (KLT) as described in modern multithreading architectures?

- (A) If a user-level thread performs a blocking system call or I/O operation, the entire parent process is blocked.
- (B) Kernel-level threads do not require any kernel mode transitions during a context switch, making them faster than user-level threads.
- (C) The OS kernel can directly schedule different user-level threads of the same process onto multiple CPU cores for true hardware parallelism.
- (D) The Many-to-One multithreading model maps each user-level thread to a dedicated kernel thread, facilitating optimal multicore utilization.

**Answer: (A) If a user-level thread performs a blocking system call or I/O operation, the entire parent process is blocked.**

**418** Consider three processes  $P_1$ ,  $P_2$ , and  $P_3$  with arrival times and burst times as follows:

- $P_1$ : Arrival = 0 ms, Burst = 9 ms
- $P_2$ : Arrival = 1 ms, Burst = 4 ms
- $P_3$ : Arrival = 2 ms, Burst = 2 ms

Using the preemptive Shortest Remaining Time First (SRTF) scheduling algorithm, what is the average turnaround time (in milliseconds) of these processes?

- (A) 6.33 ms
- (B) 7.67 ms
- (C) 8.00 ms
- (D) 9.00 ms

**Answer: (B) 7.67 ms**

**419** Imposing a strict total ordering on all resource types and requiring processes to request resources in an increasing order of their numbering is a structural deadlock prevention technique designed specifically to eliminate which Coffman condition?

- (A) Mutual Exclusion
- (B) Hold and Wait
- (C) No Preemption
- (D) Circular Wait

**Answer: (D) Circular Wait**

**420** In Inter-Process Communication (IPC), how do Shared Memory and Message Passing mechanisms compare regarding performance and kernel involvement?

- (A) Shared memory is safer because the kernel mediates and validates every single read and write operation.
- (B) Shared memory is the fastest IPC mechanism because it does not require any kernel involvement once the memory region is mapped to the processes' address spaces.
- (C) Message passing is faster than shared memory because data is copied directly between user-level registers without memory bus traffic.
- (D) Pipes are full-duplex sockets that allow concurrent, bidirectional data transfer without any synchronization overhead.

**Answer: (B) Shared memory is the fastest IPC mechanism because it does not require any kernel involvement once the memory region is mapped to the processes' address spaces.**

## Memory Management, Paging, and Disk Scheduling

Operating Systems - 8 questions

**421** A virtual memory system has a page access time (main memory access time) of  $M = 100$  nanoseconds. In the event of a page fault, it takes 10 milliseconds to service the fault. If the target effective memory access time (EAT) must be no more than 200 nanoseconds, what is the maximum permissible page fault rate  $p$ ?

- (A) 0.001% ( $10^{-5}$ )
- (B) 0.01% ( $10^{-4}$ )
- (C) 0.1% ( $10^{-3}$ )
- (D) 0.0001% ( $10^{-6}$ )

**Answer: (A) 0.001% ( $10^{-5}$ )**

**422** A system implements single-level paging where the Translation Lookaside Buffer (TLB) has a lookup time of  $T_t = 10$  nanoseconds, and the main memory access time is  $t_m = 100$  nanoseconds. The TLB hit ratio is measured as  $h = 0.9$ . According to the two-level EAT with TLB formula, what is the Effective Access Time (EAT) of the memory system in nanoseconds?

- (A) 110 ns
- (B) 120 ns
- (C) 130 ns
- (D) 150 ns

**Answer: (B) 120 ns**

**423** In a Unix-like file system, an inode is structured with 12 direct block pointers, 1 single indirect block pointer, 1 double indirect block pointer, and 1 triple indirect block pointer. If the disk block size is  $B = 1024$  bytes and each disk block pointer size is  $p = 4$  bytes, what is the maximum possible file size that can be indexed by this single inode?

- (A) 12 KB + 256 KB + 64 MB + 16 GB
- (B) 12 KB + 128 KB + 32 MB + 8 GB
- (C) 12 KB + 512 KB + 128 MB + 32 GB
- (D) 12 KB + 256 KB + 128 MB + 64 GB

**Answer: (A) 12 KB + 256 KB + 64 MB + 16 GB**

**424** Which of the following statements correctly distinguishes the hardware-supported memory management schemes Paging and Segmentation?

- (A) Paging divides memory into variable-sized logical segments and suffers from massive external fragmentation.
- (B) Segmentation divides memory into fixed-sized blocks called pages, resulting in internal fragmentation on every allocated block.
- (C) Paging divides logical memory into fixed-sized pages and maps them to physical frames, completely eliminating external fragmentation while suffering only from minor internal fragmentation on the last page.
- (D) Both paging and segmentation operate on variable physical sizes and suffer from identical levels of external fragmentation.

**Answer: (C) Paging divides logical memory into fixed-sized pages and maps them to physical frames, completely eliminating external fragmentation while suffering only from minor internal fragmentation on the last page.**



**425** Consider a reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3. If a process is allocated 3 empty physical frames in RAM, how many page faults will occur under the Least Recently Used (LRU) page replacement algorithm?

- (A) 7
- (B) 8
- (C) 9
- (D) 10

**Answer: (B) 8**

**426** A disk drive has 200 cylinders numbered 0 to 199. The cylinder requests waiting to be serviced are: 98, 183, 37, 122, 14, 124, 65, 67. The disk head is currently positioned at cylinder 53 and is moving towards larger cylinder numbers. What is the total head movement (in cylinders) required to service all requests under the LOOK disk scheduling algorithm?

- (A) 299
- (B) 331
- (C) 382
- (D) 208

**Answer: (A) 299**

**427** In contiguous physical memory allocation, which allocation scheme always searches the entire memory list to locate and allocate the smallest available free block that is large enough, and what is its primary negative side effect?

- (A) First Fit; wastes massive CPU time searching but leaves optimal block structures.
- (B) Best Fit; is slow because it searches the entire list and typically leaves tiny, unusable 'external fragmentation' holes.
- (C) Worst Fit; allocates the largest hole which completely eliminates internal fragmentation but is slow.
- (D) Dynamic Fit; requires expensive background garbage collection and causes high page fault rates.

**Answer: (B) Best Fit; is slow because it searches the entire list and typically leaves tiny, unusable 'external fragmentation' holes.**

**428** Which of the following describes the condition of 'Thrashing' in virtual memory systems, and what is the primary operating system remedy to resolve it?

- (A) The CPU scheduler spends more time context switching than executing threads; solved by increasing the Round Robin time quantum.
- (B) A process spends more time swapping pages in and out of the secondary disk than executing instructions; solved by reducing the degree of multiprogramming.
- (C) The inode allocation table overflows due to a high volume of tiny files; solved by merging file sectors.
- (D) Processes are deadlocked in a circular wait for memory pages; solved by executing the Banker's Algorithm.

**Answer: (B) A process spends more time swapping pages in and out of the secondary disk than executing instructions; solved by reducing the degree of multiprogramming.**

## C Programming and Recursion

Programming and Data Structures - 20 questions

**429** What is the output of the following C program?

```
#include <stdio.h>
int main() {
 int x = 5, y = 9;
 x = (x = x + y) - (y = x - y);
 printf("%d %d ", x, y);
 return 0;
}
```

- (A) 5 14
- (B) 5 9
- (C) 14 5
- (D) 9 5

**Answer: (D) 9 5**

**430**What is the output of the following C program?

```
#include <stdio.h>
int main() {
 int i = 5;
 printf("%d %d %d", i++, i, ++i);
 return 0;
}
```

- (A) 7 6 6
- (B) 6 7 8
- (C) 6 7 7
- (D) 5 6 7

**Answer: (C) 6 7 7**

**431**Consider the following recursive function in C:

```
int function(int x, int y) {
 if (y <= 0) return x;
 return function(y, x % y);
}
```

What does this recursive function compute?

- (A) Greatest Common Divisor (GCD) of x and y
- (B) Least Common Multiple (LCM) of x and y
- (C) x raised to the power of y
- (D) Highest Common Factor (HCF) of x and y

**Answer: (A) Greatest Common Divisor (GCD) of x and y**

**432**Which of the following describes the sequence of translation phases in a standard C compilation pipeline, and what is the typical file extension generated immediately after the compilation phase before code reaches the assembler?

- (A) Preprocessor -> Compiler (.s) -> Assembler (.o) -> Linker (.exe)
- (B) Preprocessor -> Assembler (.o) -> Compiler (.s) -> Linker (.exe)
- (C) Compiler (.s) -> Preprocessor -> Assembler (.o) -> Linker (.exe)
- (D) Preprocessor -> Linker (.exe) -> Compiler (.s) -> Assembler (.o)

**Answer: (A) Preprocessor -> Compiler (.s) -> Assembler (.o) -> Linker (.exe)**

**433**In the standard Application Memory Layout of a running C program, which memory segment stores uninitialized global and static variables, and in which direction do Stack and Heap segments grow respectively?

- (A) BSS segment; Stack grows downwards (higher to lower addresses), Heap grows upwards (lower to higher addresses)
- (B) Data segment; Stack grows upwards, Heap grows downwards
- (C) Text segment; Stack grows downwards, Heap grows upwards
- (D) Stack segment; Both Stack and Heap grow downwards into free memory space

**Answer: (A) BSS segment; Stack grows downwards (higher to lower addresses), Heap grows upwards (lower to higher addresses)**

**434**What is the structural compilation mechanism of a switch-case statement compared to an if-else block in C, and what is the outcome if a developer forgets to include a 'break;' statement at the end of a matching case?

- (A) Switch-case compiles down to a Jump Table (O(1) branching) instead of sequential branches; missing a break causes 'Fallthrough' where the CPU continues executing subsequent instructions blindly.
- (B) Switch-case is purely syntactic sugar for if-else and has identical O(N) performance; missing a break causes a syntax compilation error.
- (C) Switch-case compiles down to a binary tree of branches; missing a break results in a stack overflow.
- (D) Switch-case executes cases in parallel; missing a break makes the default case run first.

**Answer: (A) Switch-case compiles down to a Jump Table (O(1) branching) instead of sequential branches; missing a break causes 'Fallthrough' where the CPU continues executing subsequent instructions blindly.**

**435** Using the 'Spiral Rule' of C declarations, translate the following complex declaration into English:

`int (*P)(char *a[])`

- (A) P is a pointer to a function that accepts an array of character pointers as an argument and returns an integer.
- (B) P is an array of pointers to functions that return char pointer and accept integer values.
- (C) P is a function returning a pointer to an integer, which accepts an array of characters.
- (D) P is a pointer to an array of character pointers, which returns an integer.

**Answer: (A)** P is a pointer to a function that accepts an array of character pointers as an argument and returns an integer.

**436** Assume a 64-bit system where integers take 4 bytes. An integer pointer in C is initialized as: `int *p = 1000`; What is the numerical value of the expression `p + 3`, and what is the outcome of the pointer subtraction `(p + 3) - p`?

- (A) `p + 3` evaluates to 1012; Subtraction evaluates to 3 elements.
- (B) `p + 3` evaluates to 1003; Subtraction evaluates to 3 bytes.
- (C) `p + 3` evaluates to 1012; Subtraction evaluates to 12 bytes.
- (D) `p + 3` evaluates to 1003; Subtraction evaluates to 12 elements.

**Answer: (A)** `p + 3` evaluates to 1012; Subtraction evaluates to 3 elements.

**437** In C, an array name decays to a pointer to its first element. For a 1D array defined as `int arr[10]`; which of the following is syntactically INVALID in C, and why?

- (A) `arr++`
- (B) `*(arr + 5)`
- (C) `5[arr]`
- (D) `&arr[0]`

**Answer: (A)** `arr++`

**438** Consider a 2D array declared as: `int A[3][4]`; stored in row-major order. Assume a 32-bit compilation where pointer sizes are 8 bytes and integers are 4 bytes. Which of the following is correct regarding the structural offsets and scale factors of A, \*A, and &A when we increment them by 1?

- (A) A + 1 skips 16 bytes (1 row); \*A + 1 skips 4 bytes (1 int); &A + 1 skips 48 bytes (entire 2D array block).
- (B) A + 1 skips 4 bytes; \*A + 1 skips 16 bytes; &A + 1 skips 48 bytes.
- (C) A + 1 skips 16 bytes; \*A + 1 skips 16 bytes; &A + 1 skips 16 bytes.
- (D) A + 1 skips 48 bytes; \*A + 1 skips 4 bytes; &A + 1 skips 8 bytes.

**Answer: (A)** A + 1 skips 16 bytes (1 row); \*A + 1 skips 4 bytes (1 int); &A + 1 skips 48 bytes (entire 2D array block).

**439** Assume a 64-bit operating system. A developer declares two variables:

I. `int (*ptr)[3];`

II. `int *arr[3];`

What are the memory sizes allocated to 'ptr' and 'arr' respectively?

- (A) ptr: 8 bytes; arr: 24 bytes
- (B) ptr: 12 bytes; arr: 24 bytes
- (C) ptr: 12 bytes; arr: 12 bytes
- (D) ptr: 24 bytes; arr: 8 bytes

**Answer: (A)** ptr: 8 bytes; arr: 24 bytes

**440** Which of the following is TRUE regarding 'Void Pointers' and 'Function Pointers' in C programming?

- (A) A void pointer cannot be directly dereferenced without an explicit cast first; a function pointer stores the address of executable code located in the Text segment.
- (B) A void pointer automatically casts to any size on dereference; function pointers are stored on the program's Heap.
- (C) A void pointer is 0 bytes in size; a function pointer can be incremented to access local variables in stack frames.
- (D) Function pointers can only point to functions with void return types; void pointers cannot point to structures.

**Answer: (A)** A void pointer cannot be directly dereferenced without an explicit cast first; a function pointer stores the address of executable code located in the Text segment.

**441** Consider a 32-bit system (where packing/padding is ignored). A structure and union are declared as:

```
struct MyStruct { char a; int b; float c; };
```

```
union MyUnion { char a; int b; float c; };
```

What are the memory sizes of MyStruct and MyUnion respectively?

- (A) MyStruct: 9 bytes; MyUnion: 4 bytes
- (B) MyStruct: 12 bytes; MyUnion: 4 bytes
- (C) MyStruct: 9 bytes; MyUnion: 9 bytes

(D) MyStruct: 12 bytes; MyUnion: 12 bytes

**Answer: (A)** MyStruct: 9 bytes; MyUnion: 4 bytes

**442** A C program performs dynamic memory operations. What is the fundamental difference between 'malloc(size)' and 'calloc(n, size)', and what happens if a programmer fails to execute 'free(ptr)' on dynamically allocated memory?

(A) malloc allocates uninitialized memory containing garbage; calloc allocates and initializes all bits to zero; failing to free causes a 'Memory Leak'.

(B) malloc initializes memory to zero; calloc allocates uninitialized garbage; failing to free causes a dangling pointer.

(C) malloc allocates on the Stack; calloc allocates on the Heap; failing to free results in a stack overflow.

(D) There is no difference, both are identical; failing to free causes an automatic compiler warning and crash.

**Answer: (A)** malloc allocates uninitialized memory containing garbage; calloc allocates and initializes all bits to zero; failing to free causes a 'Memory Leak'.

**443** A developer writes a file handling system in C using 'FILE \*fp;'. Which of the following is correct regarding file modes and error handling?

(A) Modes 'rb' and 'wb' are used to read and write binary files; if fopen() fails to locate or open a file, it returns NULL, which must be verified to prevent crashes.

(B) Mode 'w' appends contents without clearing the file; if a file does not exist, 'r' mode automatically creates a new empty file.

(C) The EOF marker is always a positive integer ( 1); checking error is done via malloc\_error().

(D) File operations are managed entirely on the Stack, so fopen() never returns a pointer.

**Answer: (A)** Modes 'rb' and 'wb' are used to read and write binary files; if fopen() fails to locate or open a file, it returns NULL, which must be verified to prevent crashes.

**444** In C programming, how do 'static' and 'extern' storage classes differ with respect to scope, lifetime, and initial memory segments?

(A) A static local variable has local scope but retains its value across function calls (stored in BSS/Data); an extern variable is used for global variables with external linkage across different files.

(B) A static variable is allocated on the local Stack frame; an extern variable is allocated dynamically on the Heap.

(C) A static variable is visible globally across all compiled files; an extern variable is private to its declaring block.

(D) Both static and extern variables have identical block scopes and are destroyed when their declaring block terminates.

**Answer: (A)** A static local variable has local scope but retains its value across function calls (stored in BSS/Data); an extern variable is used for global variables with external linkage across different files.

**445** Consider the following macro definition in C:

```
#define SQUARE(x) x * x
```

What is the value of the expression SQUARE(3 + 2) in C, and what is the best preprocessor design practice to avoid this error?

(A) 11; Best practice is to parenthesize all arguments and the entire macro: #define SQUARE(x) ((x) \* (x))

(B) 25; Best practice is to avoid macros and always use recursive templates.

(C) 13; Best practice is to cast the macro parameters to float.

(D) 15; Best practice is to write macros in lowercase.

**Answer: (A)** 11; Best practice is to parenthesize all arguments and the entire macro: #define SQUARE(x) ((x) \* (x))

**446** What is 'Tail Recursion', and why is it considered highly advantageous in terms of system call stack space optimizations during compiler execution?

(A) It is recursion where the recursive call is the absolute last operation; the compiler optimizes it into a simple goto loop, reducing call stack space from O(N) to O(1).

(B) It is recursion where the base case is placed at the end of the file; it speeds up disk execution.

(C) It is recursion that traverses the tail of a linked list; it reduces pointer dereferencing costs.

(D) It is recursion where the return value is multiplied by a global static factor.

**Answer: (A)** It is recursion where the recursive call is the absolute last operation; the compiler optimizes it into a simple goto loop, reducing call stack space from O(N) to O(1).

**447** What is the output of the following recursive C program?

```
#include <stdio.h>
void count(int n) {
 if (n <= 0) return;
 printf("%d ", n);
 count(n - 1);
}
```

```
printf("%d ", n);
}
int main() {
count(3);
return 0;
}
```

- (A) 3 2 1 1 2 3
- (B) 3 2 1 3 2 1
- (C) 1 2 3 3 2 1
- (D) 3 2 1 0 1 2 3

**Answer: (A) 3 2 1 1 2 3**

**448** A developer compiles a C program using the register storage class: register int x = 10; Which of the following is correct regarding this variable?

- (A) The address-of operator '&' cannot be applied to 'x' because register variables may reside in CPU registers rather than RAM.
- (B) The compiler is forced to store x in CPU registers, failing compilation if no registers are free.
- (C) The variable x is placed in the read-only Text segment.
- (D) The lifetime of x is global and persists across the entire program execution.

**Answer: (A)** The address-of operator '&' cannot be applied to 'x' because register variables may reside in CPU registers rather than RAM.

## Linear Data Structures

Programming and Data Structures - 20 questions

**449** Which of the following correctly classifies the given data structures as either Linear or Non-Linear?

- (A) Linear: Arrays, Stacks, Linked Lists; Non-Linear: Trees, Graphs
- (B) Linear: Arrays, Trees, Queues; Non-Linear: Stacks, Graphs
- (C) Linear: Stacks, Graphs, Heaps; Non-Linear: Arrays, Linked Lists
- (D) Linear: Linked Lists, Queues, Graphs; Non-Linear: Stacks, Trees

**Answer: (A)** Linear: Arrays, Stacks, Linked Lists; Non-Linear: Trees, Graphs

**450** A 1-Dimensional array is declared with lower bound index -1024 and upper bound index 1024. If each element requires 4 bytes of storage, how many elements are present in the array, and what is its total memory footprint?

- (A) 2049 elements; 8196 bytes
- (B) 2048 elements; 8192 bytes
- (C) 2049 elements; 4096 bytes
- (D) 1025 elements; 4100 bytes

**Answer: (A)** 2049 elements; 8196 bytes

**451** A 2D array A[10][20] is stored in Row-Major order in memory, starting at base address 1000. If each element occupies 4 bytes, what is the memory address of element A[5][15]?

- (A) 1460
- (B) 1440
- (C) 1380
- (D) 1540

**Answer: (A)** 1460

**452** A 2D array A[10][20] is stored in Column-Major order in memory, starting at base address 1000. If each element occupies 4 bytes, what is the memory address of element A[5][15]?

- (A) 1620
- (B) 1460
- (C) 1600
- (D) 1540

**Answer: (A)** 1620

**453** Consider an empty stack implemented using an array of size 100. We perform the following sequence of operations:

Push(A), Push(B), Pop(), Push(C), Push(D), Pop(), Push(E), Pop(), Pop()

Which elements remain in the stack, and which element is at the Top of the stack?

- (A) Remaining: {A}; Top is A
- (B) Remaining: {A, B, C}; Top is C
- (C) Remaining: {A, C}; Top is C

(D) The stack is empty

**Answer: (A)** Remaining: {A}; Top is A

**454** In an array implementation of a stack of capacity MAX\_SIZE, with 'top' initialized to -1, what are the exact boundary check conditions for 'Stack Overflow' during PUSH and 'Stack Underflow' during POP respectively?

- (A) Overflow:  $\text{top} == \text{MAX\_SIZE} - 1$ ; Underflow:  $\text{top} == -1$
- (B) Overflow:  $\text{top} == \text{MAX\_SIZE}$ ; Underflow:  $\text{top} == 0$
- (C) Overflow:  $\text{top} == -1$ ; Underflow:  $\text{top} == \text{MAX\_SIZE} - 1$
- (D) Overflow:  $\text{top} == \text{MAX\_SIZE} + 1$ ; Underflow:  $\text{top} == -1$

**Answer: (A)** Overflow:  $\text{top} == \text{MAX\_SIZE} - 1$ ; Underflow:  $\text{top} == -1$

**455** What is the correct Postfix representation of the following Infix expression, assuming standard operator precedence and left-to-right associativity?

$A + B * C - D / E$

- (A)  $A B C * + D E / -$
- (B)  $A B C + * D E / -$
- (C)  $A B + C * D E / -$
- (D)  $A B C * + D / E -$

**Answer: (A)**  $A B C * + D E / -$

**456** A postfix expression '5 3 + 8 2 / \*' is evaluated using a stack. What is the final value popped from the stack upon complete evaluation?

- (A) 32
- (B) 16
- (C) 8
- (D) 64

**Answer: (A)** 32

**457** Two stacks are implemented in a single 1D array of size N (indices 0 to N-1) growing from opposite ends. Stack 1 starts at index -1 and grows upwards, while Stack 2 starts at index N and grows downwards. What is the exact mathematical condition indicating that both stacks have run out of space (Double Stack Overflow)?

- (A)  $\text{top1} + 1 == \text{top2}$
- (B)  $\text{top1} == \text{top2}$
- (C)  $\text{top1} + \text{top2} == N$
- (D)  $\text{top1} == N / 2$

**Answer: (A)**  $\text{top1} + 1 == \text{top2}$

**458** A Circular Queue of maximum capacity MAX\_SIZE is implemented in an array. If FRONT and REAR represent the indices of the front and rear elements respectively, which of the following represents the correct condition for a 'Queue Full' state, and why is a circular queue preferred over a linear array queue?

- (A)  $(\text{REAR} + 1) \% \text{MAX\_SIZE} == \text{FRONT}$ ; Circular queues solve the 'False Overflow' problem by reusing empty slots created by deletions.
- (B)  $\text{REAR} == \text{MAX\_SIZE} - 1$ ; Circular queues use less memory by utilizing bits.
- (C)  $(\text{FRONT} + 1) \% \text{MAX\_SIZE} == \text{REAR}$ ; Circular queues allow bidirectional deletion.
- (D)  $\text{REAR} == \text{FRONT}$ ; Circular queues execute enqueue operations in  $O(\log N)$  time.

**Answer: (A)**  $(\text{REAR} + 1) \% \text{MAX\_SIZE} == \text{FRONT}$ ; Circular queues solve the 'False Overflow' problem by reusing empty slots created by deletions.

**459** A Double-Ended Queue (Deque) supports insertions and deletions at both ends. Which classification of Deque permits insertions at both ends but restricts deletions to only one end?

- (A) Input-Restricted Deque
- (B) Output-Restricted Deque
- (C) Circular Deque
- (D) Priority Deque

**Answer: (B)** Output-Restricted Deque



**460**In a Singly Linked List (SLL), we maintain references to 'head' and 'tail'. Which of the following operations has a time complexity of  $O(1)$  in the worst case, and which operation is strictly  $O(N)$ ?

- (A) Insertion at tail is  $O(1)$ ; Deletion at tail is  $O(N)$
- (B) Insertion at tail is  $O(N)$ ; Deletion at tail is  $O(1)$
- (C) Insertion at head is  $O(N)$ ; Deletion at head is  $O(1)$
- (D) Insertion at tail is  $O(1)$ ; Deletion at tail is  $O(1)$

**Answer: (A)** Insertion at tail is  $O(1)$ ; Deletion at tail is  $O(N)$

**461**Which variant of Linked List is preferred by Operating System schedulers to implement CPU Round-Robin Scheduling where tasks are allocated time slices sequentially in a continuous cycle, and why?

- (A) Circular Linked List; It allows continuous loop traversal without encountering a NULL pointer, enabling seamless context-switching.
- (B) Singly Linked List; It has the lowest memory overhead.
- (C) Skip List; It allows  $O(\log N)$  express-lane context-switching.
- (D) Header Linked List; It stores scheduler metadata directly in the head node.

**Answer: (A)** Circular Linked List; It allows continuous loop traversal without encountering a NULL pointer, enabling seamless context-switching.

**462**In a Doubly Linked List (DLL), we wish to delete a node pointed to by 'temp' (which is neither the head nor the tail node) in  $O(1)$  time. What are the correct pointer assignments required to securely isolate and delete this node?

- (A) `temp->prev->next = temp->next; temp->next->prev = temp->prev; free(temp);`
- (B) `temp->next = temp->prev; temp->prev = temp->next; free(temp);`
- (C) `temp->prev->next = temp; temp->next->prev = temp; free(temp);`
- (D) `free(temp); temp->prev->next = temp->next; temp->next->prev = temp->prev;`

**Answer: (A)** `temp->prev->next = temp->next; temp->next->prev = temp->prev; free(temp);`

**463**What is a 'Skip List', what is its average search time complexity, and what mathematical structure does it emulate to bypass linear searches in linked lists?

- (A) A multi-layered linked list with 'express lane' pointers; average search complexity is  $O(\log N)$ ; emulates Binary Search.
- (B) A linked list that skips odd-indexed nodes; average search complexity is  $O(N)$ ; emulates Linear Scan.
- (C) A dynamic array that skips empty blocks; average search complexity is  $O(N \log N)$ ; emulates Interpolation Search.
- (D) A binary tree mapped into a list; average search complexity is  $O(1)$ ; emulates Hashing.

**Answer: (A)** A multi-layered linked list with 'express lane' pointers; average search complexity is  $O(\log N)$ ; emulates Binary Search.

**464**What is the primary architectural purpose of a 'Header Linked List' containing a dummy header node at the beginning, and how does it optimize linked list operations?

- (A) It stores metadata (such as total node count) and guarantees that the list is never physically empty, eliminating edge-case checks for NULL pointers in insertion and deletion algorithms.
- (B) It speeds up searches by indexing nodes using a hash function.
- (C) It compresses list data to reduce memory usage.
- (D) It converts singly linked lists into circular ones automatically.

**Answer: (A)** It stores metadata (such as total node count) and guarantees that the list is never physically empty, eliminating edge-case checks for NULL pointers in insertion and deletion algorithms.

**465**Which of the following describes the correct steps to iteratively reverse a Singly Linked List using three pointer variables: prev, curr, and next?

- (A) Initialize `prev = NULL, curr = head`; inside loop: `next = curr->next; curr->next = prev; prev = curr; curr = next`; finally update `head = prev`;
- (B) Initialize `prev = head, curr = head->next`; inside loop: `curr->next = prev; prev = curr; curr = curr->next`;
- (C) Initialize `prev = NULL, curr = head`; inside loop: `curr->next = next; next = curr; curr = prev`;
- (D) Initialize `prev = head, curr = NULL`; inside loop: `next = curr->next; prev->next = curr; curr = next`;

**Answer: (A)** Initialize `prev = NULL, curr = head`; inside loop: `next = curr->next; curr->next = prev; prev = curr; curr = next`; finally update `head = prev`;

**466**A system implements a Queue using two Stacks (S1 and S2). Stack S1 is used for enqueueing, and S2 is used for dequeueing. If we wish to dequeue an element when S2 is completely empty, what is the sequence of operations required, and what is its amortized time complexity?

- (A) Pop all elements from S1 and push them into S2, then pop from S2; amortized complexity is  $O(1)$ .
- (B) Pop 1 element from S1 and push it to S2; complexity is  $O(N)$ .
- (C) Clear S1 completely and return its bottom; complexity is  $O(1)$ .
- (D) Throw an underflow error immediately; complexity is  $O(1)$ .

**Answer: (A)** Pop all elements from S1 and push them into S2, then pop from S2; amortized complexity is  $O(1)$ .

**467**In terms of memory efficiency and caching behaviors, which of the following represents a major disadvantage of Linked Lists compared to contiguous Arrays?

- (A) Linked lists store elements non-contiguously in memory, leading to high cache-miss rates, and incur pointer storage overhead (sizeof(pointer) per node).
- (B) Linked lists require dynamic sizing which uses double the memory on stack frames.
- (C) Linked lists are slow because they are always stored in read-only Text segments.
- (D) Linked lists cannot store structures or floating-point elements.

**Answer: (A)** Linked lists store elements non-contiguously in memory, leading to high cache-miss rates, and incur pointer storage overhead (sizeof(pointer) per node).

**468**In a Josephus Problem simulation, N people stand in a circle and every M-th person is eliminated recursively until only one remains. Which data structure is mathematically optimal to implement this simulation with minimum boundary resets?

- (A) Circular Linked List
- (B) Doubly Linked List
- (C) Singly Linked List
- (D) Static 1D Array

**Answer: (A)** Circular Linked List

## Non-Linear Data Structures

Programming and Data Structures - 20 questions

**469**In any strictly binary tree (where every non-leaf node has exactly two child nodes), what is the relationship between the number of leaf nodes (L) and the number of internal nodes (I)?

- (A)  $L = I + 1$
- (B)  $L = 2 * I$
- (C)  $L = I - 1$
- (D)  $L = I / 2$

**Answer: (A)**  $L = I + 1$

**470**What is the maximum number of nodes in a perfect binary tree of height h (where height of root node is 0)? If a perfect binary tree has a height of 4, what is the total number of nodes present?

- (A) 31 nodes; formula is  $2^{(h+1)} - 1$
- (B) 15 nodes; formula is  $2^h - 1$
- (C) 32 nodes; formula is  $2^{(h+1)}$
- (D) 63 nodes; formula is  $2^{(h+2)} - 1$

**Answer: (A)** 31 nodes; formula is  $2^{(h+1)} - 1$

**471**What is the minimum height (h) of a binary tree containing exactly N nodes, and what is the minimum height possible for a tree with N = 31 nodes (assuming height of root is 0)?

- (A) 4; formula is  $\text{ceil}(\log_2(N + 1)) - 1$
- (B) 5; formula is  $\text{floor}(\log_2(N))$
- (C) 4; formula is  $\text{ceil}(\log_2(N)) - 1$
- (D) 3; formula is  $\text{floor}(\log_2(N + 1))$

**Answer: (A)** 4; formula is  $\text{ceil}(\log_2(N + 1)) - 1$

**472**Consider the following Inorder and Preorder traversal of a binary tree:

Inorder: D B E F A G H C

Preorder: A B D E F C G H

What is the Postorder Traversal of the same tree?

- (A) D F E B H G C A
- (B) D F E B H C G A
- (C) D F E B G H C A
- (D) D F E H B G C A

**Answer: (A)** D F E B H G C A

**473**Suppose the following numbers are inserted in sequence into an empty Binary Search Tree (BST):

50, 70, 30, 40, 20, 80, 60, 90, 85

How many nodes will be present in the Left Sub-Tree (LST) and the Right Sub-Tree (RST) of the root node respectively?

- (A) LST - 3, RST - 5
- (B) LST - 4, RST - 4
- (C) LST - 3, RST - 6

(D) LST - 5, RST - 3

**Answer: (A) LST - 3, RST - 5**

**474** What is the balance factor of a node in an AVL Tree, and what is the range of permissible balance factors for any node in a valid AVL Tree to remain balanced?

- (A) Height(Left Subtree) - Height(Right Subtree); Allowed range is  $\{-1, 0, 1\}$
- (B) Height(Left Subtree) + Height(Right Subtree); Allowed range is  $\{0, 1, 2\}$
- (C) Height(Right Subtree) - Height(Left Subtree); Allowed range is  $\{-2, 0, 2\}$
- (D) Nodes(Left Subtree) - Nodes(Right Subtree); Allowed range is  $\{-1, 0, 1\}$

**Answer: (A) Height(Left Subtree) - Height(Right Subtree); Allowed range is  $\{-1, 0, 1\}$**

**475** In an AVL tree, we insert a node that causes a balance violation of +2 at a node 'A'. If the insertion was made in the Right Sub-tree of A's Left Child, which sequence of rotations must be executed to restore AVL balance?

- (A) Left-Right (LR) double rotation: Left rotation on A's left child followed by a Right rotation on A
- (B) Right-Left (RL) double rotation: Right rotation on A's right child followed by a Left rotation on A
- (C) Single Right (RR) rotation on A
- (D) Single Left (LL) rotation on A

**Answer: (A) Left-Right (LR) double rotation: Left rotation on A's left child followed by a Right rotation on A**

**476** Which self-balancing binary search tree is preferred to implement the standard map container in C++ (std::map) due to having a higher tolerance for minor height imbalances (fewer rotations on insertion/deletion) than AVL trees, and what is its path length constraint?

- (A) Red-Black Tree; The longest path from root to a leaf is at most twice the length of the shortest path.
- (B) AVL Tree; Every path from root to leaf must have identical heights.
- (C) Threaded Binary Tree; All paths are bound to  $O(1)$  space.
- (D) Extended Binary Tree; The average path length is strictly  $\log_2(N)$ .

**Answer: (A) Red-Black Tree; The longest path from root to a leaf is at most twice the length of the shortest path.**

**477** An Extended Binary Tree (or 2-Tree) is created by replacing every NULL pointer in a binary tree of N internal nodes with dummy external nodes. How many external nodes are present in this Extended Binary Tree?

- (A)  $N + 1$
- (B)  $N - 1$
- (C)  $2 * N$
- (D)  $2^{(N - 1)}$

**Answer: (A)  $N + 1$**

**478** What is the primary objective of a 'Threaded Binary Tree', and how does it optimize traversals in resource-constrained hardware systems?

- (A) It reuses NULL leaf pointers to store direct links to Inorder Predecessors and Successors, enabling stack-free and recursion-free inorder traversals in  $O(1)$  auxiliary space.
- (B) It links all leaf nodes together to form a Circular Queue.
- (C) It compresses tree nodes into thread-level CPU registers.
- (D) It converts a binary tree into a binary heap by swapping parent keys.

**Answer: (A) It reuses NULL leaf pointers to store direct links to Inorder Predecessors and Successors, enabling stack-free and recursion-free inorder traversals in  $O(1)$  auxiliary space.**

**479** A Binary Heap is mapped into a contiguous 1D array, 0-indexed. If a parent node is located at array index 'i', what are the mathematical formulas used to locate its Left Child, Right Child, and Parent nodes respectively?

- (A) Left:  $2*i + 1$ ; Right:  $2*i + 2$ ; Parent:  $\text{floor}((i - 1) / 2)$
- (B) Left:  $2*i$ ; Right:  $2*i + 1$ ; Parent:  $\text{floor}(i / 2)$
- (C) Left:  $2*i + 2$ ; Right:  $2*i + 3$ ; Parent:  $\text{floor}((i - 2) / 2)$
- (D) Left:  $2*i - 1$ ; Right:  $2*i$ ; Parent:  $\text{floor}(i / 2) + 1$

**Answer: (A) Left:  $2*i + 1$ ; Right:  $2*i + 2$ ; Parent:  $\text{floor}((i - 1) / 2)$**

**480** We perform insertion of an element into a Max Heap of size N. What is the worst-case time complexity of this operation, and what is the underlying heapify bubble-up mechanism?

- (A)  $O(\log N)$ ; The new element is appended at the end of the array and bubbled-up by comparing and swapping with its parent recursively until heap property is restored.
- (B)  $O(1)$ ; The new element is placed at the root and child nodes are shifted down.
- (C)  $O(N)$ ; The array must be fully re-sorted using bubble sort.
- (D)  $O(N \log N)$ ; The heap is completely rebuilt from scratch.

**Answer: (A)  $O(\log N)$ ; The new element is appended at the end of the array and bubbled-up by comparing and swapping with its parent recursively until heap property is restored.**

**481** Which of the following traversals of a Binary Search Tree (BST) is guaranteed to always produce the keys in strictly sorted ascending order?

- (A) Inorder Traversal
- (B) Preorder Traversal
- (C) Postorder Traversal
- (D) Level Order Traversal

**Answer: (A) Inorder Traversal**

**482** Which of the following represents a fundamental structural constraint of a 'Spanning Tree' created from a connected, undirected graph  $G = (V, E)$  of  $V$  vertices and  $E$  edges?

- (A) It must contain all  $V$  vertices of  $G$ , contain exactly  $V - 1$  edges, contain no cycles, and remain fully connected.
- (B) It must contain  $V - 1$  vertices and  $V$  edges, forming a single closed cycle.
- (C) It must be a complete graph where every vertex has degree  $V - 1$ .
- (D) It must have a height of exactly  $\log_2(V)$ .

**Answer: (A) It must contain all  $V$  vertices of  $G$ , contain exactly  $V - 1$  edges, contain no cycles, and remain fully connected.**

**483** A connected undirected graph  $G$  has 12 vertices and 25 edges. How many edges must be deleted from  $G$  to obtain a spanning tree of  $G$ ?

- (A) 14 edges
- (B) 13 edges
- (C) 11 edges
- (D) 12 edges

**Answer: (A) 14 edges**

**484** Using Cayley's Formula, what is the total number of distinct spanning trees that can be constructed from a labeled complete graph of 5 vertices ( $K_5$ )?

- (A) 125
- (B) 25
- (C) 10
- (D) 64

**Answer: (A) 125**

**485** Which graph traversal algorithm uses a Queue (FIFO) to explore nodes level-by-level, and is mathematically guaranteed to find the shortest path (in terms of minimum number of edges) from a starting vertex to any other vertex in an unweighted graph?

- (A) Breadth-First Search (BFS)
- (B) Depth-First Search (DFS)
- (C) Dijkstra's Algorithm
- (D) Kruskal's Algorithm

**Answer: (A) Breadth-First Search (BFS)**

**486** A developer executes Depth-First Search (DFS) on a directed graph. During traversal, an edge is encountered that points from a vertex to one of its ancestors in the DFS activation tree. What is this type of edge called, and what structural property of the graph does its presence reveal?

- (A) Back Edge; Indicates the presence of a cycle in the directed graph.
- (B) Forward Edge; Indicates that the graph is a Directed Acyclic Graph (DAG).
- (C) Cross Edge; Indicates that the graph is fully connected.
- (D) Tree Edge; Indicates that the graph has a spanning tree.

**Answer: (A) Back Edge; Indicates the presence of a cycle in the directed graph.**

**487** A complete undirected graph with  $N$  vertices has how many total edges, and what is the degree of each vertex? If a complete graph has 8 vertices, how many edges are present?

- (A) 28 edges; degree is 7
- (B) 56 edges; degree is 8
- (C) 28 edges; degree is 8
- (D) 16 edges; degree is 7

**Answer: (A) 28 edges; degree is 7**

**488**In the build\_heap algorithm (Floyd's Heap Construction), an unordered array of size N is transformed into a Max Heap by executing max-heapify bottom-up on all non-leaf nodes. What is the worst-case time complexity of this build\_heap operation, and why is it more optimal than N insertions?

- (A)  $O(N)$  linear time; It is more optimal because the height of the subtrees decreases exponentially as we go down, and the majority of nodes are leaves requiring 0 swaps.
- (B)  $O(N \log N)$  time; It is identical to N sequential insertions.
- (C)  $O(\log N)$  logarithmic time; It only examines the root node.
- (D)  $O(N^2)$  quadratic time; It performs sequential bubble swaps across all levels.

**Answer: (A)**  $O(N)$  linear time; It is more optimal because the height of the subtrees decreases exponentially as we go down, and the majority of nodes are leaves requiring 0 swaps.

## SDLC, Requirements, and Design

Software Engineering - 5 questions

**489**Which software development life cycle (SDLC) model is specifically characterized by an explicit focus on risk analysis and management at each iteration?

- (A) Waterfall Model
- (B) Spiral Model
- (C) Iterative Waterfall Model
- (D) Rapid Application Development (RAD)

**Answer: (B)** Spiral Model

**490**Which of the following is classified as a NON-FUNCTIONAL requirement in a Software Requirements Specification (SRS) document?

- (A) The system must allow users to reset their passwords via email.
- (B) The system must generate weekly sales report PDFs for administrators.
- (C) The system must process all online transactions within 2 seconds.
- (D) The system must authenticate users using OAuth 2.0.

**Answer: (C)** The system must process all online transactions within 2 seconds.

**491**For a robust, maintainable, and modular software design, which of the following characteristics should be prioritized?

- (A) High coupling and low cohesion
- (B) Low coupling and high cohesion
- (C) High coupling and high cohesion
- (D) Low coupling and low cohesion

**Answer: (B)** Low coupling and high cohesion

**492**Which of the following types of coupling is the LEAST desirable (i.e., worst) in software design, indicating the tightest interdependence?

- (A) Data Coupling
- (B) Stamp Coupling
- (C) Control Coupling
- (D) Content Coupling

**Answer: (D)** Content Coupling

**493**Which type of cohesion is exhibited when a module performs several tasks that must occur in a specific chronological sequence, where the output of one task serves as the input to the next task?

- (A) Temporal Cohesion
- (B) Procedural Cohesion
- (C) Sequential Cohesion
- (D) Functional Cohesion

**Answer: (C)** Sequential Cohesion

## Metrics, Testing, and SCM

Software Engineering - 5 questions

**494**A control flow graph (CFG) of a program module has 14 edges and 10 nodes. What is McCabe's Cyclomatic Complexity,  $V(G)$ , of this program, assuming it has a single connected component?

- (A) 4
- (B) 6
- (C) 8
- (D) 12

**Answer: (B)** 6

**495**In the Basic COCOMO model, a software project with a small team, stable requirements, and familiar, simple programming environments is classified as:

- (A) Organic
- (B) Semi-Detached
- (C) Embedded
- (D) Iterative

**Answer: (A) Organic**

**496**Which of the following testing methodologies is classified as a WHITE-BOX testing technique?

- (A) Equivalence Partitioning
- (B) Boundary Value Analysis
- (C) Decision Coverage Testing
- (D) State Transition Testing

**Answer: (C) Decision Coverage Testing**

**497**What is the relationship between statement coverage and branch/decision coverage in white-box testing?

- (A) 100% statement coverage guarantees 100% branch coverage.
- (B) 100% branch coverage guarantees 100% statement coverage.
- (C) Statement coverage and branch coverage are completely independent and have no relationship.
- (D) Both statement and branch coverage are mathematically equivalent to cyclomatic complexity.

**Answer: (B) 100% branch coverage guarantees 100% statement coverage.**

**498**In the Capability Maturity Model Integration (CMMI), at which maturity level is process performance managed quantitatively, using statistical and quantitative techniques?

- (A) Level 2 (Managed)
- (B) Level 3 (Defined)
- (C) Level 4 (Quantitatively Managed)
- (D) Level 5 (Optimizing)

**Answer: (C) Level 4 (Quantitatively Managed)**

## Finite Automata and Regular Languages

Theory of Computation - 5 questions

**499**What is the minimum number of states in a Deterministic Finite Automaton (DFA) that accepts all binary strings containing the substring '110'?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

**Answer: (C) 4**

**500**Which of the following classes of languages is NOT closed under the complementation operation?

- (A) Regular Languages
- (B) Context-Free Languages (CFL)
- (C) Recursive Languages
- (D) Context-Sensitive Languages

**Answer: (B) Context-Free Languages (CFL)**

**501**The language  $L = \{a^n b^n \mid n \geq 0\}$  is which type of language?

- (A) Regular Language
- (B) Context-Free Language (CFL)
- (C) Context-Sensitive Language
- (D) Unrestricted Language

**Answer: (B) Context-Free Language (CFL)**

**502**Which of the following regular expressions represents the set of all binary strings that end with '01'?

- (A)  $(0+1)^*01$
- (B)  $(01)^*(0+1)^*$
- (C)  $(0+1)^*01^*$
- (D)  $0^*1^*01$

**Answer: (A)  $(0+1)^*01$**



**503**The Pumping Lemma for Regular Languages is primarily used to:

- (A) Prove that a language is regular
- (B) Prove that a language is non-regular
- (C) Minimize the number of states in a DFA
- (D) Convert an NFA into an equivalent DFA

**Answer: (B)** Prove that a language is non-regular

## CFGs, Pushdown Automata, and Undecidability

Theory of Computation - 5 questions

**504**Which of the following problems is DECIDABLE?

- (A) Whether a given Turing Machine halts on all inputs
- (B) Whether a given Context-Free Grammar is ambiguous
- (C) Whether two given regular expressions represent the same language
- (D) Whether a given Turing Machine accepts any string at all

**Answer: (C)** Whether two given regular expressions represent the same language

**505**What is the relationship between deterministic and non-deterministic machines for Context-Free Languages (CFL) and Regular Languages?

- (A) NFA = DFA in power; NPDA = DPDA in power
- (B) NFA = DFA in power; NPDA is strictly more powerful than DPDA
- (C) NFA is strictly more powerful than DFA; NPDA = DPDA in power
- (D) NFA is strictly more powerful than DFA; NPDA is strictly more powerful than DPDA

**Answer: (B)** NFA = DFA in power; NPDA is strictly more powerful than DPDA

**506**A grammar is defined as  $G = (V, \Sigma, R, S)$  where  $V = \{S\}$ ,  $\Sigma = \{a, b\}$ , and rules  $R$  are:

$S \rightarrow aSb \mid SS \mid \epsilon$

What language does this grammar generate?

- (A) All strings with equal number of a's and b's
- (B) All balanced strings of parentheses where 'a' represents '(' and 'b' represents ')'
- (C) The language  $\{a^n b^n \mid n \geq 0\}$
- (D) All binary strings ending with 'ab'

**Answer: (B)** All balanced strings of parentheses where 'a' represents '(' and 'b' represents ')'

**507**According to Rice's Theorem, any non-trivial semantic property of a recursively enumerable language is:

- (A) Decidable
- (B) Undecidable
- (C) Context-Free
- (D) Regular

**Answer: (B)** Undecidable

**508**Which of the following classes of languages is accepted by a Linear Bounded Automaton (LBA)?

- (A) Regular Languages
- (B) Context-Free Languages
- (C) Context-Sensitive Languages
- (D) Recursively Enumerable Languages

**Answer: (C)** Context-Sensitive Languages

## Web Protocols, Services, and Cloud Models

Web Technologies - 5 questions

**509**What is the primary operational difference between cookies and sessions in modern web development?

- (A) Cookies are stored server-side, whereas sessions are stored client-side.
- (B) Cookies are stored client-side (in the browser), whereas sessions are stored server-side.
- (C) Cookies can store binary files, while sessions can only store text.
- (D) Cookies do not expire, whereas sessions always expire within 10 minutes.

**Answer: (B)** Cookies are stored client-side (in the browser), whereas sessions are stored server-side.

**510** Which of the following is a key architectural constraint of REST (Representational State Transfer) web services that distinguishes it from SOAP?

- (A) REST requires stateful client-server connections.
- (B) REST enforces strict XML-only messaging standards.
- (C) REST relies on standard HTTP methods (GET, POST, PUT, DELETE) and is designed to be completely stateless.
- (D) REST requires an explicit WSDL file to define available methods.

**Answer: (C)** REST relies on standard HTTP methods (GET, POST, PUT, DELETE) and is designed to be completely stateless.

**511** According to NIST, a cloud computing model where a customer can deploy consumer-created or acquired applications onto the cloud infrastructure using programming languages, libraries, services, and tools supported by the provider is classified as:

- (A) Infrastructure as a Service (IaaS)
- (B) Platform as a Service (PaaS)
- (C) Software as a Service (SaaS)
- (D) Database as a Service (DBaaS)

**Answer: (B)** Platform as a Service (PaaS)

**512** Which of the following web application security vulnerabilities occurs when an attacker forces an authenticated user's browser to execute an unwanted, unauthorized command on a web application where the user is currently logged in?

- (A) Cross-Site Scripting (XSS)
- (B) SQL Injection (SQLi)
- (C) Cross-Site Request Forgery (CSRF)
- (D) Directory Traversal

**Answer: (C)** Cross-Site Request Forgery (CSRF)

**513** In an XML document, what is the role of a DTD (Document Type Definition)?

- (A) To compile the XML code into an executable format
- (B) To style the XML elements and define color/layout in the browser
- (C) To define the legal structure, elements, attributes, and nesting rules of the XML document
- (D) To encrypt the XML document for secure transmission

**Answer: (C)** To define the legal structure, elements, attributes, and nesting rules of the XML document

## HTML, CSS, and JavaScript Core Concepts

Web Technologies - 5 questions

**514** What is the output of the following JavaScript code snippet?

```
``js
var x = 10;
function test() {
 console.log(x);
 var x = 20;
}
test();
```
```

- (A) 10
- (B) 20
- (C) undefined
- (D) ReferenceError

Answer: (C) undefined

515 In the standard CSS Box Model, how does setting 'box-sizing: border-box;' affect the calculation of an element's total width and height?

- (A) The width and height properties include content, padding, and border, but exclude margins.
- (B) The width and height properties only include the content area, excluding padding, borders, and margins.
- (C) The width and height properties include content, padding, border, and margin.
- (D) The width and height properties are calculated automatically based on the viewport size.

Answer: (A) The width and height properties include content, padding, and border, but exclude margins.

516 Consider the following JavaScript asynchronous execution block. In what order will the letters be logged to the console?

```
``js
console.log('A');
setTimeout(() => console.log('B'), 0);
Promise.resolve().then(() => console.log('C'));
console.log('D');
``
```

- (A) A -> B -> C -> D
- (B) A -> D -> B -> C
- (C) A -> D -> C -> B
- (D) A -> C -> D -> B

Answer: (C) A -> D -> C -> B

517 In DOM (Document Object Model) event handling, what is the distinction between event bubbling and event capturing (trickling)?

- (A) Bubbling propagates the event from the target element up to the root, while capturing propagates the event from the root down to the target element.
- (B) Capturing propagates the event from the target element up to the root, while bubbling propagates the event from the root down to the target element.
- (C) Bubbling works on text nodes, whereas capturing works only on element nodes.
- (D) Bubbling is supported only in old browsers, while capturing is the modern standard.

Answer: (A) Bubbling propagates the event from the target element up to the root, while capturing propagates the event from the root down to the target element.

518 When parsing XML documents in web applications, what is the primary operational difference between a DOM parser and a SAX (Simple API for XML) parser?

- (A) DOM is an event-driven parser that reads XML line-by-line, while SAX loads the entire document into memory as a tree.
- (B) DOM loads the entire XML document into memory to build a tree structure, while SAX is a pull/event-driven parser that parses XML sequentially/line-by-line without loading the whole file.
- (C) DOM is only used for JSON, while SAX is used for XML.
- (D) DOM is synchronous, whereas SAX is completely asynchronous.

Answer: (B) DOM loads the entire XML document into memory to build a tree structure, while SAX is a pull/event-driven parser that parses XML sequentially/line-by-line without loading the whole file.

